

GE Healthcare

Discovery MR450 1.5T Magnet Enclosure Upgrade

Technical Publication



5500111
Rev 1

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

Call 1-800-548-3366 and use option 8.

Contact your local service coordinator for more information on this process.

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Discovery MR450 1.5T Magnet Enclosure Upgrade

Table of Contents

Table of Contents	3
Language Warning.....	4
Revision History.....	6
Upgrade Process	7
Processing Notes.....	9
Secure Hardware Confirmation.....	9
Equipment Notes.....	9
Tools, Consumables	10
Upgrade Components for M7000MH.....	11
1. Magnet Ground	17
2. Passive Shim Removal.....	18
3. Enclosure Bracket Install.....	20
4. I/O Assembly and Receiver Optic Cable Guide Installation	21
5. Side Tray Electronics Installation.....	23
6. Bracket Shelf Guide installation:	27
7. Turtle Ring Install	28
8. Gradient Coil Installation.....	32
9. XRGB Gradient Mounting Kit for LCC 1.5T Magnet.....	37
10. Installing the XRGB in the Magnet.....	39
11. Rear Foam Air Seal Install.....	49
12. Bus-Bar Assembly Bracket Installation.....	50
13. Gradient Cable Bus-bar Assembly Install	52
14. RF Installation.....	60
15. Rear End Bell Install	61
16. RF Tuning Process Check	62
17. Front Endbell Installation	63
18. Final RF Tuning Checks	64
19. Bridge Installation Preparation	65
20. Bridge Installation	66
21. Bridge Cover Installation.....	67
22. Enclosure Tools and Parts for Installation:.....	68
23. In Room Display Assembly Process (32 channel only)	69
24. PAC / SRI Sled Assembly	77
25. Install MR450 Modules on Side of Magnet	79
26. Install Cables on Side Electronic Modules	80
27. Cable Installation to magnet I/O Bracket	81
28. Install the PAC / SRI Sled.....	85
29. Magnet Bore Cable Installation.....	87
30. Installation of cables for MAG- DOCK_L.....	90
31. Front and Rear Arc Installation.....	91
32. Front and Rear Cover Installation	92
33. Front Cover Trim Installation	95
34. Logo Installation.....	96
35. Side Cover Assembly	97

Discovery MR450 1.5T Magnet Enclosure Upgrade

Language Warning

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

ATTENTION

- CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAINER CHEZ LE TECHNICIEN, L'OPERATEUR OU LE PATIENT DES BLESSURES DUES A DES DANGERS ELECTRIQUES, MECANIQUES OU AUTRES.

ATENCION

- ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.
- SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION
- NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELECTRICAS, MECANICAS O DE OTRA NATURALEZA.

WARNUNG

- DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

Discovery MR450 1.5T Magnet Enclosure Upgrade

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO
- IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- このサービスマニュアルは英語版しかありません。
- GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳業者はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意

- 本维修手册仅存有英文本。
- 非GE Healthcare公司的维修员要求非英文本的手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员、操作员或病人造成触电，机械伤害或其他伤害。

Discovery MR450 1.5T Magnet Enclosure Upgrade

Revision History

Date	Rev	Change Description
April 30, 2009	1.0	Initial Release of Document

Discovery MR450 1.5T Magnet Enclosure Upgrade

Upgrade Process

Task	Who	Time needed to complete
Magnet ramp-down - Cryogen Compressor with water supply must be kept running.	GE Service	1.5 days
Equipment to be Retained on Site: 1) MR450 compatible Coils 2) Operator Workspace Table 3) Surface coils that use the A port 4) Cryo gas lines 5) Cryo Compressor 6) Cryo Compressor cabling 7) Power cables for coldhead 8) Phantoms (including positioners, loaders and cart) 9) Pads 10) Save Info MOD or DVD 11) Operator chair 12) Options such as Music In Bore and Injector 13) Dock Bracket 14) Dock floor bolt/screw 15) Magnet and Vibromat (if applicable) 16) Magnet ground cable 17) Magnet Rundown Unit 18) Oxygen Monitor and cable (if applicable) 19) RF Door Switch and cable	GE Service/Installation Services	
Equipment to Temporarily Keep on Site 1) MDP 2) Magnet Monitor 3) Pen panel for cold head connections 5) Service platform	GE Service/Installation Services	
Equipment to be Removed 1) RFS Cabinet 2) HFD Cabinet 3) Monitor 4) non-MR450 Coils such as quad head. 5) Penetration Panel 6) All cables except those identified above 7) Gradient coil 8) Enclosure covers and end bells 9) Rear Pedestal	GE Service/Installation Services	
Remove existing magnet covers, bridge, endbells; Drain Gradient Coil coolant	GE Service	1 day
Remove existing gradient coil/RF coil	Florence & 2 GE Service reps	0.5 day
Remove Dock; Leave Dock Bracket for new install; Leave floor stud	GE Service	.2 days
Remove Passive shims from magnet	Florence & 2 GE Service reps	.2 day
"Remove all cabling w/exception of items listed in Equipment to be Retained on Site and Equipment to Temporarily Keep on Site - Magnet Room - Equipment Room - OW"	GE Service/Installation Services	1.5 days
Ship parts/cables back to Goldseal	GE Service	1 day

Discovery MR450 1.5T Magnet Enclosure Upgrade

Task	Who	Time needed to complete
Room Modifications for Dv	Hospital	Site Dependent
"Coordinate Pre-install start/end; travel arrangements; Communicate timing to customer Communicate ship date to Florence for Upgrade Kit Communicate need date for Service Tools (Service team)"	OFS/PJM	
"Order Tools for Mechanical Install - Gradient Cart - Gradient Insertion Tool - Turtle Ring Alignment Tool - FE Tool Kit"	GE Service	
Install M7000MH - Gradient coil - RF Coil - Magnet electronic components - Magnet cables - Enclosure covers	Florence & GE Service reps	
Turtle Ring Installation	Florence & 1 GE Service rep	0.5
XRMB Coil Installation	Florence & 2 GE Service reps	0.5
Gradient Busbar Installation	Florence & 1 GE Service rep	0.2
Magnet Component installation	Florence & 1 GE Service rep	0.5
Magnet I/O cabling	Florence & 1 GE Service rep	1
PAC / SRI installation	Florence & 1 GE Service rep	0.2
End-bell installation	Florence & 1 GE Service rep	0.5
Magnet Cover Installation	Florence & 1 GE Service rep	0.6
Bridge Installation	Florence & 1 GE Service rep	0.1

Discovery MR450 1.5T Magnet Enclosure Upgrade

Processing Notes

- Personal Protective Equipment relevant to the processing steps should be worn at all times.
- Cotton, vinyl or latex gloves must be worn during all process steps to protect hands and maintain cleanliness of copper parts.
- Check all threads on bolts and parts for damage before use.
- Check all o-rings for damage before installation.
- Check all threads on bolts and parts for damage before use.
- Assure that all void areas during the coil build are filled with glass filler material. Keep all voids to less than 1.0 in² and no thicker than 3mm (0.125").
- Conventions for the angular orientation of the XRMB coil is 0 degrees on top. While viewing the coil from the service end 90 degrees is CW and when viewing the coil from the patient end 90 degrees is CCW.
- The Y-axes are centered on 0 and 180 degrees, the X-axes on 90 and 270 degrees.
- Cleanliness is critical, both for insuring gradient coil performance is not degraded as well as for insuring that the mold seals properly for impregnation. Constant care must be taken to keep the coil, boards, and components clean during storage and assembly. Take care also to clean all o-ring surfaces thoroughly. Do not allow vacuum grease to contact coil surfaces.

Secure Hardware Confirmation

- Confirmation that all hardware mounted to the structure (component) is secure and tight is required.
- Each BOLT / NUT / SCREW HEAD should be inspected for full engagement (secure & tight); once full engagement is confirmed, each BOLT / NUT / SCREW HEAD should be marked with a visible **BLUE** dot or mark on the exposed surface of the BOLT / NUT / SCREW HEAD.

Equipment Notes

- Photographs shown are typical for all systems – however, valve location and/or panel fronts and/or control dials may differ slightly.
- Valves & control dials are labeled according to processing steps; the appropriately labeled valve or control dial should be in the same relative area as that which is shown in the corresponding photo.

Discovery MR450 1.5T Magnet Enclosure Upgrade

Tools, Consumables

TOOLS

Slotted screwdriver	
Gradient Install Tool	2164744-6
Ring Alignment Fixture	2299496
Crescent Wrench	
Gradient Install Tool Plate	5175963
Vacuum cleaner	
Wire Tie Cutters	

CONSUMABLES

Isopropyl Alcohol	
Loctite	271& 268
Clean, Lint-Free towels (Kimwipes™)	
Never-seize	46-294151P8
Cable Ties	46-252283P68
RTV Adhesive	46-294151P48
Engineering Bill of Materials: Assembly	5151372-2 rev 2:

Discovery MR450 1.5T Magnet Enclosure Upgrade

Upgrade Components for M7000MH

The following table contains a list of assemblies and components that are delivered to the site for upgrading an existing LCC magnet to Discovery MR450 enclosure. The instructions for installing the parts are provided on the following pages.

Name	Level	Qty	Description
5336312	1	1.0	collector, xrbm gradient mounting, 1.5t LCC magnet
5170015	2	2.0	positioning block, LCC radial, xrbm gradient
5170098	2	4.0	positioning block, LCC axi-radial, xrbm gradient
5170312	2	5.0	mounting wedge, LCC inward, xrbm gradient
5170035	2	4.0	mounting wedge, LCC outward, xrbm gradient
5326293-4	2	15.0	screw, m10x30 hex head cap, brass uns c27000, per astm f468m
5146169	2	15.0	M10x24mm Washer
46-294151P11	2	1.0	LOCKTITE 271, LOCKTITE NO. 27121, 10 ML. BOTTLE
5307840	1	1.0	collector, xrbm busbar assembly replacement
5305786	2	1.0	assembly, machined leadboard , braided cable version, xrbm gradient
46-294167P96	2	8.0	bushing, leadboard vibration damping, xrbm gradient
5268439	2	12.0	shim, power connection leadboard axial adjustment, xrbm gradient
5214524-6	2	20.0	washer, m10 large od nordlock, a4 stainless
2341885	2	22.0	NUT M10 FULL BRASS
5321557	2	1.0	crate, cable leadboard assembly, xrbm gradient
5308479	2	1.0	tool, leadboard installation gage
2144093-11	1	1.0	collector, xrbm fru gradient coil with hos, gray cart with cradle
2134810	2	1.0	BRM CART FOR A STANDARD ROOM
2134810-3	2	1.0	ALUMINUM CRADLE FOR BRM CART
5307835-2	2	1.0	collector, bay tested xrbm gradient fru with hos
5308479	2	1.0	tool, leadboard installation gage
5321375	2	4.0	label, hdv magnet or xrbm fru gradient shipping temperature
5313540	2	1.0	Foam Seal Rear Gradient Coil
5326377	2	1.0	XRMB FRU Ship Packaging Collector
5326101	2	1.0	FRU Return Packaging Collector
2208999-14	1	1.0	DVMR Body RF Coil - Orange & Black Case
5168509	2	1.0	HDv 1.5T RF body coil assembly
2208999-14	3	1.0	DVMR 1.5T Body RF Coil - Orange & Black Case
2226313-9	2	1.0	1.5T HDv RF COIL FRU CASE, ORANGE
2226312-5	2	1.0	2226312-2 & -3 SHIP~G SKID W/STRAPS
5147904	1	12.0	assembly, passive shim tray, xrbm gradient
5147899	2	1.0	tray, passive shim, xrbm gradient
5147900	2	1.0	fitting, shim tray retaining, xrbm gradient
5306007-3	2	1.0	setscrew, m8 x 12 flat slotted flat point, short glass filled nylon
5155291-2	1	1.0	HDv Service Platform
5193795	1	1.0	DVMR Magnet Monitor Cable Kit
5197321-2	2	1.0	Run M3023 SPW-J117 to Mag-FJ2,14000mm
5194040	2	1.0	Run E3023 SPW-J117 to MON-J8,17000mm

Discovery MR450 1.5T Magnet Enclosure Upgrade

Name	Level	Qty	Description
5172921	2	1.0	Run E3022,MON-J7 to SPW-J116, Instrumentation Box
5169162	2	1.0	Run E3024, MON-J9 to MON-FJ4 and MON-FJ3, Flow Monitoring
5175597	2	1.0	Run E3026, HEC-FL-Temp to MON-FJ3, Compressor Interface
5169404-7	2	1.0	Run E3020 Pen-J13 to Mon-J10,17000mm
5174110	2	1.0	Run E3025, CRY to MON-FJ4, Flow Monitor-Sumitomo
5169277-12	2	1.0	Run M3022, SPW-J116 to MAG-P403, 14000 mm
5167907-5	2	1.0	Run E3034 MON-J5 to Network, 30000mm
5115594-3	2	1.0	POWER CORD IEC 60320 C13 to NEMA 5-15P 72 inches long
2222797	2	1.0	(830) CABLE, MAGNET ROOM INTERFACE
5245331	1	1.0	Side Electronics Tray Assembly
5244552	2	1.0	Side electronics Mounting plate 16-32 channel
5245338-2	2	1.0	side electronics isolation gasket rear
5245338	2	1.0	electronics tray isolation gasket front
5245332	2	1.0	side electronics foam isolator
5245332-2	2	1.0	side electronics foam isolator
5266914	2	6.0	Micro Blind Hole Side Entry Wire Saddle Richco MBHWSSE-1-01
5263916	1	2.0	Kit, 3.0T HDv Field Passive Shim
5159201-3	2	60.0	passive shim, half width, .020 inch thick, xrm b gradient
5159201-2	2	40.0	passive shim, .020 inch thick, xrm b gradient
5159201	2	30.0	passive shim, .031 inch thick, xrm b gradient
2379197-4	2	30.0	SHIM, 2.0MM, HALF WIDTH, M4 THRU
2379197	2	30.0	SHIM, 2.0MM, M4 THRU
2379197-5	2	30.0	Shim,Carbon steel, 3mm thk, M4 thru
5212343	2	100.0	square nut, passive shim, xrm b gradient
5214479	2	120.0	screw, passive shim, xrm b gradient
5265652	1	1.0	Kit, Field Spares, 3.0T HDv
46-281101P9	2	2.0	1.487 ID X .103 THK., O-RING #2-128
46-281101P5	2	2.0	5.737 ID X .103 THK, O-RING #2-162
46-281101P6	2	2.0	9.487 ID X .103 THK, O-RING #2-177
46-281101P10	2	2.0	6.237 ID X .103 THK., #2-164
46-294104P1	2	2.0	NUT, VACUUM QUICK COUPLING
46-294105P1	2	2.0	RING, QUICK COUPLING RETAINING
46-260389P1	2	4.0	SILICONE, PARKER#2-116
46-260389P2	2	2.0	SILICONE, PARKER #2-014
46-260902P5	2	1.0	PLUG, 1/4 OD TUBE FITTING, BRASS SWAGELOK CAT #B-400-P
2203803	2	2.0	SPLIT O-RING
46-318068P1	2	6.0	BRONZE WOOL USED FOR RF SHIELDING CDA 649 BRONZE 4 TO 5 IN. WIDE X 12 IN. LONG MINIMUM.
46-252283P58	2	4.0	CABLE TIE WRAP, PANDUIT PLT7LH-L
46-260342P10	2	5.0	QUICK SEAL CONNECTOR O-RING SILICONE PARKER TYPE 2-112
46-252065P24	2	1.0	VACUUM GREASE,DOW CORNING
46-260389P4	2	1.0	O-RING, SILICONE, PARKER 2-205
46-252065P141	2	1.0	AEROQUIP O-RING, FLAT, RUBBER
5310483	1	1.0	Kit, HDv Enclosures Upgrade
5151771-2	2	1.0	Side Cover Assembly, Right, White
5169762	2	1.0	Front Bridge Cover Assembly, XRMB
5173750	2	1.0	Rear Cover Assembly, Left
5175740	2	1.0	Rear Cover Assembly, Right
5177586-2	2	1.0	Front Trim Ring, Right, Assembly

Discovery MR450 1.5T Magnet Enclosure Upgrade

Name	Level	Qty	Description
5177586	2	1.0	Front Trim Ring, Left, Assembly
5180596	2	1.0	Front Cover for LCD, Right, Assembly
5212760	3	1.0	Vacuum Port Cover Assembly, LCD Front Cover
5180596-2	2	1.0	Front Cover for LCD, Left, Assembly
5182894	2	1.0	Laser Cover, Top, Assembly
5191116	2	1.0	Rear Left Arc Cover Assembly, HDv
5191677	2	1.0	Rear Right Arc Cover Assembly, HDv
5194882	2	1.0	Front Right Arc Cover Assembly, HDv
5194883	2	1.0	Front Left Arc Cover Assembly, HDv
2219096	3	1.0	ECG EXTENSION CABLE_MG2A11-ECG
2219097	3	1.0	EXTEN. CABLE_MG2A11 PERIPHERAL PULSE
46-317758P18	3	2.0	10 foot tubing, pneumatic, with bulb bonded to tubing
5310485	1	1.0	Kit, HDv Bracket Upgrade
5160992	2	1.0	Rear Bracket Assembly, Left
5160992-2	2	1.0	Rear Bracket Assembly, Right
5181498	2	1.0	I_O Assembly for DVMR
5195461	2	1.0	Front Bracket Assembly, Left
5195461-2	2	1.0	Front Bracket Assembly, Right
5224568	2	1.0	HDV RECEIVER OPTIC CABLE GUIDE ASM
5232034	2	1.0	Side Cover Bracket Asm RH
5232034-2	2	1.0	Side Cover Bracket Asm LH
5133768-3	2	1.0	In Room Display Mount
5178328	2	1.0	Laser Mounting Bracket, Top, Assembly
5182762	2	1.0	Monitor Interface Cover
5310486	1	1.0	Bridge Kit, HDv Upgrade
2293540	2	2.0	TWIN SUPPORT LOCK TOP
2293541	2	2.0	TWIN BRIDGE SUPPORT SLIDER
5168288	2	1.0	SUPPORT BRIDGE FRONT
5181729	2	1.0	BRIDGE FRONT DVMR ASM
5310487	1	1.0	End Bell Kit, HDv Upgrade
5173510	2	1.0	Rear End Bell XRMB Assembly
5183628	3	1.0	Run M3347 connect to M3348, Comfort Module Speaker Link Cable, 916mm
5173983	2	1.0	Front End Bell XRMB Assembly
5183293	3	1.0	Run M3345,XRMB MICROPHONE ASM
5336331	1	1.0	Turtle Bracket Kit, HDv Upgrade
5191537	2	1.0	Turtle Bracket Asm Rear XRMB, 1.5T
5191982	2	1.0	Turtle Bracket Asm Front XRMB, 1.5T, With LCD
5336332	1	1.0	Cable Kit, HDv Upgrade
5149629	2	1.0	Run M3366, Cable, Scan Room Monitor Control Y
5166749-4	2	1.0	Run M1317, MAG-J8 to PED-HUB-Slot 11-J17, Rx Lo - Exciter Reference Clock, Twinax, 8300mm
5169566	2	1.0	Run M3367, MAG-TDM-J1 to MAG-RX1 , 44-pin HD Sub-D, 560mm
5170095	2	1.0	RUN M3337 MAG-SRI-J4 to MAG-LCP-J11, 5000mm and MAG-RCP-J11, 6600mm
5170192	2	1.0	Run M3369, MAG-J3 to MAG-SRI-J21, SRI Signal, 5600 mm
5170365	2	1.0	Run M3336, MAG-J4 to MAG-SRI-J1, SRI Power, 5600mm
5171941-2	2	1.0	Run M3339, MAG-J1 to MAG-SRI-J33, Console Intercom, 5600mm
5172496	2	1.0	Run M3353, MAG-J15 to PED-HUB-SLOT11-J9, Transmt Swtch, 8300mm

Discovery MR450 1.5T Magnet Enclosure Upgrade

Name	Level	Qty	Description
5172512	2	1.0	Run M3352, MAG-J7 to PED-HUB-SLOT11-J10, Unblank, 8300mm
5172512-2	2	1.0	Run M3340, MAG-J17 to PED-TIF-J4, Halls Signal, 8900mm
5172679-8	2	1.0	Run M1313 MAG-J9 to MAG-BORE-J6,6250mm
5172679-9	2	1.0	Run M1314, MAG-J10 to MAG-BORE-J7,6250mm
5172955-4	2	1.0	Run M1316, MAG-J13 to PED-Hub-SLOT11-J16, NB Exciter, 8300mm
5172994	2	1.0	Run M3364, MAG-J5 to MAG-DCK_L-J3, Dock Power, 4400mm
5174620	2	1.0	RUN M2301, MAG-RX1-J9 to MAG-SRI-J17, Dab Out, Simplex F0, Plastic, 2000mm
5176536	2	1.0	Run M1325, MAG-J12 to PED-IO-J2 , A RF Transmit 3747mm Ref, Phase Length Controlled
5176536-2	2	1.0	Run M1326, MAG-J14 to PED-IO-J1, P1-RF Transmit, 4088mm, Ref Dimension Only
5177909	2	1.0	Run M3357 MAG-J16 to PED-J1, Long Drive Power, 8600mm
5178261	2	1.0	Run M1319, Body Transmit, MAG-HYB-J1 to RUN M1001, 1.5T, 3065mm
5180907	2	1.0	Run M3350 MAG-SRI-J10 to MAG-BORE, Bore Temperature, 3000mm
5180923	2	1.0	Run M3342, MAG-SRI-J23 to MAG-BORE, Patient Alignment Lights
5181092	2	1.0	Run M3344, MAG-SRI-J22 to MAG-BORE-GRAD_TEMP and MAG-BORE-LIGHT, Gradient Temp 2000mm and Bore Light 3500mm
5183773	2	1.0	Run M3348 MAG-SRI-J34 to Run M3347, Comfort Module Speaker Cable, 4600mm
5189419	2	1.0	Run M3370 MAG-SRI-J3 to MAG-DCK_L-J2, Dock Control, 2200mm
5194965	2	1.0	Run M3360 MAG-J6 to MAG-PAC-J2, PAC, HDv Option, 5600mm
5330595	2	1.0	Run M1322 MAG-HYB-J4 to MAG-BORE-J2, Body Transmit, 3446mm
5330595-2	2	1.0	Run M1323 MAG-HYB-J3 to MAG-BORE-J3, Body Transmit, 3446mm
5172679-18	2	1.0	Run M1343 MAG-J118 to MAG-BORE-J8, 6250mm
5172679-19	2	1.0	Run M1344 MAG-J119 to MAG-BORE-J9, 6250mm
5169277-5	2	1.0	Run M3341 MAG-SRI-J20 to MAG-DCK_L-J1, Landmark, Touch and Go, 2500mm
5169566-2	2	1.0	Run M3368, MAG-TDM-J2 to MAG-RX2, 44-pin HD Sub-D, 1360mm
5310490	1	1.0	Gradient Installation Kit, HDv Upgrade
5189608	2	1.0	cover, gradient to power supply leadboard terminal safety, xrbm
5197228	2	1.0	bracket, gradient power connection leadboard, xrbm gradient
5197229	2	1.0	bottom clamp, gradient power cable strain relief, xrbm gradient
5197230	2	3.0	top clamp, gradient power cable strain relief, xrbm gradient
5309075	2	1.0	base, power connection safety cover, xrbm gradient
5309076	2	1.0	cover, power connection safety, xrbm gradient
5338538	1	1.0	Hardware Kit, MR450 Upgrade
2109866-26	2	10.0	SCREW HEXAGON HEAD 10 MM 80 MM
2109866-50	2	3.0	SCREW HEXAGON HEAD
2109867-13	2	4.0	SCREW HEX SOCKET CAP M6X16 A2-70
2109867-21	2	11.0	SCREW HEX SOCKET CAP M10X16 A2-70
2109867-23	2	58.0	SCREW HEX SOCKET CAP M10X25 A2-70
2109867-24	2	14.0	SCREW HEX SOCKET CAP M10X30 A2-70
2109867-6	2	4.0	SCREW HEXAGON SOCKET_M4 X 20 SST
2109872-12	2	4.0	SCREW BUTTON HEAD
2109872-6	2	5.0	SCREW BUTTON HEAD 6 MM 12 MM
2109872-60	2	2.0	M6x35 Hex Socket Button Head Cap Screw, Stainless Steel
2109873-14	2	1.0	SCREW PAN HEAD Z M4X10 A2-70
2109873-24	2	8.0	SCREW PAN HEAD 6 MM 12 MM

Discovery MR450 1.5T Magnet Enclosure Upgrade

Name	Level	Qty	Description
2109873-28	2	4.0	SCREW PAN HEAD Z M6X30 A2-70
2109875-4	2	14.0	NUT HEXAGON M10 A2-70
2109878-2	2	4.0	WASHER PLAIN - NORMAL 4.3 MM 9 MM
2109878-4	2	18.0	WASHER FLAT ISO7089 M10 A2
2109880-2	2	4.0	WASHER LOCK - SPRING 4.1 MM 7.6 MM
2116325-2	2	2.0	STUD SST M10 X 65
2167321	2	2.0	WASHER, NEOPRENE RUBBER
2229794	2	4.0	M6 NYLON WASHER
2293956	2	4.0	HEX NUT, NYLON, M10X1.5MM, 9.8 HIGH
2370892	2	2.0	G10 FIBERGLASS FLAT WASHER
2381227-7	2	2.0	SCREW, SS HEX SOCKET CAP, M5 X 30
2381267-9	2	11.0	WASHER, SS SPRING LOCK, M10
46-136246P11	2	1.0	5/8~ WIDE, BLK ADH BACK, VELCRO HOOK
46-136246P12	2	1.0	0.625 WIDE X 25 YD,BLK,ADH BACK,LOOP (REV TK, 2/92).
46-208746P9	2	1.0	CABLE TIE PANDUIT TM3510C
46-208747P1	2	10.0	ADHESIVE MOUNT - CABLE TIE PANDUIT CORP 5.5~ X .140~ SELF-LOCKING CABLE TIE. 1/16~ TO 1-1/4~ WIRE
46-208758P2	2	15.0	BUNDLE RANGE. 30LB TENSILE STRENGTH, NYLON MATR~L. 11.08~X .140~ SELF-LOCKING CABLE TIE 1/16~ TO 3~ WIRE BUNDLE
46-208758P9	2	5.0	RANGE. 40LB TENSILE STRENGTH, NYLON MATR~L.
46-220191P1	2	15.0	CLAMP FOR FLAT CABLE
46-252283P68	2	50.0	CABLE TIE, .190 WIDE X 14~LG, NYLON
46-294167P73	2	60.0	FLAT WASHER, METRIC, NYLON, M10 SCREW, 10.5MM ID, 22MM OD, 2.5MM THK MCMASTER #95610A200
46-312342P48	2	1.0	M6 X 1.0 X 12MM PHILLIPS CROSS RECESSED PAN HEAD STAINLESS STEEL MACHINE SCREW
46-318508P23	2	6.0	HEX HEAD CAP SCREW, M10 X 40, FULL THREAD.
46-318508P25	2	2.0	HEX HEAD CAP SCREW, M10 X 50, FULL THREAD.
46-318508P27	2	4.0	HEX HEAD CAP SCREW, M12 X 20, FULL THREAD.
5119801-5	2	4.0	nut, stainless steel nylon lock, a2, m10 x 1.50
5125751-2	2	1.0	100MM GE Logo stamped aluminum
5146165	2	4.0	M10 Washer Phenolic, 11mm ID, 50mm OD, 8mm THK
5146169	2	16.0	M10x24mm Washer
5182822	2	2.0	label, electrical shock and general hazard, xrbm gradient
5194162-2	2	4.0	Flanged Bearing 12 x 18 x 10 non-metallic
5194695-2	2	8.0	#3-56 x .31 SS Phlps Pan Head Screw
5197019	2	1.0	label, passive shim trays, xrbm gradient
46-252283P70	2	25.0	CABLE TIE MOUNT, PANDUIT ABM3H-A-L
2116325-14	2	12.0	STUD, M10 X 50MM, STAINLESS STEEL
2332574	2	1.0	ADHESIVE DOUBLE BEAD SYRINGE
5199899	2	8.0	Nylon Push Rivet, white, 5.1 dia mtg hole grip range 7,5-8,5
5265376	2	1.0	DISPENSER, EPX PLUS II, APPLICATOR Mixing Nozzle 9742, 3M Scotch-Weld EPX applicator, 50 ml, pn 62974299356
46-252065P154	2	1.0	62974299356
2109878-3	2	2.0	WASHER PLAIN - NORMAL 6.4 MM 12 MM
5323651	2	2.0	Foam End Bell Side
5323652	2	1.0	End Bell Foam Center
2109867-15	2	4.0	SCREW HEX SOCKET CAP M6X25 A2-70
5309674	2	12.0	PAD-RADIAL
5288468-4	2	2.0	Screw M6 x 16 slotted pan head, SST A2 or A4
46-318508P24	2	20.0	M10 x 45 Hex Cap Screw, SST, Full thread
5335926	2	1.0	Base, power connection safety cover

Discovery MR450 1.5T Magnet Enclosure Upgrade

Name	Level	Qty	Description
5310482	1	1.0	Kit, HDv Upgrade - Electronics
5264147	2	1.0	Pac Tray Asm
5245152		31.0	TRAY SRI PAC MTG
5268331		32.0	Pac Tray Slide Bar
5110749-2		316.0	3M Dual Lock Reclosable Fastener SJ3560
5270612	2	1.0	Pac Shelf Guide Asm
5245151		31.0	BRACKET SHELF GUIDE
2109873-24		32.0	SCREW PAN HEAD 6 MM 12 MM
5304072		31.0	Pac Guide Angle RH
5110749-2		36.0	3M Dual Lock Reclosable Fastener SJ3560
5112668-2	2	1.0	PHYSIOLOGICAL ACQUISITION CONTROLLER WITH INDEPENDENT VECTOR GATING
5160929	2	1.0	SRI4 with Remote Intercom Board 3
5165957	2	1.0	TDM Module
5168798	2	1.0	1.5T Hybrid Body Hybrid ASM with Bracket
5174169	2	1.0	HDv Remote Receiver Module
5273219	2	2.0	AIR FLOW SENSOR FRONT DVMR
5266150		31.0	AIR FLOW SENSOR BLOCK DVMR
5270967		31.0	AIR FLOW SENSOR CLAMP
2109873-14		32.0	SCREW PAN HEAD Z M4X10 A2-70
5266979	2	1.0	AIRFLOW SENSOR
5270612-2	2	1.0	Pac Shelf Guide Asm LH
5245151-2		31.0	Pac Shelf Guide LH
2109873-24		32.0	SCREW PAN HEAD 6 MM 12 MM
5304072-2		31.0	Pac Guide Angle LH
5110749-2		36.0	3M Dual Lock Reclosable Fastener SJ3560
5318637	1	1.0	Shipping crate, xrb gradient fru
5325871	1	2.0	Time-Temperature Sensitive Product Placard

Discovery MR450 1.5T Magnet Enclosure Upgrade

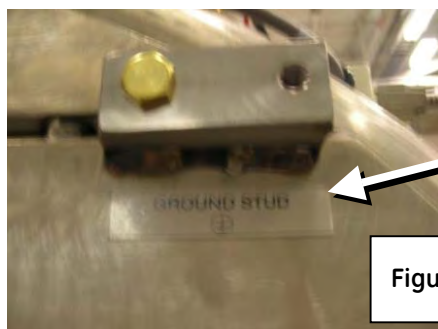
Process Description

Note: *This Magnet Enclosure Upgrade procedure assumes the magnet enclosure covers, gradient coil, and all associated electronic equipment and cables have been removed and returned to GE Healthcare.*

1. Magnet Ground

The magnet must be grounded using either of the following steps.

- A. Route and connect new magnet ground cable to Ground Block located at upper rear of magnet and connect with M10 Brass Screw (5110638-61) and M10 Washer (2381267-9). **(FIG 1).**



NOTE: Make sure that the grounding block has been labeled.

Figure 1

- a) Make sure there is no paint on any of the contact surfaces. If paint is present, it must be removed to make sure there is good contact for grounding. **(FIG 1).**
 - b) Label the grounding block.
- B. Reuse, and reroute if necessary, the existing ground cable and connect to ground stud on inside of right rear magnet leg on the side opposite of the SRI/PAC assembly. **(Fig 2)**

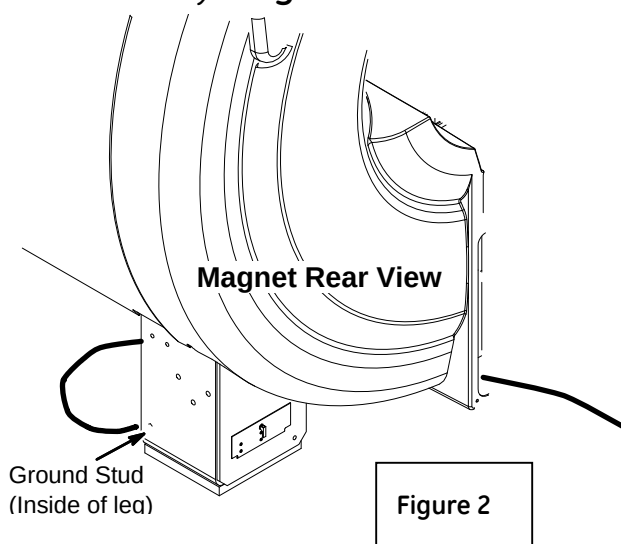


Figure 2

Discovery MR450 1.5T Magnet Enclosure Upgrade

2. Passive Shim Removal

CAUTION

Magnet MUST be ramped down in order to safely remove Passive Shims.

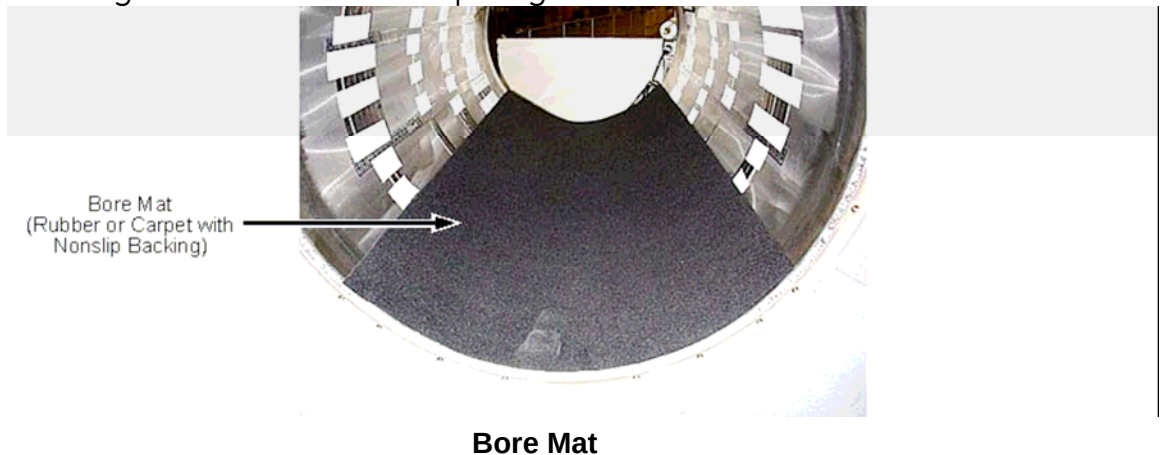
WARNING

PASSIVE SHIMS HAVE SHARP EDGES THAT MAY CAUSE INJURY.

- **WEAR LEATHER GLOVES, A LONG SLEEVE SHIRTS AND PANTS WHILE REMOVING PASSIVE SHIMS.**
- **USE A MAT TO PROTECT YOURSELF AND YOUR CLOTHS.**
- **DO NOT HOLD PASSIVE SHIMS EDGES. HOLD THEIR SURFACES INSTEAD.**
- **MAINTAIN CONTROL OF SHIMS AND SECURELY HOLD TOOLS.**

A. Remove all individual passive shims from the magnet bore.

- a) Place a bore mat of rubber or carpet with non-slip backing inside the bore. See illustration below. The mat is to protect yourself and your clothing from cuts from sharp-edged shims.



Discovery MR450 1.5T Magnet Enclosure Upgrade

- b) Use a small titanium screwdriver or a brass putty knife to loosen one end of each existing passive shim. See illustration below.
- c) Use titanium needle nose pliers or titanium channel lock pliers while pulling free a loosened shim. Securely control the pliers and passive shim as each shim becomes completely unglued from the magnet bore.
- d) Remove all old shims from the magnet room.



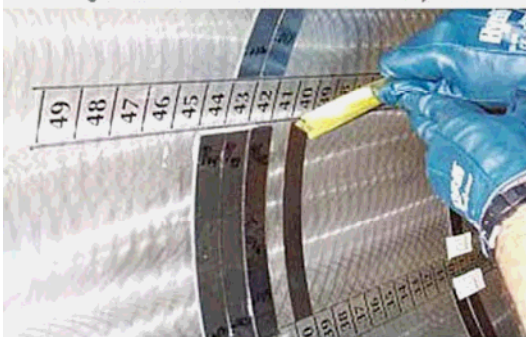
CAUTION

Potential Skin Damage

Sharp shim edges and uncontrolled tools can cut or damage skin.

- Wear leather gloves while removing old shims.
- Do not hold shim edges while removing old shims.
- Hold tools securely.

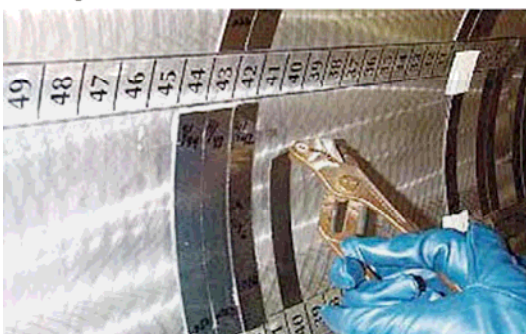
Removal of Old Shims
Using Titanium Screwdriver or Brass Putty Knife*



Do Not Hold Shim **EDGES**
While Removing Old Shims!



Removal of Old Shims
Using Titanium Needle Nose or Channel Lock Pliers



Do Hold Shim **SURFACES**
While Removing Old Shims!



* Brass Putty Knife to be used on magnets $\leq 1.5T$ only.
Titanium tools must be used with magnets $> 1.5T$.

26 Oct 2006

Removing Existing Passive Shims

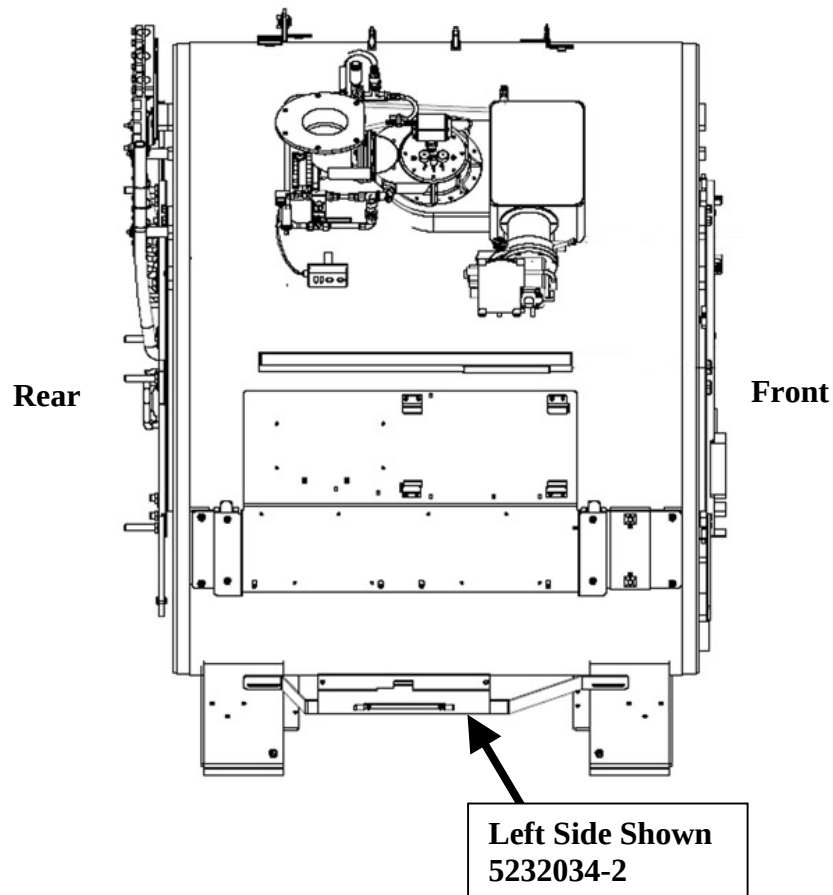
Discovery MR450 1.5T Magnet Enclosure Upgrade

3. Enclosure Bracket Install

A. Components:

- a) Left bracket 5232034-2
- b) Right bracket 5232034
- c) M10 X 25 SHCS 2109867-23
- d) M10 Nylon Washer 46-294167P73

- B. Install Left (5232034-2) and Right Side (5232034) Enclosure Bracket to upper holes in magnet feet using (2) M10 Screws and Washers at each end. (FIG 11)



Discovery MR450 1.5T Magnet Enclosure Upgrade

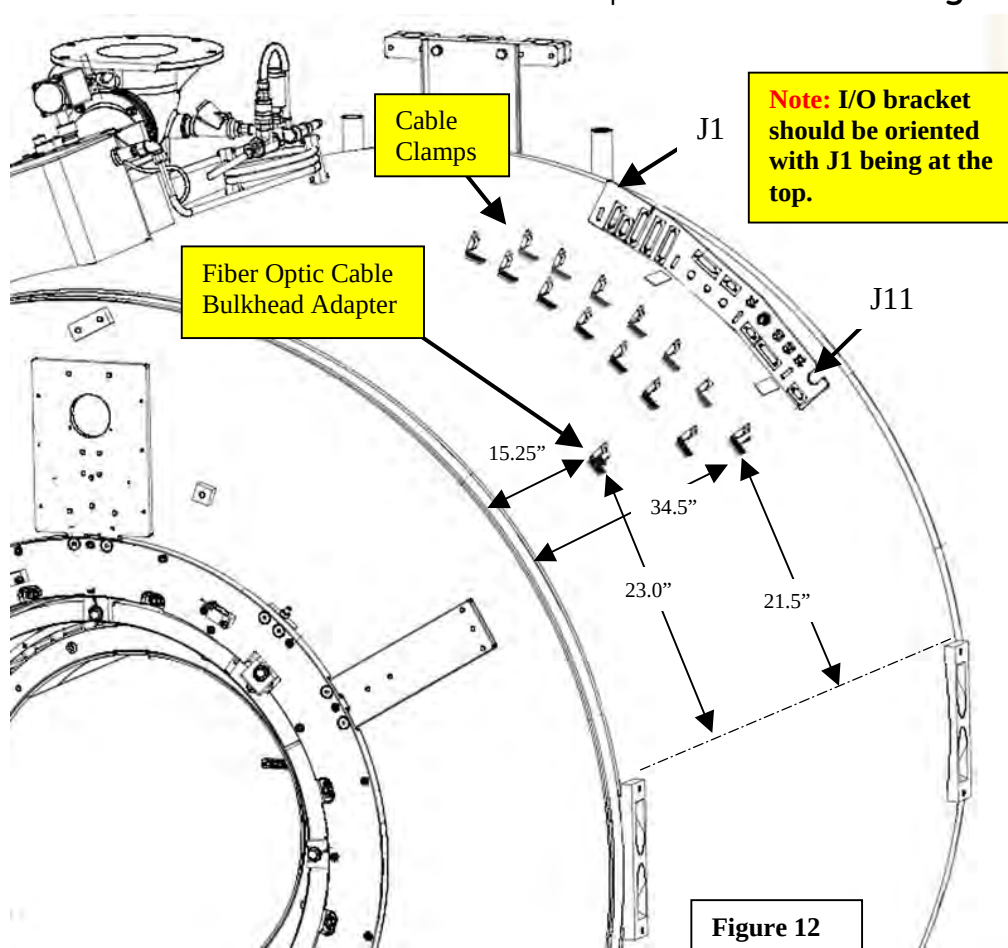
4. I/O Assembly and Receiver Optic Cable Guide Installation

A. Components

- | | |
|---------------------------------------|-------------|
| a) 3M Dual Lock | 3M SJ3560 |
| b) I/O Cable Guide Asm | 5181498-2 |
| c) Fiber Optic Cable Bulkhead Adapter | 5122507 |
| d) Cable Clamps | 46-220191P1 |

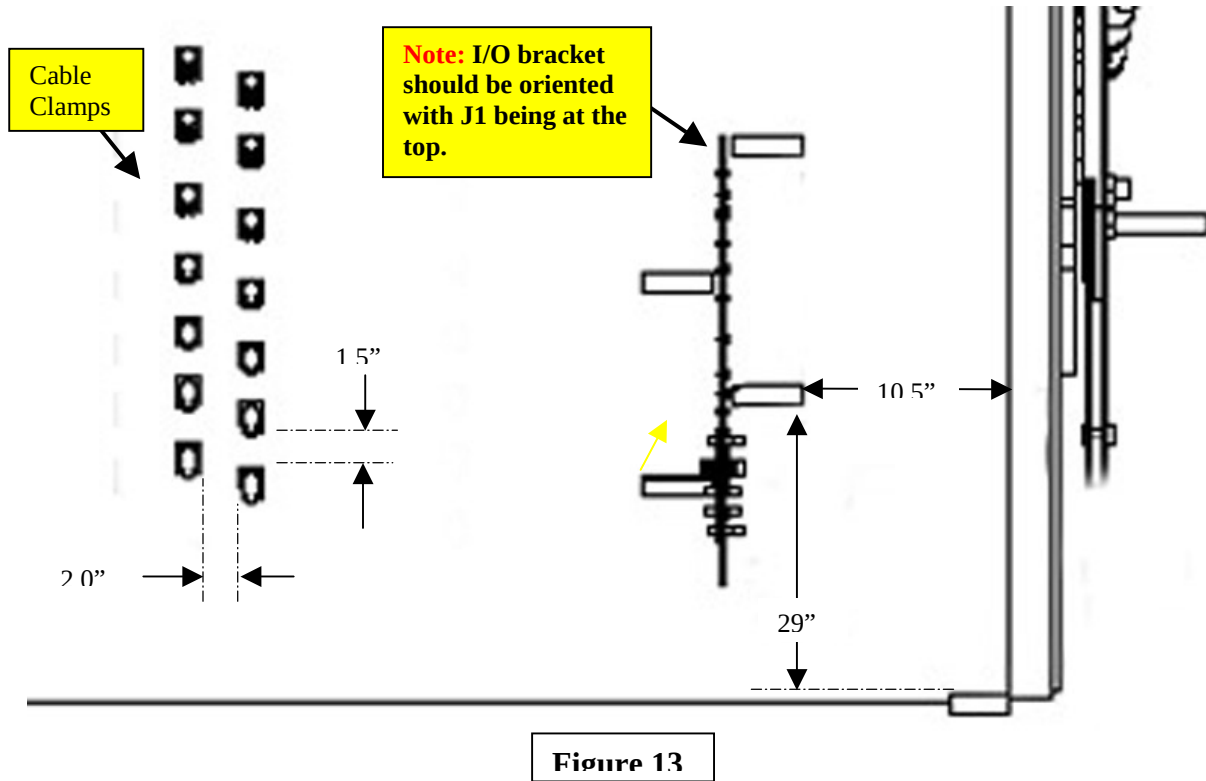
B. Position and install as dimensioned the Fiber Optic Bulkhead Adapter (5122507) using cable clamp (46-220191P1).

C. Position and attach initial Cable Clamp as dimensioned. **(Fig 12)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

- D. Position and attach the I/O Cable guide (5181498-2) using Dual Lock (3M SJ3560) IB **(Fig 13)**.
- E. Attach the remaining 13 Cable Clamps using the spacing dimensions as indicated. **(Fig 13)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

5. Side Tray Electronics Installation

A. Components

a) Side Electronics Tray Asm	5316129
b) Side Electronics Bracket Front	5316115
c) Side Electronics Bracket Rear	5316116
d) Side Electronics Cable Guide	5321625
e) (4) Isolation Pads	5316121
f) (2) Isolation Sheets	5316122
g) (2) Cable Tie Mount, Panduit	46-252283P70
h) (8) M10 Nylon Flat Washer	46-294167P73
i) (8) M10 x 25 Hex Socket Cap Screw	2109867-23
j) (2) M6 x 12mm Hex Socket Screw	2109867-12
k) (4) Polymer Flanged Bushing	5316128

- B. Check if the threads of the mounting holes in the radial support pads are damaged. Re-tap the holes as necessary (*Fig 14*).

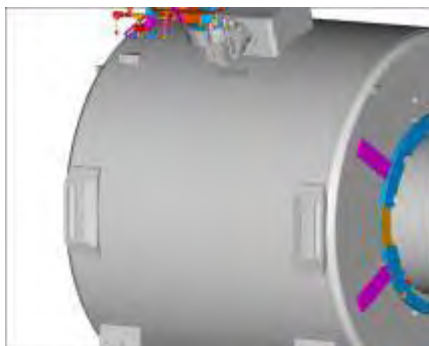
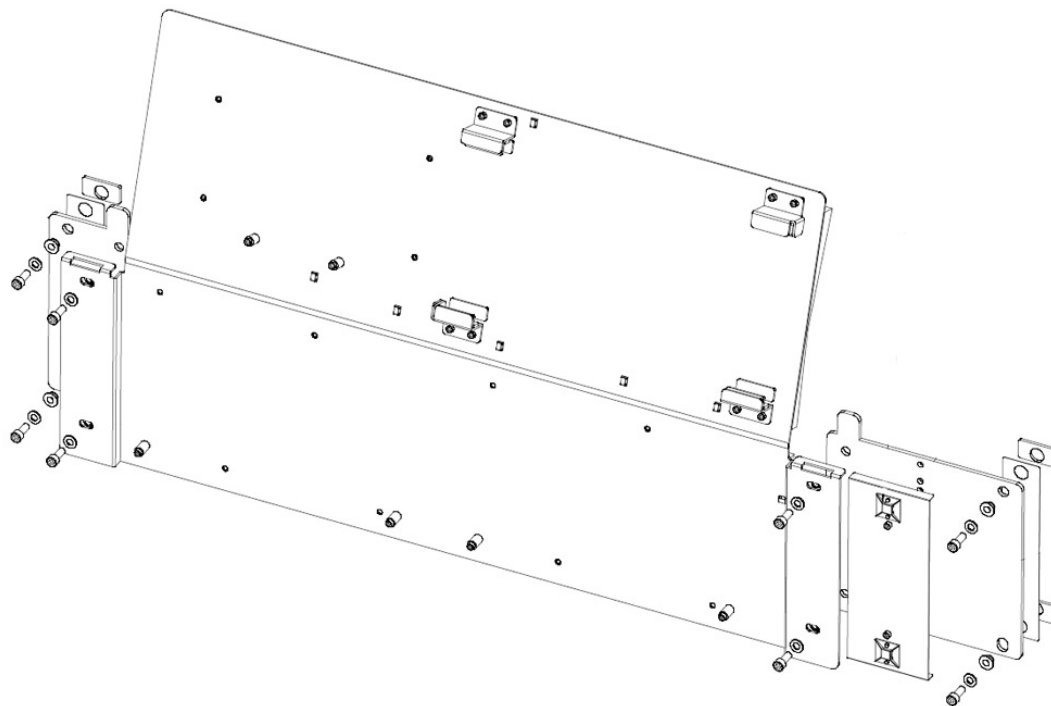


Figure 14

Discovery MR450 1.5T Magnet Enclosure Upgrade

C. Side Electronics Tray Components for assembly.



D. Peel the backing of the isolation sheet (5316122) and apply it onto the magnet side of the rear and front attachment brackets as shown. Peel the adhesive backing from the isolating pads (5316121) over each of the attachment hole locations. **(Fig 7)**

Rear Bracket – 5316116

Front Bracket 5316115

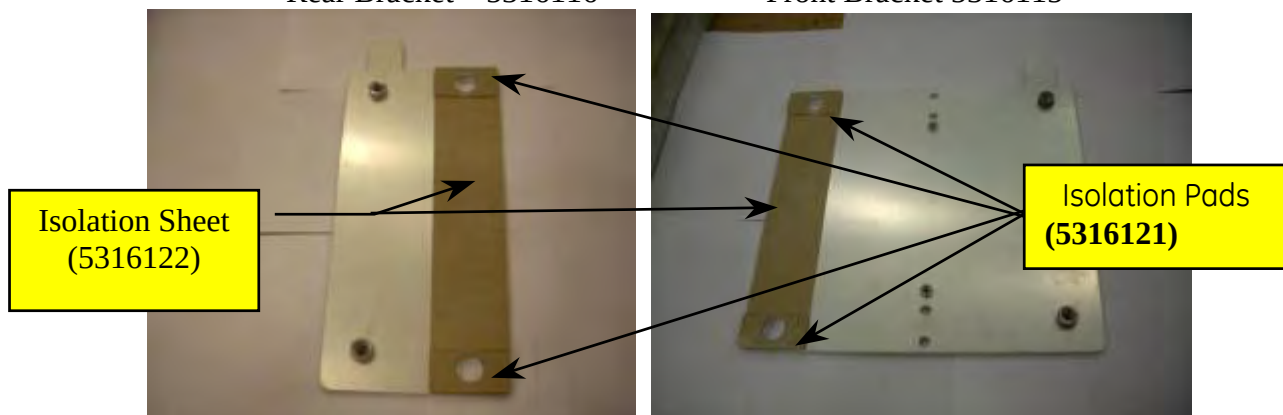


Figure 7

Discovery MR450 1.5T Magnet Enclosure Upgrade

- E. On the front attachment bracket install the Cable guide bracket (5321625) using 2 M6X12 socket head cap screws (2109867-12). Peel off the adhesive backing from the Cable tie mounts (46-252283P70) and align the holes in the mount with the small holes on the outer side of the front bracket as shown. **(Fig 8)**

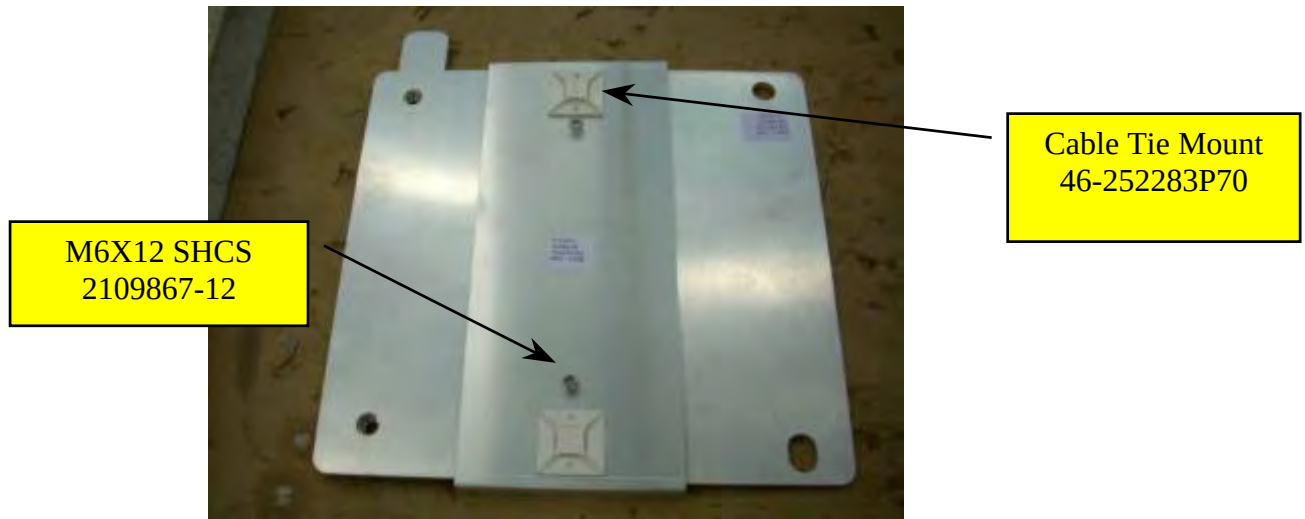


Figure 8

- F. Install Side Electronics Tray Assembly support Brackets at each of the radial pad locations as shown using M10X25 Socket head cap screws, a 10 mm flanged bushing (5316128) and a nylon washer as shown. **(Fig 9 & 10)**

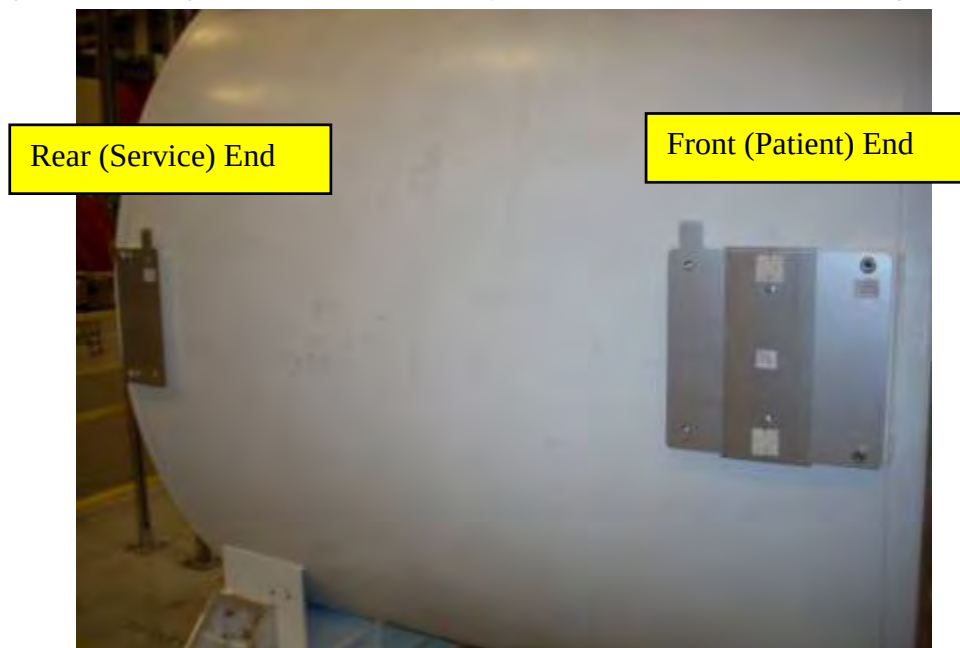


Figure 9

Discovery MR450 1.5T Magnet Enclosure Upgrade



Figure 10

- G. Mount the side electronics assembly (5316129) on cold head side of Magnet using (4) M10 Hex Head Cap screws and nylon washers (46-294167P73) at each of the mounting holes in the front and rear attachment brackets (*Fig 11*).

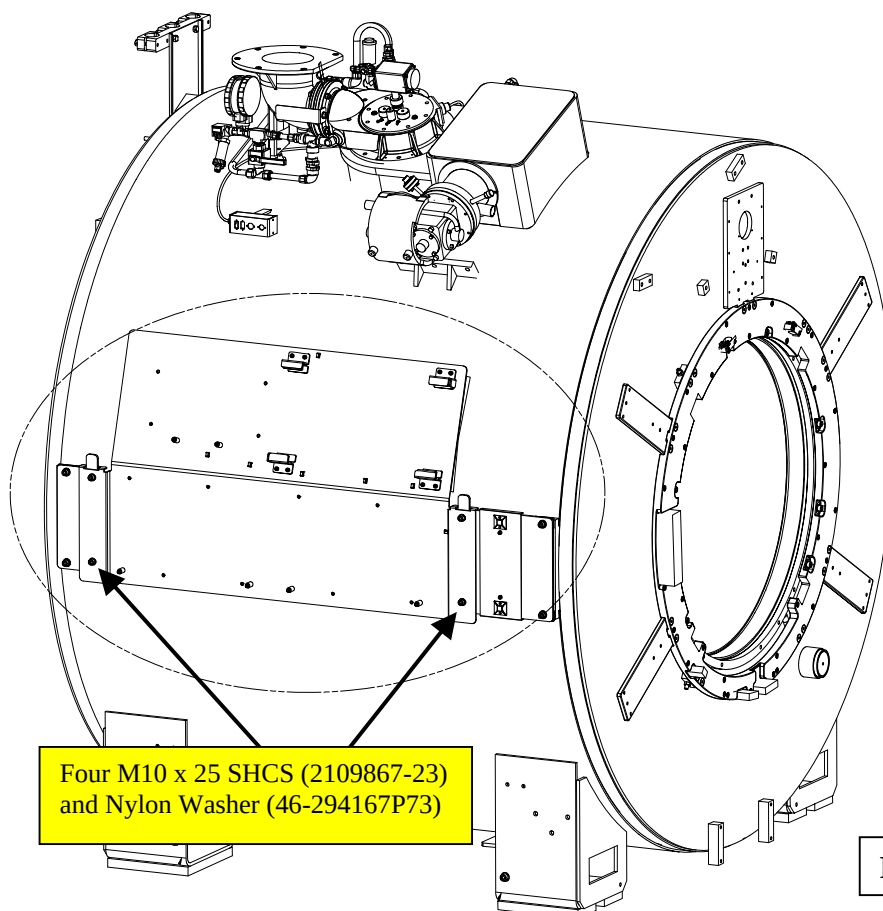


Figure 11

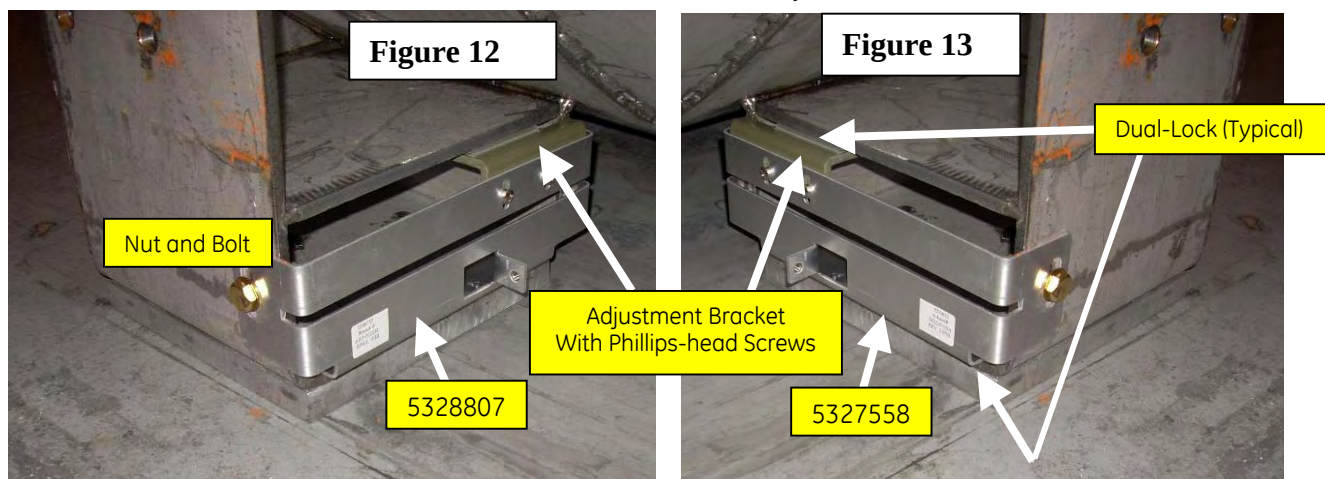
Discovery MR450 1.5T Magnet Enclosure Upgrade

6. Bracket Shelf Guide installation:

A. Components

- a) Left PAC/SRI Shelf Guide Asm 5328807
- b) Right PAC/SRI Shelf Guide Asm 5327558

- B. If the grounding bolts and nuts are already installed remove them from the magnet feet on the cold head side of the magnet. **(Fig 12 & 13)**
- C. Peel off the backing for the dual-lock on the bottom and adjustment bracket of the Bracket Shelf Guide. Position near edge of feet, top and bottom.
- D. Loosen the Phillips-head screws of the adjustment bracket and slide the front of the bracket onto the magnet foot as shown.
- E. Install the grounding bolt, washer and nuts and tighten with a wrench.
- F. Slide the adjustment bracket to the bottom of the magnet foot and stick the dual-lock.
- G. Tighten the Phillips-head screws to retain the bracket position.
- H. Remove the "ground" label at the bolt-hole being utilized.
- I. DO NOT install the PAC/SRI shelf assembly at this time.



NOTE: Make sure no grounding label is applied to magnet feet. This bolt is now considered to be load bearing and cannot be used as a ground.

Discovery MR450 1.5T Magnet Enclosure Upgrade

7. Turtle Ring Install

Note: The front Turtle-bracket assemblies must be matched to the gradient type being assembled.

A. Components

a) M10 Socket Screw	2109872-23
b) Flat washer	2109878-4
c) Never-Seize	46-294151P8
d) Turtle Ring	5191537 Rear, 5191982 Front
e) M10x35 SHCS	2109866-22
f) Washer	2184009
g) Lock nut	2109875-4
h) M10X25MM Socket screw	2109867-23
i) M10 Fiberglass washer	2370892

Note: For Step B, The Turtle Ring shipped to MR450 forward production sites has radial pads on radial screws. HDx forward production sites upgrading to MR450 may not have the radial pads. If so, Step B must be completed.

- B. Thread radial pads (5309674) onto the radial screws on the front and rear turtle ring assemblies. These pads are threaded on one end and should be screwed into the side of the radial screws that will contact the magnet interface ring. (There will be 4 on the front turtle ring bracket and 4 on the rear bracket.) **(Fig 18)**

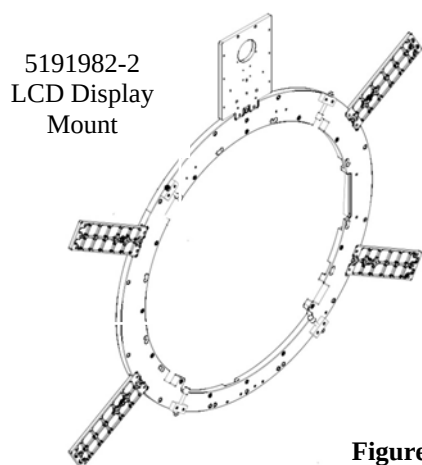
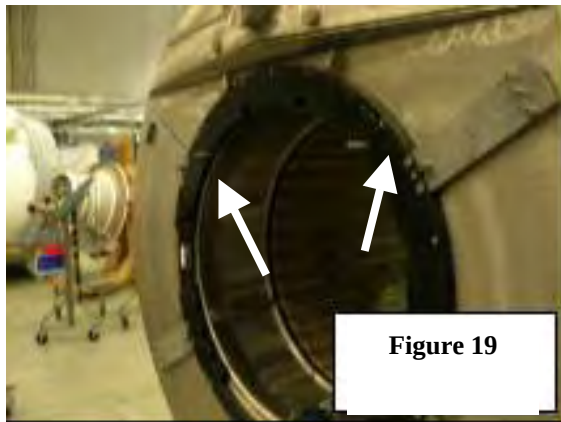


Figure 18

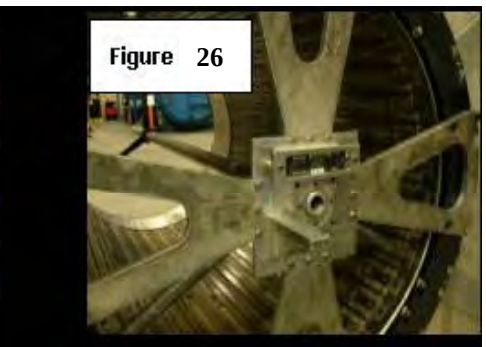
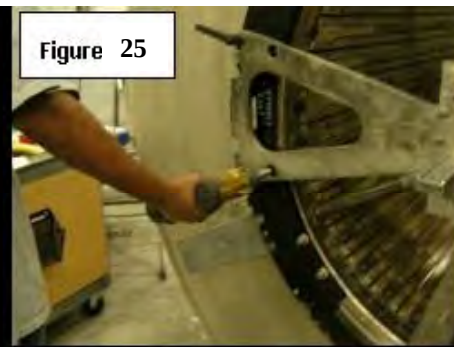
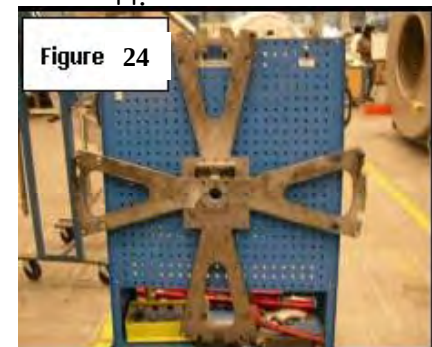
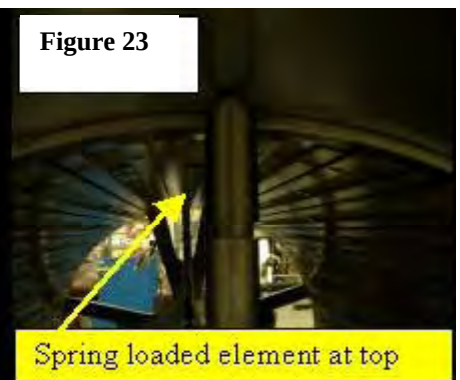


Discovery MR450 1.5T Magnet Enclosure Upgrade

- C. Install 2 M10 studs into the top to holes in the magnet interface ring at each end (FIG 19).
- D. Using screws in Figure 30, hang the turtle rings on the magnet at both ends. (FIG 20)



- E. Install the spider / shaft assembly (2290365 and 5178531) (FIG 21) in the bore of the magnet. Support legs about 6" inside the bore weld (FIG 22).
- F. Make sure the spring-loaded arm of the spider is at the top (FIG 23).
- G. Install the alignment fixture (5141975) (FIG 24) on the ring and adjust the position so that the fixture engages the shaft of the spider/shaft assembly. Turn the radial adjustment screws to center the ring until the shaft slips freely into the fixture hub. When centered, the small shaft of the spider/shaft assembly will pass through the bushing in the end plate freely without binding. Check for a minimum clearance of 0.5 mm between the pilot diameter of the shaft and the hub of the turtle bracket alignment fixture and adjust as necessary. (FIG 25, 26)



Discovery MR450 1.5T Magnet Enclosure Upgrade

- H. Adjust the turtle ring clock angle to 0 +/- 0.25 degrees. Finger-tighten the (4)radial screws to hold the magnet turtle ring in place. (FIG 27)

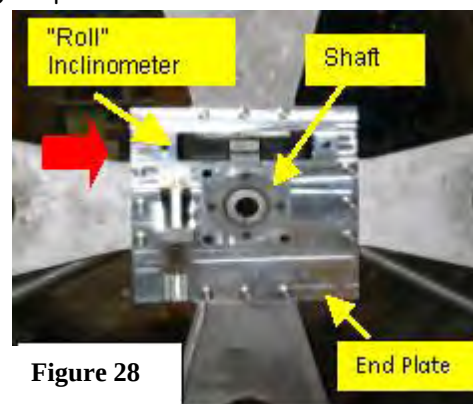


Figure 28

Note: Both End Rings may be installed simultaneously.

- I. Coat 14 M10x25 socket head cap screws with Never-Seize (FIG 29). Put flat washers on the screws and install them through the turtle ring and into the magnet interface ring (8 rear, 6 front). Do not install the top two cap screws on the front turtle ring at this time. (FIG 30)



M10X25MM Socket screw
2109867-23
M10 Fiberglass washer
2370892
Flat washer 2109878-4

Figure 30



- J. Do not tighten the screws at this time.
- K. Check Adjustment of turtle ring centering.
- L. Draw in the Turtle-ring mounting screws to seat the turtle-ring onto the magnet. Tighten the Turtle-ring mounting screws in a cross pattern to ensure even loading of the ring.

Discovery MR450 1.5T Magnet Enclosure Upgrade

M. Once seated, tighten each of the mounting screws (8 front, 6 rear). (FIG 31)

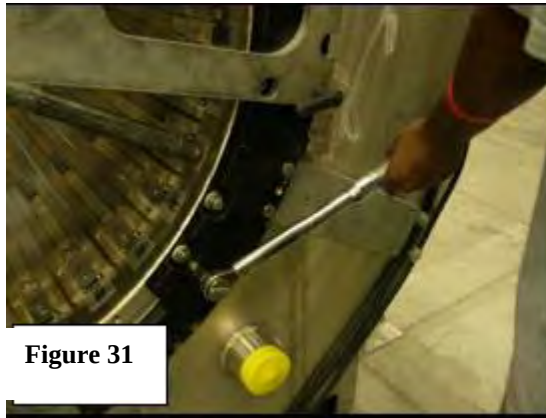


Figure 31

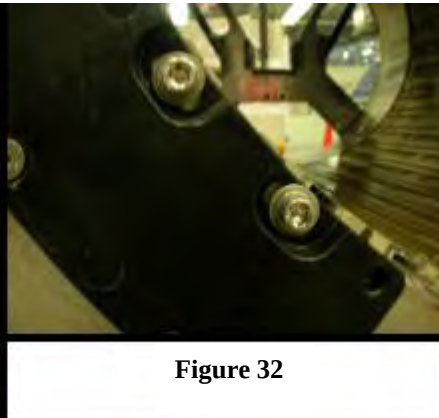


Figure 32



Radial Support screw

Figure 51
Figure 33

- N. Tighten the radial support screws and snug the lock nuts. (FIG 33)
- O. Remove the ring alignment fixtures from the magnet.
- P. Vacuum the inside of bore for debris prior to installing the gradient.

Discovery MR450 1.5T Magnet Enclosure Upgrade

8. Gradient Coil Installation

A. Tools and Consumables

Item	Qty	Part#	Manufacturer
Non-absorbant protective clothing (long sleeve shirt and pants); one set per person	1		
Nonferrous Safety Shoes; one pair per person	1		
Safety Glasses; one pair per person	1		
General-duty gloves; one set per person			
Floor Sign, Warning: Authorized Personnel Only (included in Safety Signage Kit 46-258770G4 and in 3.0T Warning Sign and Label Kit 2379494)			
Gradient Coil Insertion Kit		2164744-6	
Gradient Cable Splice Kit			
Cable Crimper-Stripper Kit			
Gradient Cable Splice Kit			
HDv Passive Shim tray extraction tool	1	5172460	
Non-Magnetic Tools Kit	1		
5 Gallon Pail	1	2239133	

Item	Qty	Effectivity	Part#	Manufacturer
Isopropyl Alcohol	1	-		
T-Wraps	1			
Red Loctite #271	1	-		
XRMB Coolant Fluid (1 gallon container) *	15		5174313	
XRMB Coolant Fluid (5 gallon container) *	3		5174313-2	

* **Note: approximately 15 gallons of coolant fluid is needed for each XRMB installation. FEs will decide which type of container to order.**

Important: XRMB Coil is temperature sensitive.

Do not expose the coil to temperatures below 32°F (0°C). Exposure to temperature below 32°F (0°C) can cause coil damage. Ensure the temperature data logger is shipped back with the coil. The coil has to be attached to the temperature data logger. Arrange for temperature-controlled transportation prior to removal of the XRMB coil.

For international shipments, if the coil has been exposed below 32°F (0°C), please return the defective gradient coil in the original crate. For domestic shipments, please use the temperature-controlled mode of transportation to return the defective coil.

Discovery MR450 1.5T Magnet Enclosure Upgrade

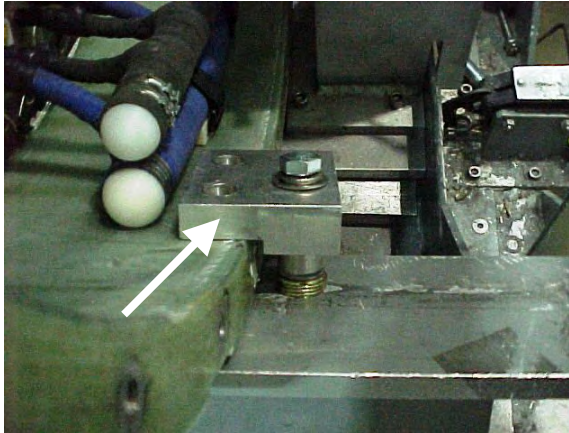
- B. Wipe the magnet bore using a clean towel and isopropyl alcohol **(FIG 34)**.

Figure 34



- C. Exam the electrical connection terminals at the XRMB coil service-end. If necessary, clean up the terminal surface with isopropyl alcohol.
- D. Remove the shipping brackets that secure the XRMB to the cart. **(FIG 35)**

Figure 35



- E. Remove the Gradient Coil cradle fasteners, two per side, from the cradle. **(FIG 36)**

Figure 36



Discovery MR450 1.5T Magnet Enclosure Upgrade

- F. Use a vacuum cleaner to clean up the XRGB coil in the out. Remove any metallic dust/shavings in the coil.
- G. Obtain the two Tube Guide Roller Assemblies (5308402) and XRGB Mounting Plates (5161983, 5161985) from the Gradient Coil Insertion and Lift Kit's shipping crate. Make sure the Tube Guide Roller Assemblies are attached to the XRGB Mounting (**FIG 37**).

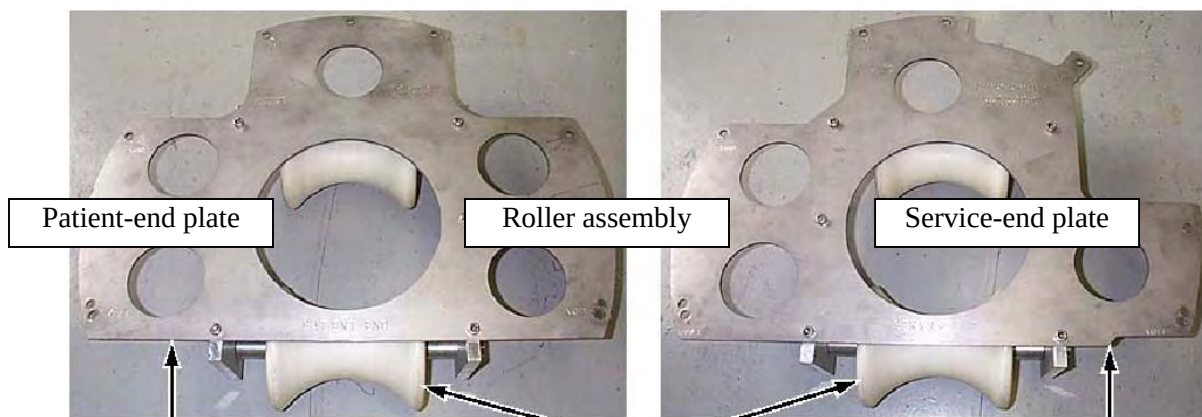


FIGURE 37 – Roller assembly and Mounting Plate

- H. Obtain the Male Insertion Tube (2284929) and XRGB Tube Standoff (5191626). Attach the Tube Standoff to the Male Insertion Tube using the 4-inch, 0.375-16UNC Hex Socket Screw (5303993) included in the Gradient Coil Insertion and Lift Kit. Once the Standoff is secured to the Tube, attach the Tube Pilot Shaft to the Standoff (**FIG 38**).



FIGURE 38 – Attaching Standoff to Insertion Tube

Discovery MR450 1.5T Magnet Enclosure Upgrade

- I. Remove the PVC shield from the brass thread of the Male Insertion Tube (**FIG 39**).



FIGURE 39 – PVC shield for brass thread

- J. Mounting Plate Installation
 - a) Install the Patient End Mounting Plate 5161985 (with a Tube Guide Roller Assembly) on the gradient coil's patient end and the Service End Mounting Plate 5161983 (also with a Tube Guide Roller Assembly) on the gradient coil's service end. Use the M10 x25 SS Hex Cap screws (46-318508P20) included in the Gradient Coil Insertion and Lift Kit. (**FIG 40**)

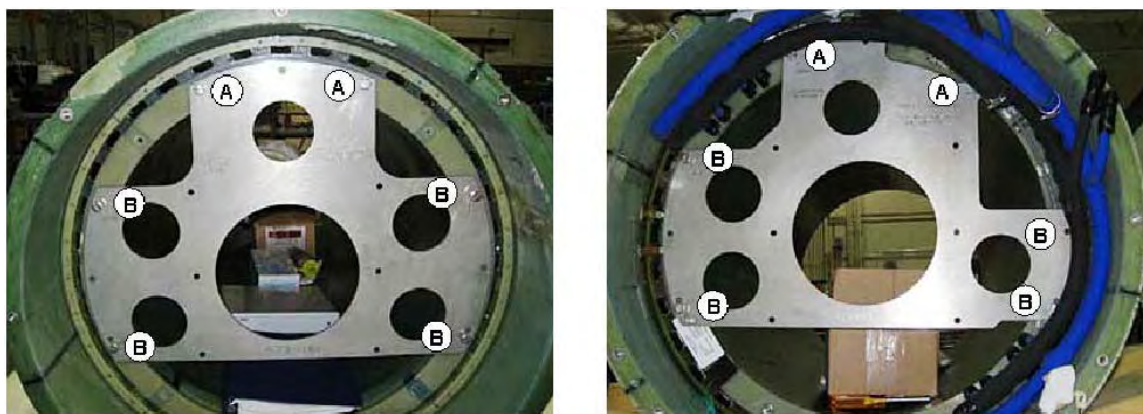


FIGURE 40 – Mounting plates with Roller Assemblies

Discovery MR450 1.5T Magnet Enclosure Upgrade

- b) Obtain Tube Support Plate Assembly 2284928-2 from the Gradient Coil Insertion and Lift Kit's shipping crate.
- c) Adjust the Tube Support Plate Assembly's Horizontal Adjustment Screws until the Tube Support Bearing is centered left-to-right. **(FIG 41).**

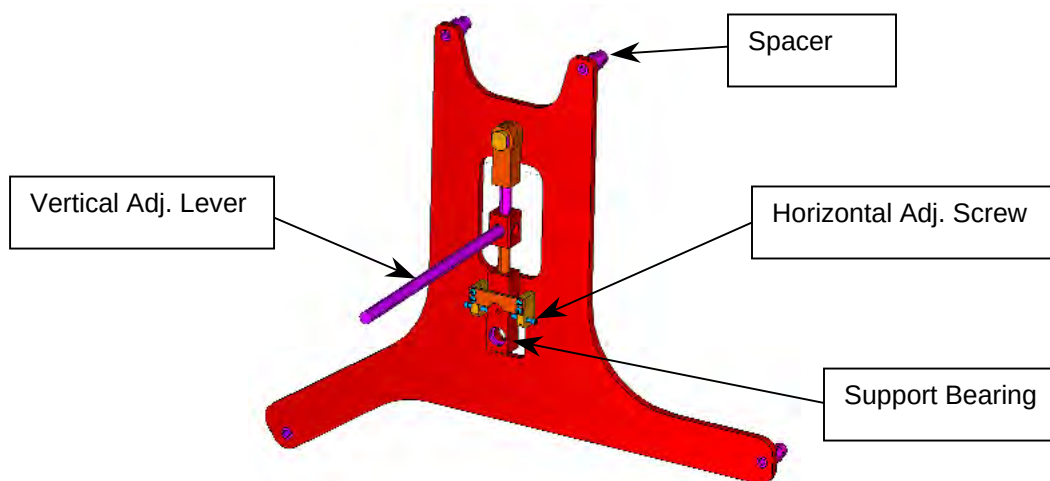


FIGURE 41 – Support Plate

- d) Attach the Tube Support Plate to the magnet **service-end** using the 100mm M10x25 SS studs (5303994), SS M10 nuts and washers included in the Gradient Coil Insertion and Lift Kit. Make sure the four extension spacers face towards the Magnet Interface Ring. **(FIG 42).**

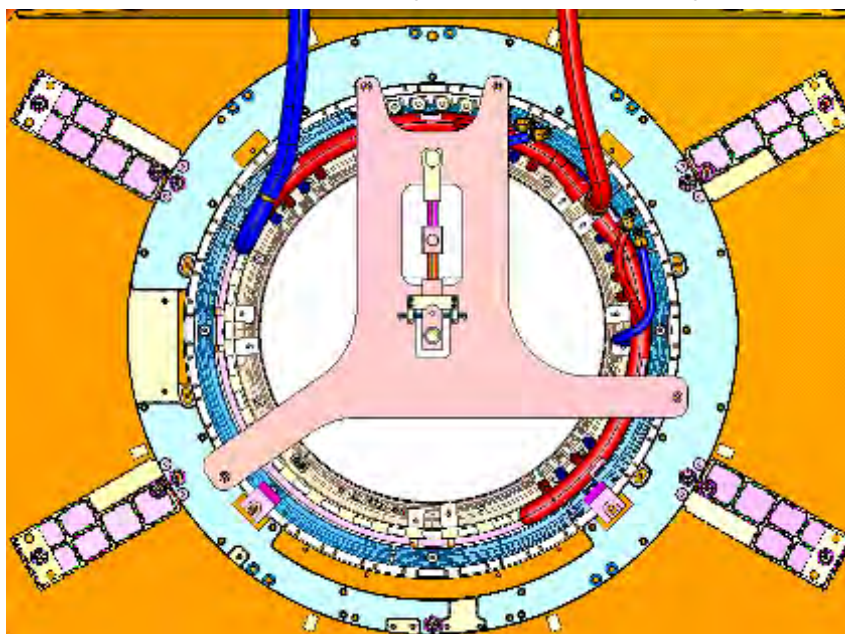


FIGURE 42 – Support plate attached on the Interface Ring

Discovery MR450 1.5T Magnet Enclosure Upgrade

9. XRMB Gradient Mounting Kit for LCC 1.5T Magnet

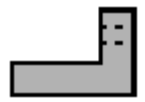
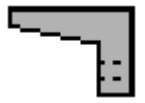
Locate the 2 (two) applicable Gradient Mounting Kits which contains the following components:

Table 1 - XRMB Gradient Mounting Kit for LCC (1.5T) Magnet

GE PN	Description	Quantity
5170035 DDW	Wedge, LCC Gradient, 45°, 315	4
5170312 DDW	Wedge, LCC Gradient, 0°, 90° & 270°	5
5170098 DDW	Wedge, LCC Gradient, 135° & 225°	4
5170015 DDW	Wedge, LCC Gradient, 180°	2
5325293-4	M10-1.5 x 30mm Brass Hex Hd Bolt C27000	15
5146169 DDW	G10 Washer ID=10.5mm, OD=24mm	15
46-170684-P2	Loctite 271 red	As Req'd

A. Refer to Table 3 and **Figures 43 and 44** for support block information.

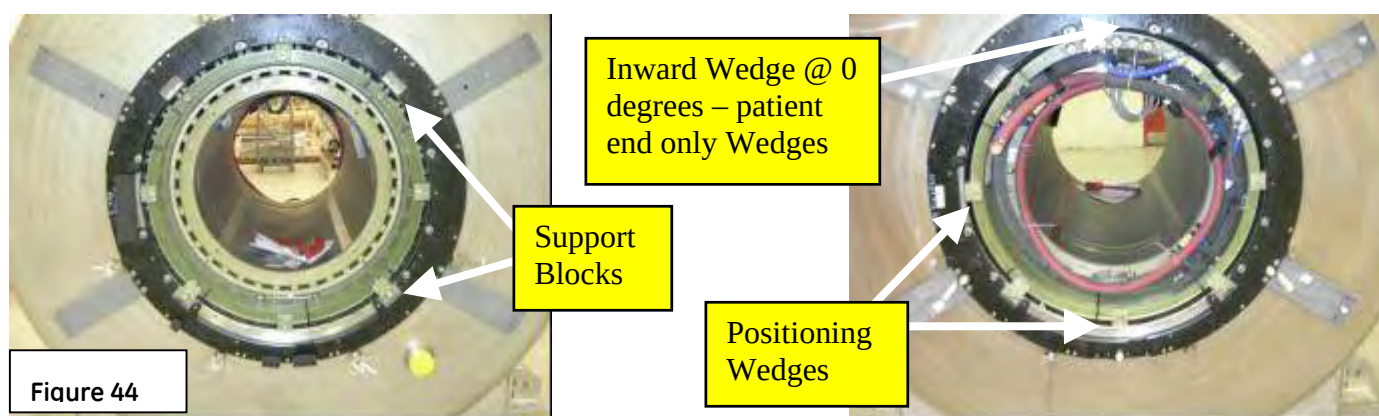
Table 2 - XRMB Wedges Installation Configuration

Item	GE PN	Profile	Angle	Description	Hardware
A	5170098 (LCC)		135° & 225°	Outwards (attached to the Magnet) Longitudinal (Z axis) positioning	2 (two) G10 washers / 1 brass bolt
B	5170015 (LCC)		180°	Inwards (attached to the XRMB) Vertical (Y axis) support	1 (one) G10 washers / 1 brass bolt
C	5170312 (LCC)		0°, 90° & 270°	Inwards (attached to the XRMB) Horizontal (X axis) support	1 (one) G10 washers / 1 brass bolt
D	5170035 (LCC)		45° & 315°	Outwards (attached to the Magnet) X-Y support	2 (two) G10 washers / 1 brass bolt

Discovery MR450 1.5T Magnet Enclosure Upgrade



Note: Use M10 x 30mm bolts and one G10 washer on all wedges.



Discovery MR450 1.5T Magnet Enclosure Upgrade

10. Installing the XRMB in the Magnet

- A. Attach the 135° and 225° XRMB Wedges (5154485) to the service-end magnet flange using 1 (one) M10x30mm brass bolt (5110638-64) and 2 (two) G10 flat washers (5146169). **(FIG 45)** These two wedges will serve as the positioning blocks for XRMB longitudinal (Z axis) alignment. To allow future adjustment, tighten the bolt only enough to secure the wedges. Do not apply loctite at this moment.

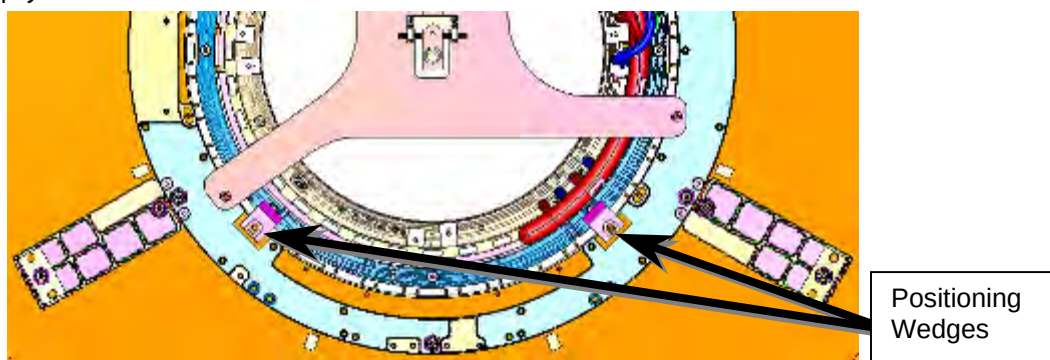


Figure 45 - Service-end positioning wedges

- B. Maneuver the cart and XRMB coil to the patient-end of the magnet. Position the cart in front of the magnet with a gap of at least 2 inches (50.8 mm) between the cart and the Magnet Interface Ring. The 2-inch gap is required for lower wedges installation. **(FIG 46)**

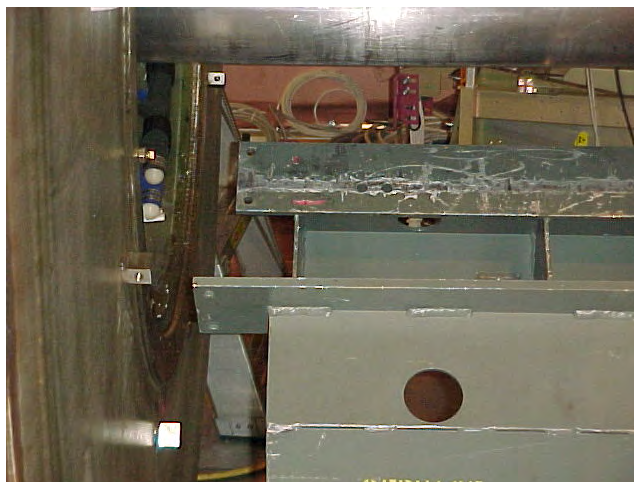


Figure 46 – 2-inch gap between magnet and cart

Discovery MR450 1.5T Magnet Enclosure Upgrade

- C. Center the cart left-to-right with respect to the magnet bore.
- D. Release the hand lever on the cart handle to set the brakes. This will keep the cart from moving after the cart is aligned with the bore.
- E. Raise or lower the cradle of the Cart as required to align the two support tub sections (Tube Support Plate at the magnet and Gradient Mounting Plates at the XRGB) vertically. **(FIG 47)**



Figure 47 – Adjusting the height of the cart

- F. Place Tube Jack Assembly (5191132) in the Cart. Make sure the back of the Jack Assembly fits properly into the Cart. **(FIG 48)**



Figure 48 – Jack assembly placement

Discovery MR450 1.5T Magnet Enclosure Upgrade

- G. Slide the Male Insertion Tube, with the Standoff attached, through the Tube Guide Roller Assemblies at both sides of the XRMB. Make sure the end with the Standoff and Pilot Shaft faces towards the magnet.
- H. With one person holding the Male Insertion Tube to keep it from rolling, two persons obtain the Female Insertion Tube from the shipping crate and thread the two Tubes together. **(FIG 49)**



Figure 49 – Setting up the Insertion Tube Assembly

NOTE: Do not force the threads. Instead, make vertical or horizontal adjustments to the Cart or Tube Support Bearing, as necessary, to make the threading easier.

Discovery MR450 1.5T Magnet Enclosure Upgrade

- I. Push the complete Insertion Tube Assembly (**FIG 50**) into the magnet bore. Get the Tube Pilot Shaft as close to (but not touching) the Tube Support Plate as possible.



Figure 50 – Complete Insertion Tube Assembly

- J. Align the Support Plate mounting bearing with the Tube Pilot Shaft by adjusting the vertical height of the Support Plate bearing using the stainless steel lever, and/or raising or lowering the Cart.
- K. Meanwhile recheck the centering of the Cart left-to-right with respect to the magnet bore. Adjust cart longitudinal alignment if required.
- L. Push the Pilot Shaft through the Support Plate mounting bearing. Insert a nonmagnetic safety pin through the hole in the Pilot Shaft. (**FIG 51**)

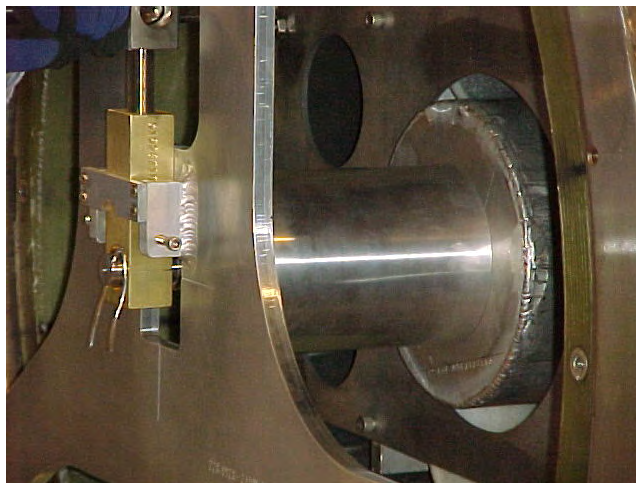


Figure 51 – Tube Pilot Shaft and Support Plate

Discovery MR450 1.5T Magnet Enclosure Upgrade

- M. Raise the Tube Jack and Support Plate mounting bearing to lift the XRMB coil up. Keep the coil approximately level with the magnet bore during the lifting. Coil should be able to rotate freely once completely lifted off the cart. **(FIG 52)**



Figure 52 – Raising the XRMB coil

- N. Rotate the XRMB, if necessary, until the electrical connection terminals at the XRMB's service-end is oriented upwards.
- O. Slowly push the XRMB into the magnet bore while watching the clearances to the left, right, top and bottom. Adjust the Tube Jack and Support Plate mounting bearing as required to keep the gradient coil from touching the magnet bore surface.
- P. Slowly push the XRMB until stopped by the two positioning wedges (135° and 225°) that are already attached to the magnet service-end flange. Push the XRMB against the two wedges so that the coil's end surface is in good contact with the vertical side of the 2 wedges. Raise the Support Plate bearing to gain more clearance between the coil and the lower wedges if necessary. **(FIG 53)**

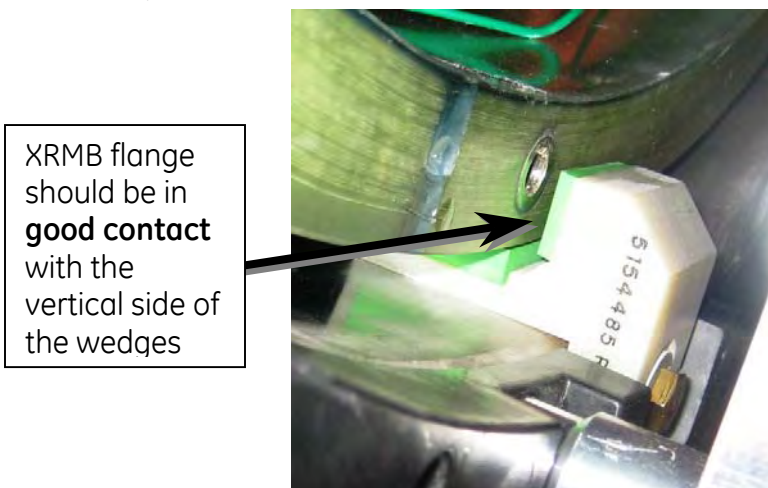


Figure 53 – Longitudinal Positioning Wedge

Discovery MR450 1.5T Magnet Enclosure Upgrade

- Q. At the magnet patient-end, attach the XRMB Radial Alignment Tool (5266408) to the magnet interface ring at 0° (top) using M10 studs and nuts. **(See Figure 54 for reference)**
- R. Slowly rotate the XRMB coil to align its 0° bolt hole with the slot opening in the Radial Alignment Tool. Once aligned, put the thumbscrew through the Alignment Tool slot and thread it into the 0° hole in XRMB. **(Fig 54)**



Figure 54 – Using the Radial Alignment Tool

- S. Apply red loctite (#271) to 135° and 225° hole in the patient-end magnet flange. Attach the 135° and 225° wedge (5154485) to the magnet flange using 1 (one) M10x30 brass bolt (5110638-64) and 2 (two) G10 flat washers (5146169). Tighten the brass screws firmly to secure the wedges. **(Fig 55, A)**
- T. Apply red loctite (#271) to 180° hole in XRMB coil. Attach the 180° wedge (5155280) to the magnet flange using 1 (one) M10x30 brass bolt (5110638-64) and 1 (one) G10 flat washers (5146169). Tighten the brass screws firmly to secure the wedge. **(Fig 55, B)**

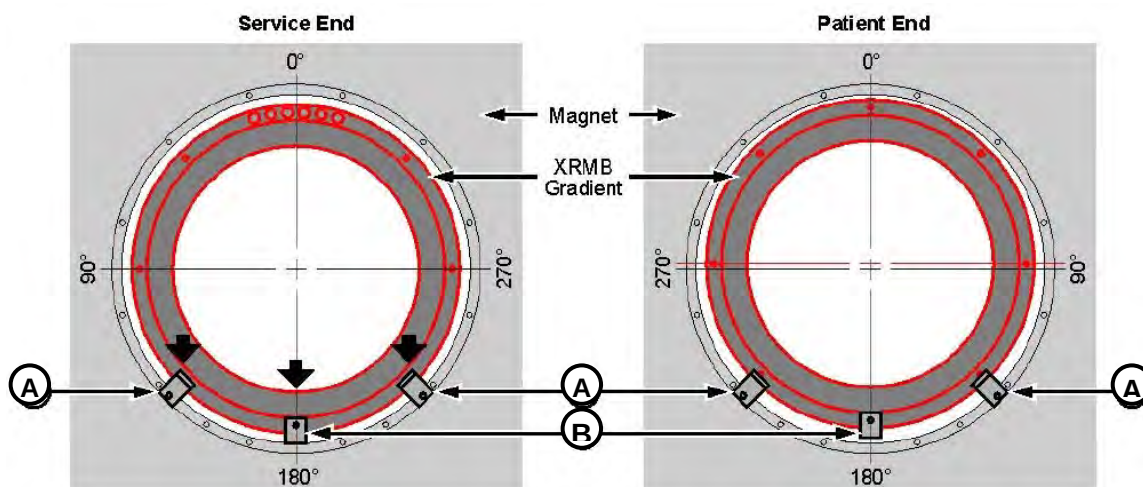


Figure 55 – Installing the Lower Wedges

Discovery MR450 1.5T Magnet Enclosure Upgrade

- U. Repeat step "S" and "T" at the service-end to secure the 135°, 225° and 180° wedges at this end.
- V. Slowly lower Support Plate bearing and Tube Jack. The full weight of the XRMB is now loaded onto the six lower wedges. Take care not to rotate the coil. Once on the support blocks use an outward wedge to check the alignment (when bolted to the I/F ring, the wedge lines up with the pocket) **(Fig 56)**

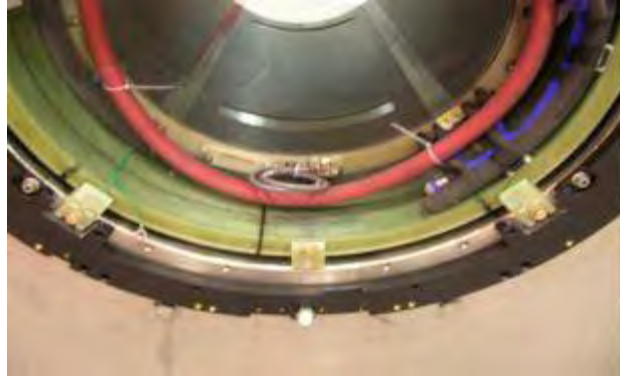


Figure 56 – Positioning of the Lower Wedges

- w. Keep the Radial Alignment Tool attached and secured during this step. **(Fig 57)**

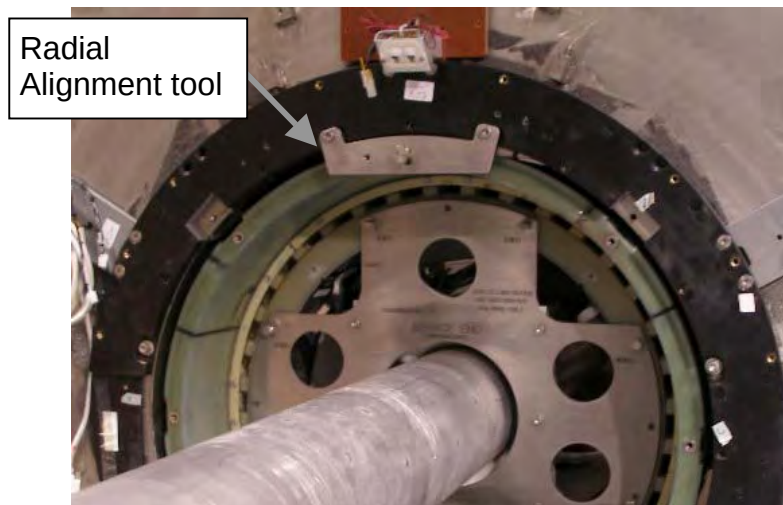


Figure 57 – Lowering the XRMB with the Radial Alignment Tool attached

Discovery MR450 1.5T Magnet Enclosure Upgrade

- X. Use the same method as in step "t", install 90° and 270° wedge to both ends of the XRMB. Push the wedge into the bore until snug. Working at one end (patient or service), tighten the bolt onto the wedge, alternating between the 90 degrees and 270 degrees wedge to maintain the coil in the center of the bore. Repeat this step on the other end. **(Fig 57, C)**

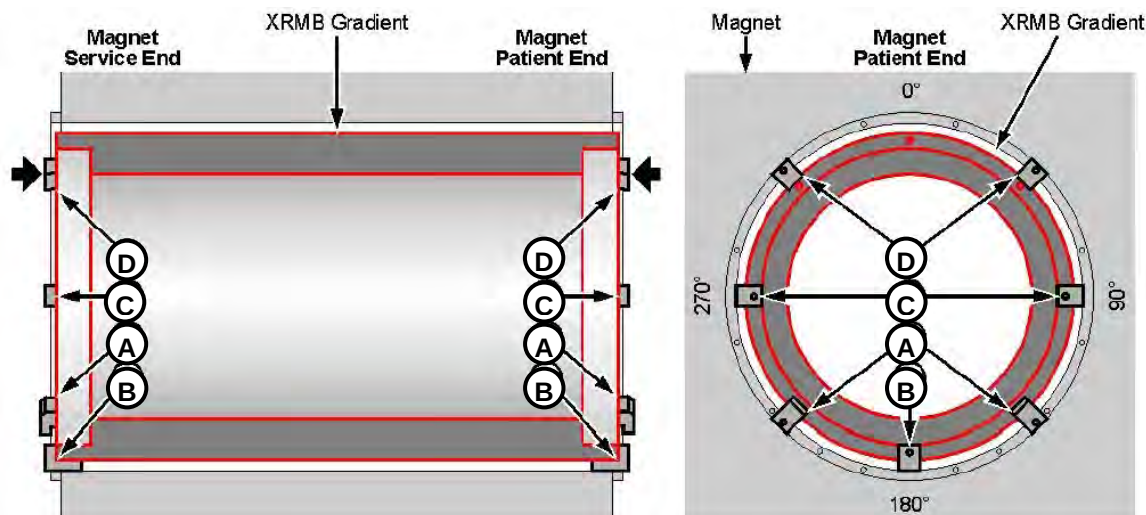


Figure 57 – Installing the middle Wedges

- Y. Use the same method as in step "s", install 45° and 315° wedge to both ends of the magnet. Push the wedge into the bore until snug. Working at one end (patient or service), tighten the bolt onto the wedge, alternating between the 90 degrees and 270 degrees wedge to maintain the coil in the center of the bore. Repeat this step on the other end. **(Fig58, D)**

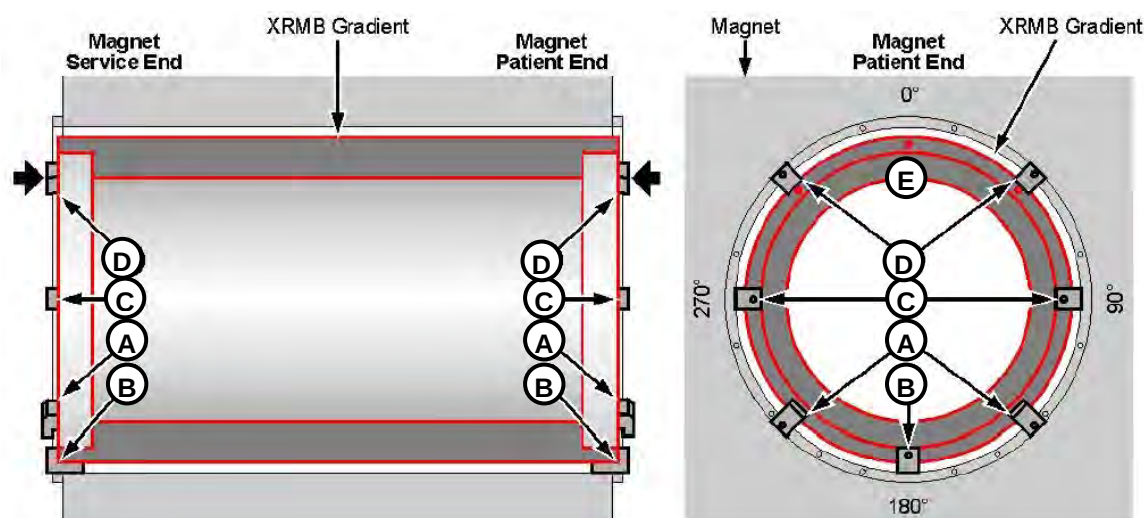


Figure 58 – Installing the upper Wedges

Discovery MR450 1.5T Magnet Enclosure Upgrade

- Z. Remove the Radial Alignment Tool. Restore the original screws to the Magnet Interface Ring
- AA. Use the same method as in step "t", install 0° wedge to the patient-end of the XRMB. (there is no wedge at 0° at the service end). **(Fig 58, E)**
- BB. Check that all the wedges have no greater than a 6 mm (3 +/- 3) gap between the wedge and either gradient or magnet interface ring. **(Fig 59)**

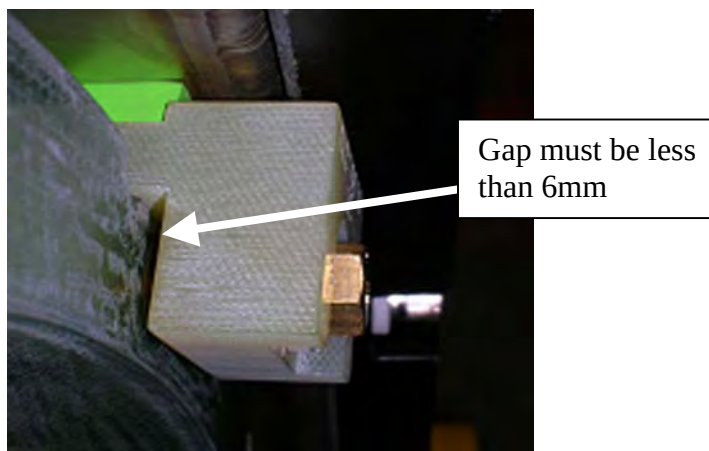


Figure 59 – Check gap of Wedges

- CC. Restoring Insertion Tool Kit
 - a) Move the cart out of the magnet room.
 - b) With one person at the service-end holding the Male Insertion Tube, two persons work together to un-thread the Female Insertion Tube from the assembly and place it back to the shipping crate.
 - c) Remove the safety pin from the Pilot shaft of the Male Insertion Tube. Remove the Male Insertion Tube from the magnet.
 - d) Disassemble the Standoff from the Male Insertion Tube. Restore the Pilot
 - e) Shaft to the Tube. Place all components back to the shipping crate.
 - f) Remove the two mounting Plates with the Tube Guide Roller Assembly from the XRMB and place them back to the shipping crate.
 - g) Remove the Tube Support Plate and studs from the magnet. Place them back to the shipping crate.

Discovery MR450 1.5T Magnet Enclosure Upgrade

DD. Install the RF Shield Ground wire on the rear magnet interface ring adjacent to the 225° support block using an M10X14 hex head capscrew (2109866-50) and M10 stainless lockwasher (2381267-9). **(Fig 60)**

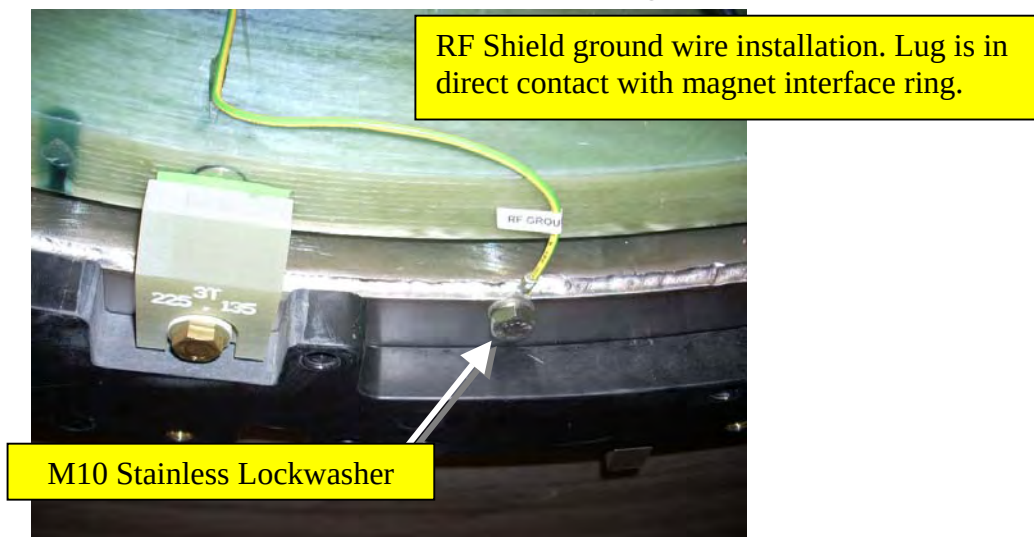


Figure 60 – Connecting RF Shield Ground Wire
S.

EE. Install the acoustic foam strips in the locations used for gradient handling. **(Fig 61)**

FF. Components

- | | |
|-----------------------------|----------------|
| a) Acoustic foam strip | 5193573 rev. 3 |
| b) M10 Hex Head Screw | 2109866-50 |
| c) M10 Stainless Lockwasher | 2381267-9 |

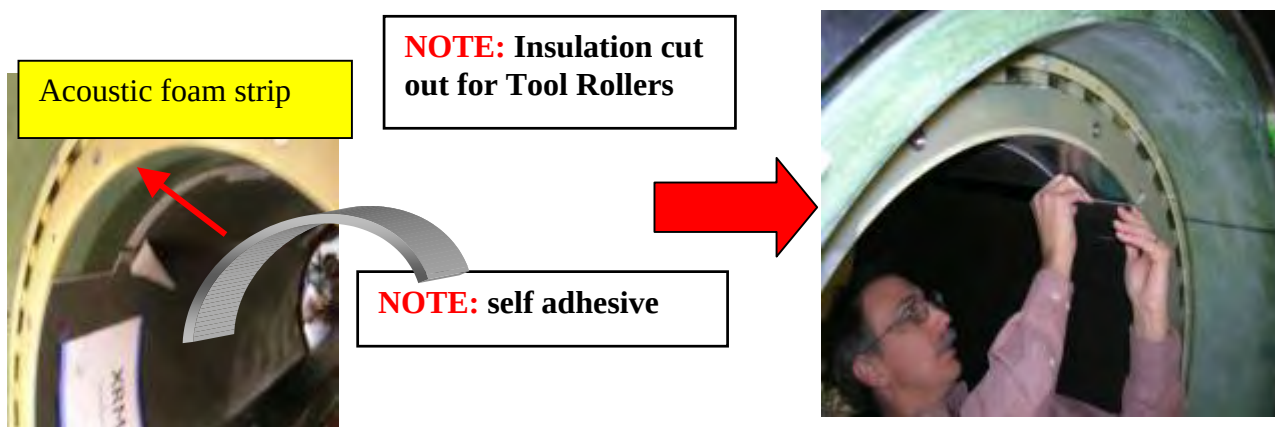


Figure 61 – Installing Acoustic Foam Strips

Discovery MR450 1.5T Magnet Enclosure Upgrade

11. Rear Foam Air Seal Install

- A. Peel back the adhesive backing from the rear foam seal (5313540) and attach it to the ID of the inner gradient coil as shown. **(Fig 62)** Be sure to keep the foam approximately 1/8" below the ID of the gradient coil (Fiberglass). Trim and butt the ends of the foam together. Locate the joint away from features that would project from the surface of the inner coil.

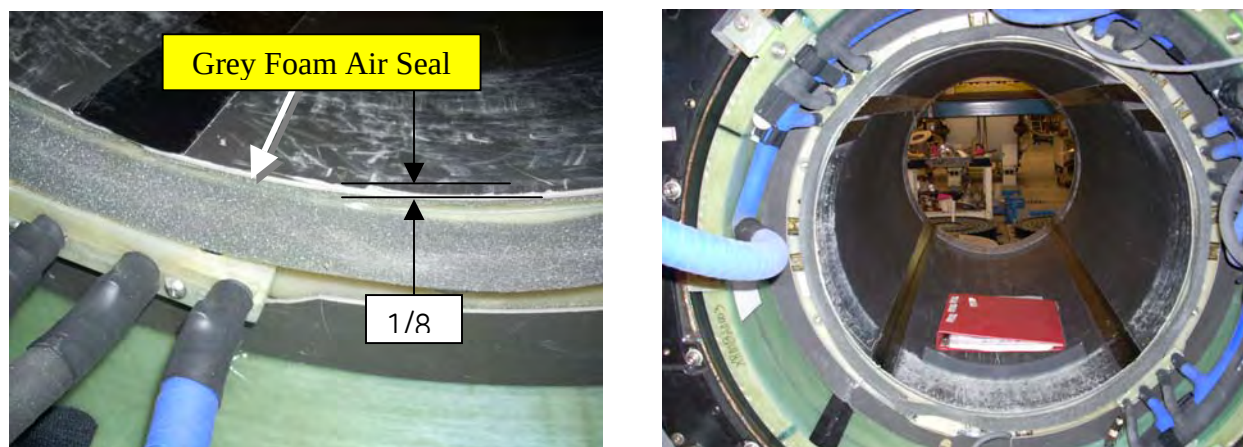
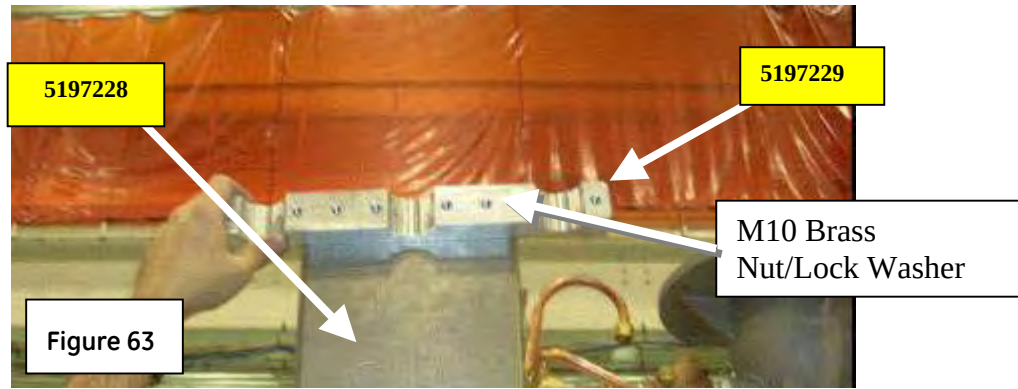


Figure 62 – Installing Rear Foam Air Seal

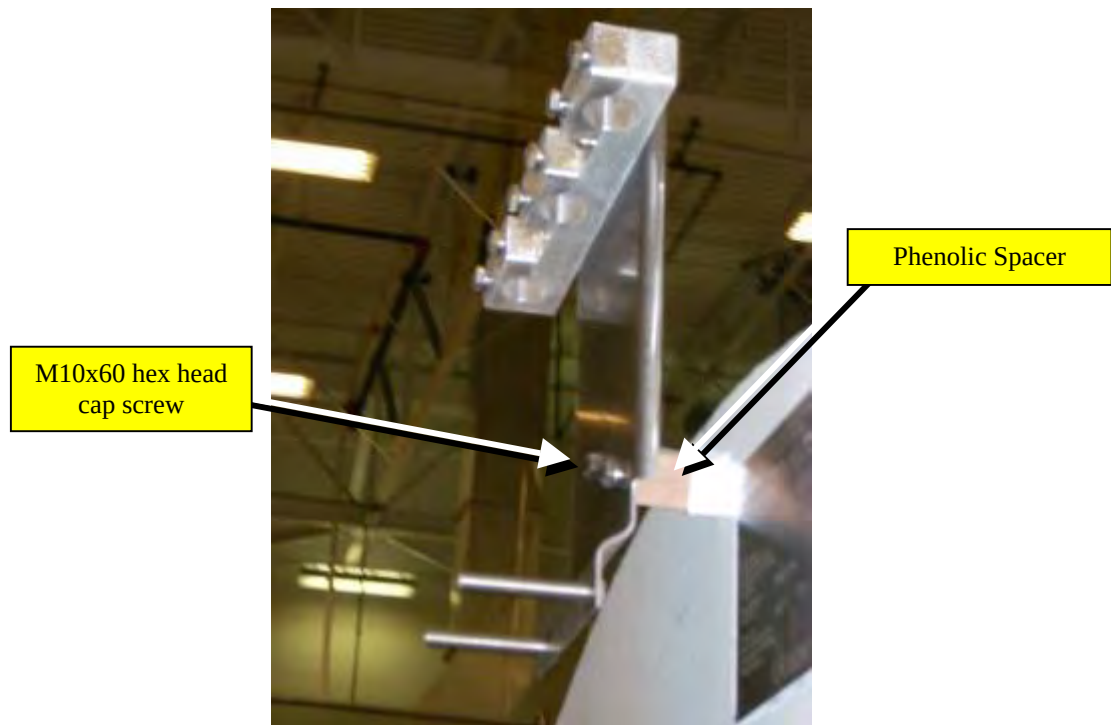
Discovery MR450 1.5T Magnet Enclosure Upgrade

12. Bus-Bar Assembly Bracket Installation

- A. Attach the power cable support bracket (5197229) to the bus bar assembly support bracket (5197228) using two SS M10X50 full thread bolts (46-318508P25) lock washers (one each side) and 2 (two) brass nuts (2341885) on the cable support bracket side as shown **(FIG 63)**.

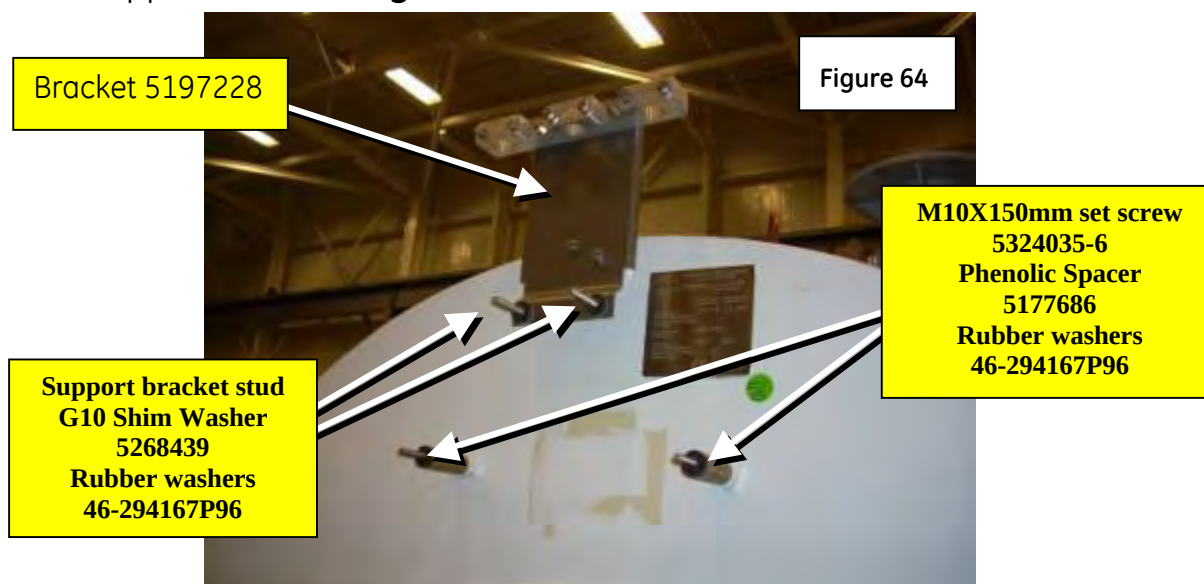


- B. Attach the bus-bar plate support bracket (5197228) to the top mounting block (5323771) located on the magnet flange (service end) using (2) M10 hex head cap screws (2109866-25) and lock washers (2381267-7) as shown **(FIG 64)**. Do not tighten the screws to allow future adjustment of the bus-bar plate if needed.

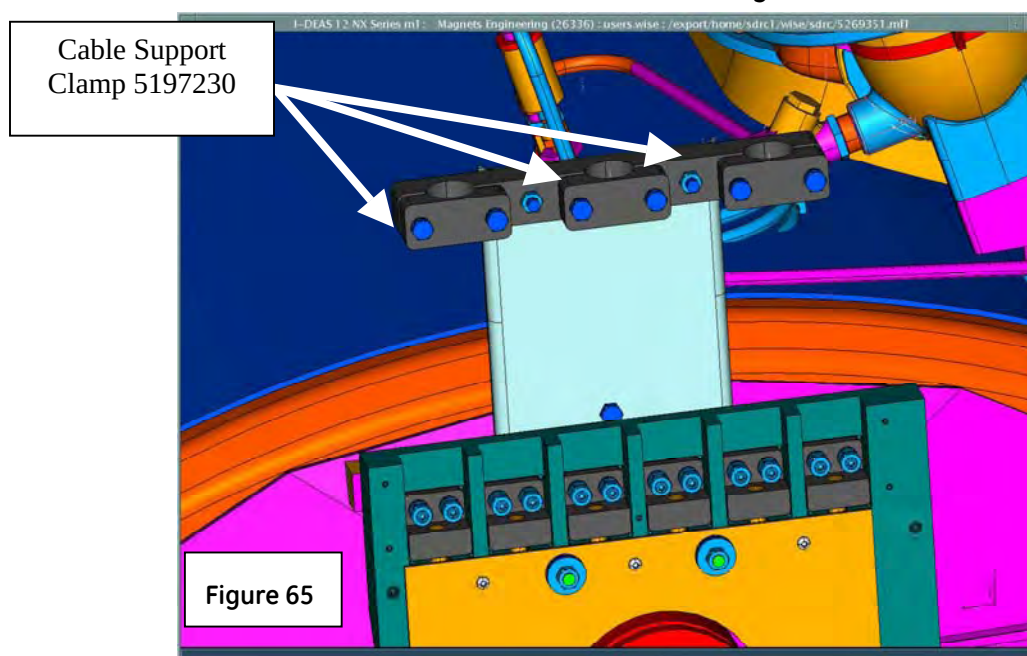


Discovery MR450 1.5T Magnet Enclosure Upgrade

- C. Install 2 M10 x 150mm set screws (5324035-6) into the mounting points for the bus assembly using Loctite 242. Install an M10x1 ¼ G-10 spacer and a Blue Rubber washer (46-294167P96) with the pilot diameter facing outward over each of the mounting studs for the bus assembly on both the magnet and support bracket. **(Fig 64).**



- D. Install cable support clamps (5197230, 3X) using M10x40 screws (46-318508P23) and lock washers (2381267-9). **(Fig 65)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

13. Gradient Cable Bus-bar Assembly Install

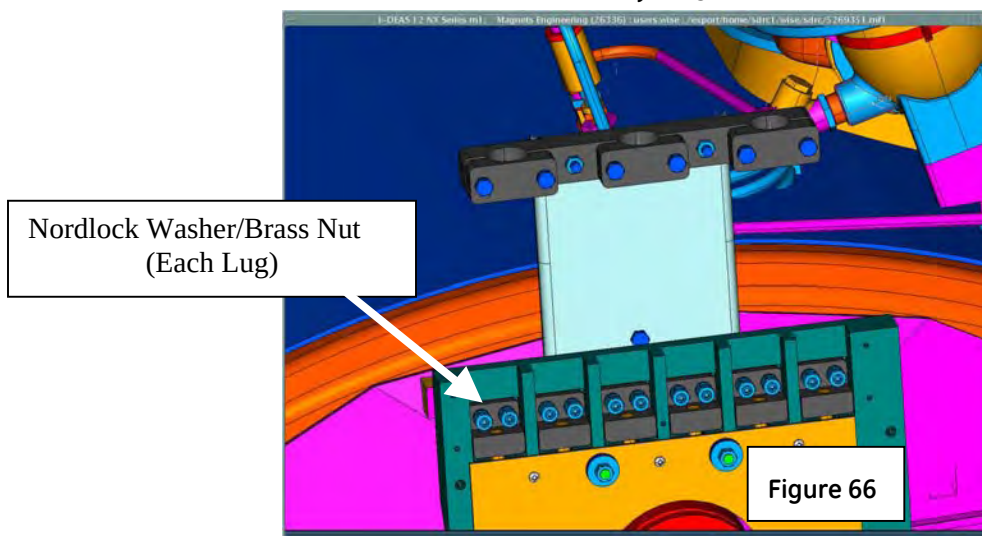
Note: The busbar assembly comes with the safety cover nuts and Nordlock washers installed. If this is not the condition steps “C” and “D” should be performed

A. Components

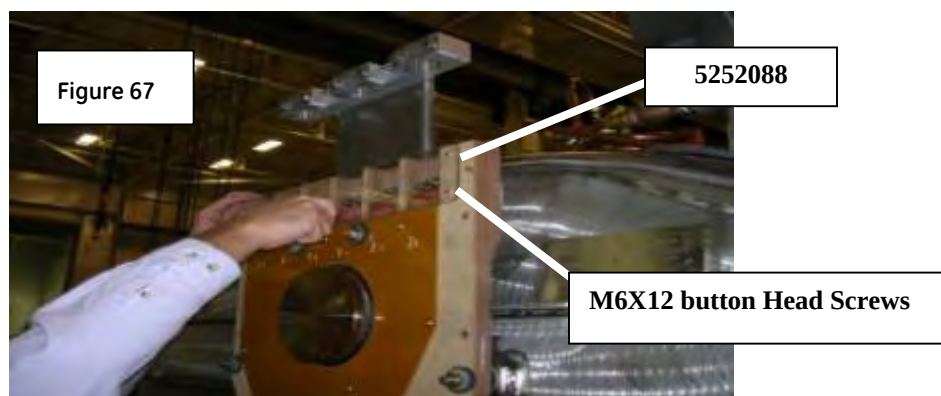
a) Bus Bar Bracket	5197228
b) Lead Board Assy.	5198564
b) M10 X 40 Screws	46-318508p2
c) M10 SS Nordic Washers	5214524-6
d) Rubber Washers	46-294167p96
e) M10 Lock Washers	2381267-9
f) M5 X 30	2381227-1
g) M10 X 75 Studs	2116325-10
h) Power cable support bracket	5197229
i) M10 X 50 Screws	46-318508p25
j) Brass Nuts M10 X 30	2341885
k) M5 X 30 Screws	5110638-64
l) Clamp Bracket	5197230
m) G10 Shim Washer	5268439
n) M10 X 24 washer	5146169
o) Safety cover	5189608
p) Safety cover	5252088
q) Nylon washer	46-294167P73
r) G10 Washer	2370892

Discovery MR450 1.5T Magnet Enclosure Upgrade

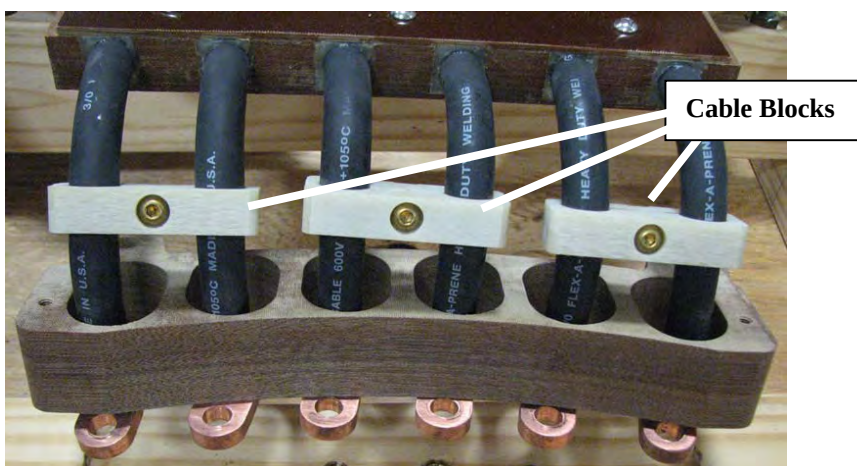
- B. Install a Nordlock washer (5214524-6) and brass nut to each of the upper terminals (12X) on the bus assembly (Fig 66).



- C. Attach the power cable safety cover (5252088) using 5 (five) SS M6x12 button head screws (FIG 67).

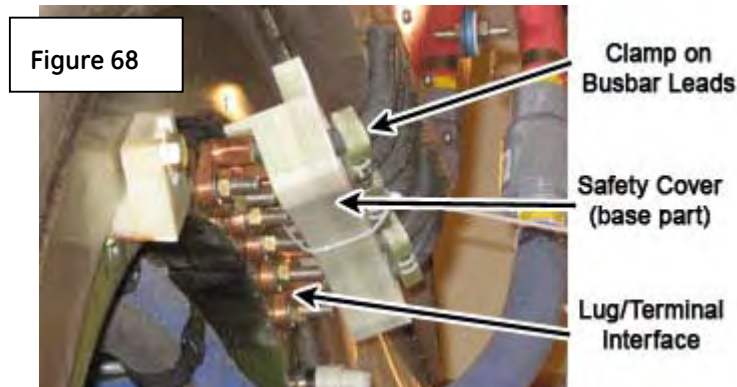


- D. Do not remove cable blocks.



Discovery MR450 1.5T Magnet Enclosure Upgrade

- E. Install the base part of the gradient terminal safety cover (5335926) loosely on the bus-bar lugs. **(Fig 68)**



Nordlock Washer Note: These SHALL NOT be taken apart, install between the bus-bar lug and a nut. All nuts are M10 brass nuts.

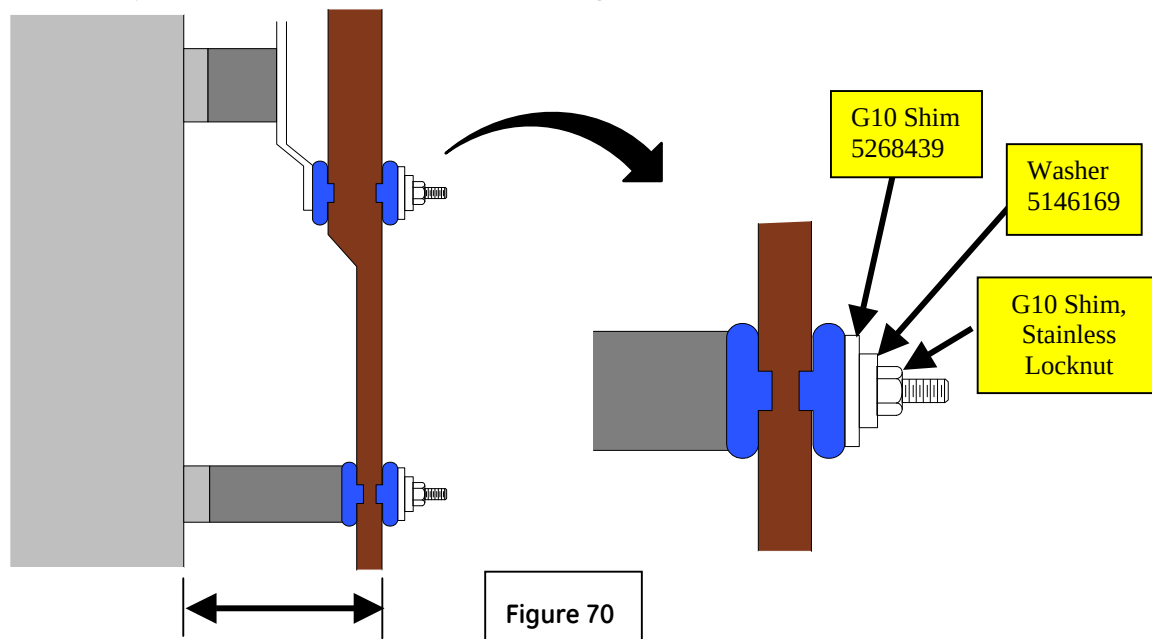
- F. Install the busbar assembly on the mounting studs of the making sure that bus-bar lugs engage the studs at the service end of the gradient. **(Fig 69).**



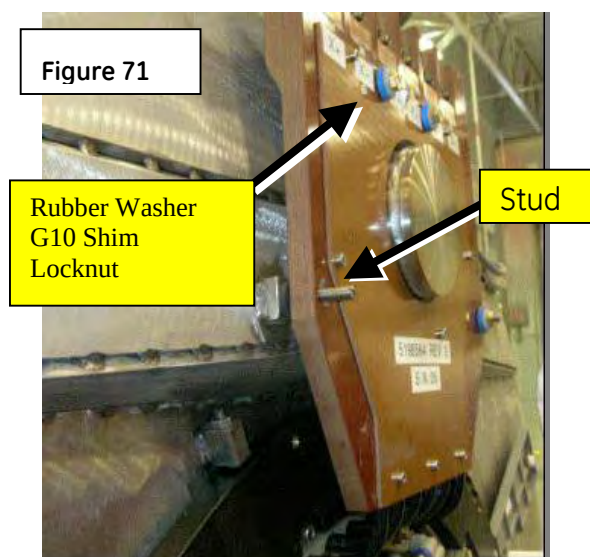
Note: The bus assembly should be parallel with the main surface of the magnet end flange. The gap between the back surface of the bus assembly and the surface of the mounting points should be equal at upper and lower mounting points.

Discovery MR450 1.5T Magnet Enclosure Upgrade

- G. Position the busbar so that the Lug surface is parallel to the surface of coil terminal. Inspect the alignment between:
- a) **Bus-bar Lugs and the gradient coil**
 - b) **Measure distance from magnet to outer surface of Busbar assembly.**
- H. If the upper and lower gaps are not equal add a G10 Shim Washer between bus bar support bracket and the blue rubber washers or between the phenolic spacer and rubber washer. **(Fig 70).**

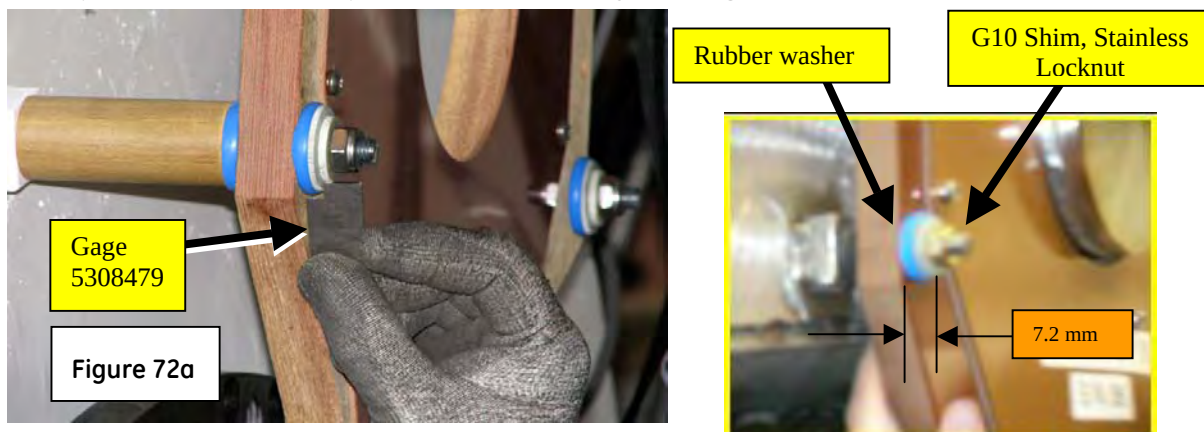


- I. Install rubber washers with the pilot diameter facing the bus assembly, a G10 shim washer (5268439) and a Stainless Locknut (5119801-5) as shown (4 places). Apply Loktite 242 as a lubricant to the threads of the nut to prevent galling. **(Fig 71).**

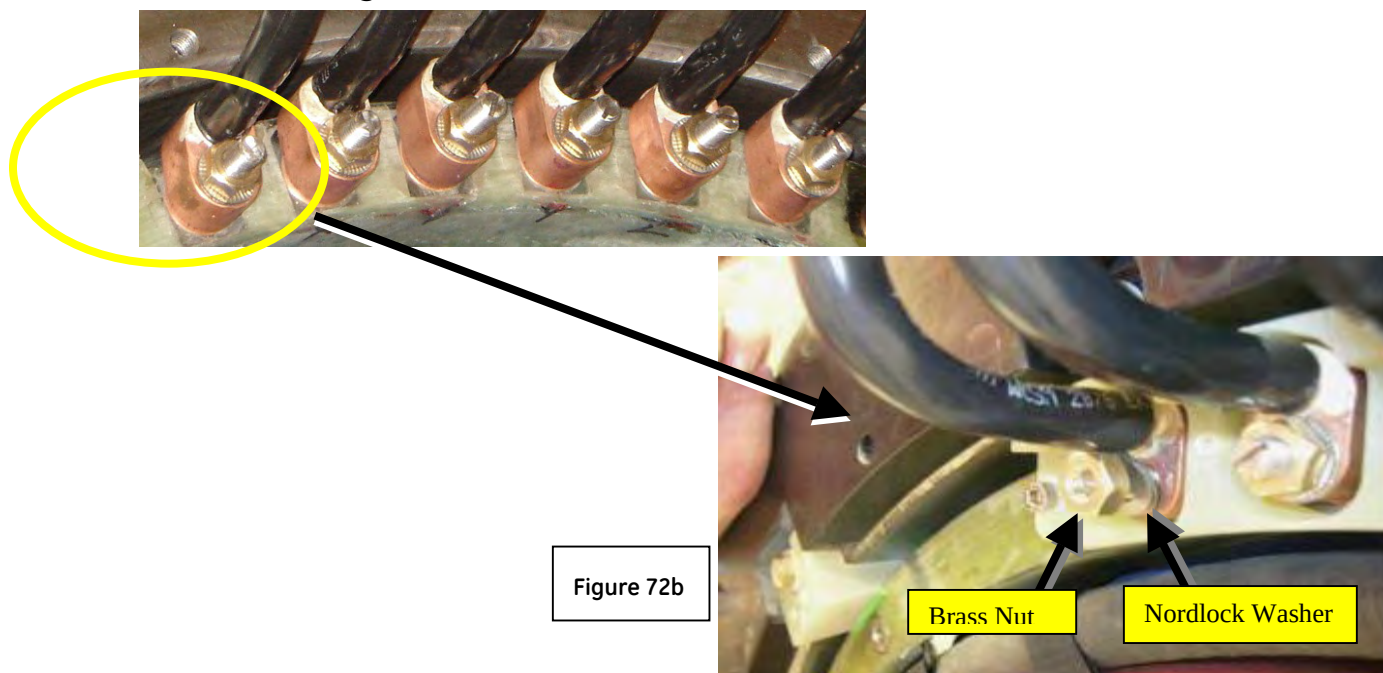


Discovery MR450 1.5T Magnet Enclosure Upgrade

- J. Tighten the nut so the height of the G10 washer is 7.2 mm off from the surface of the bus assembly. The lead board installation gage (5308479) may be used to verify the 7.2 mm height. **(Fig 72a)**.



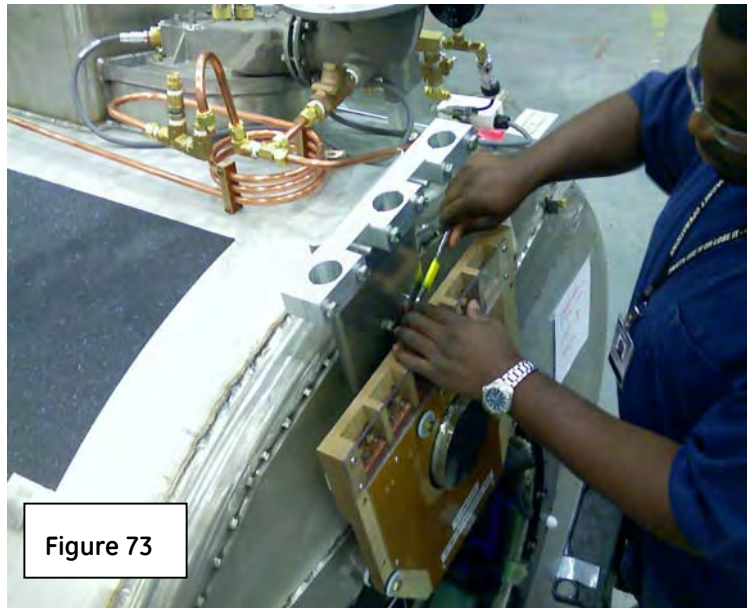
- K. Install a Nordlock washer (5214524-6) and brass nut at each of the six terminal studs at the rear end of the gradient. Tighten using a 17 mm wrench. **(Fig 72b)**



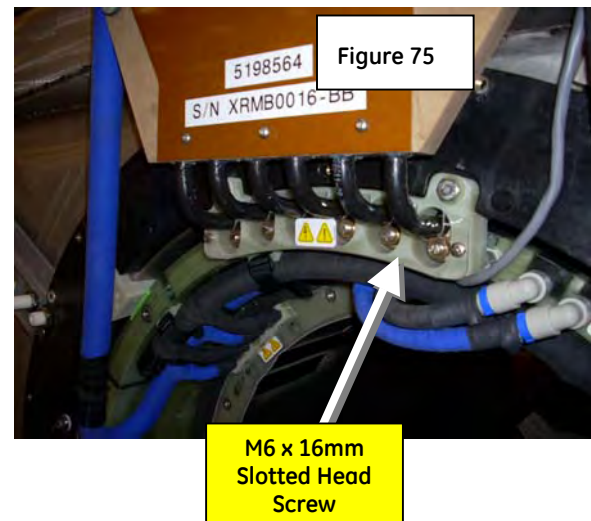
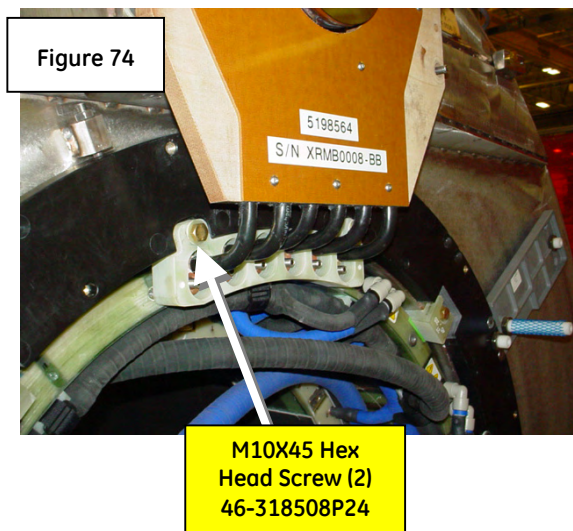
Warning. Care should be given to ensure that the Nordlock washers are two part with correct sides touching each other. Generally, when it comes from the manufacturer, the two parts are glued together the proper way. If both pieces of the washer are separated, DO NOT USE the separated washer. Use a new washer that has two parts held together.

Discovery MR450 1.5T Magnet Enclosure Upgrade

- L. Once the plate is secure and bus bars are connected to the terminal blocks on the gradient coil, tighten the bolts (2) holding the bus-bar support bracket. **(Fig 73).**

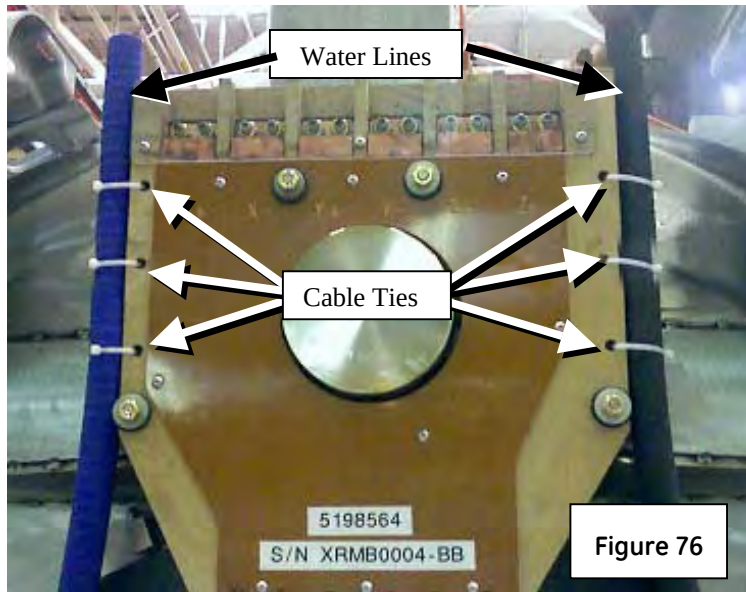


- M. Attach the gradient bus bar safety cover (5335926) using 2 (two) M10x45 hex-head screws (46-318508P24) and G10 washers (5146169) inserted into the top two holes of the rear turtle bracket as shown. **(Fig 74)** Torque the screws to 10 ft-lb. Install the clear safety cover (5309076) using 2 M6X16 Phillips-head screws (5288468-4). **(Fig 75).**

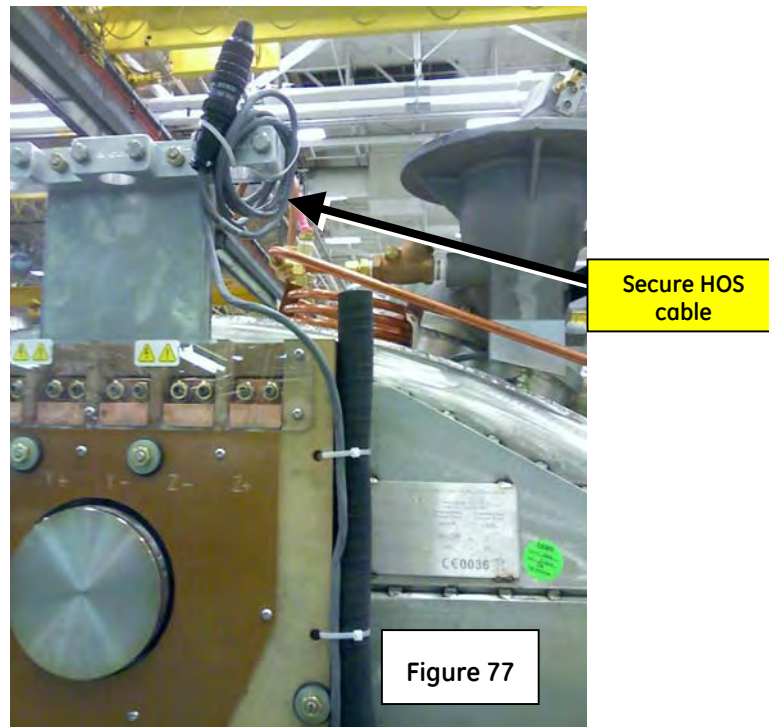


Discovery MR450 1.5T Magnet Enclosure Upgrade

- N. Attach manifold supply and return tubes to the sides of the bus-bar assembly using plastic ties (*FIG 76*)



- O. For Gradients with High Order Shim (5240893-2) route and secure the HOS cable along the black (nearest) water tube. Loop and secure the excess cable at the gradient cable support clamp as shown. (*FIG 77*)



Discovery MR450 1.5T Magnet Enclosure Upgrade

- P. Apply the electrical shock warning label (5182822) to the safety covers on the bus-bar and gradient as shown. *(FIG 78)*



Discovery MR450 1.5T Magnet Enclosure Upgrade

14. RF Installation

- A. Install the RF coil into the bore of the gradient from the front. Before inserting completely, install the Bore Temperature Sensor in the center hole of the RF coil (10 o'clock position). Use thermal join compound on the sensor and hole. Tape the sensor cable to wall of the gradient coil.

Note: Remember to leave enough cable for when the RF Tube is in its normal position.

- B. Estimate the position of the front endbell based on the front pad on the gradient. Clock the RF into place and check with a level on the RF cable cover. The RF should be level within $\pm 1^\circ$.
- C. Install the front endbell (this is used for proper location only at this time and will be removed later in the process). Set the front endbell in place. Start all M10 x 25 screws and washers in place (2109867-23 and 46-294167P73). Starting at the top, adjust the endbell to be flush with the RF surface. Begin tightening all screws.
- D. Continue to adjust the sides, working around the endbell in a clockwise fashion and tightening screws ~ 2 turns per rotation so the RF and endbell surfaces are flush. Tighten all screws using a T-handle tool (not the battery powered guns) Note: There may be some variation on the bottom surface. The top and sides need to be as even as possible. Adjust the gap between the front endbell and the RF to a target value of 2mm ($\pm \frac{1}{2}$ mm). The gap can be increased or decreased by sliding the RF coil for adjustment.
- E. Apply air seal (part number 5313540, part of XRMB FRU) along the ID on the XRMB endplate (Service-end ONLY). **(Fig 104)** Refer to Section 11.

Apply the air seal (5313540) along the ID as far away as possible radially

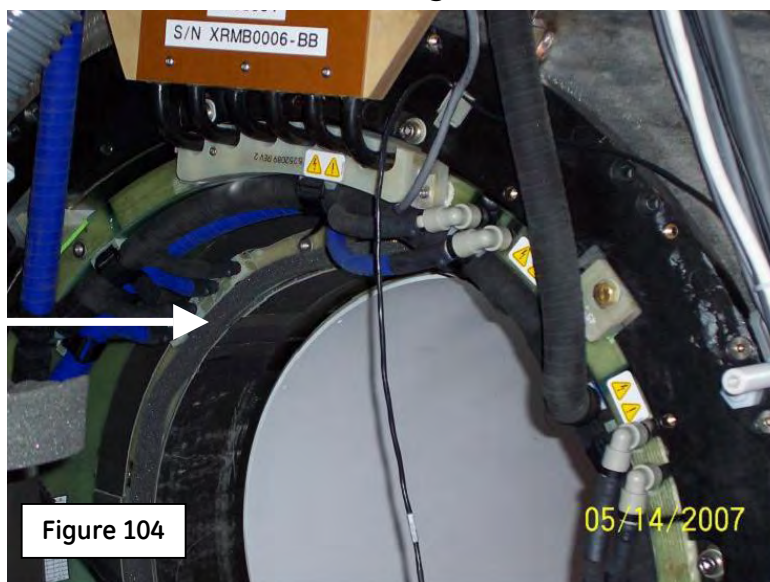


Figure 104

Discovery MR450 1.5T Magnet Enclosure Upgrade

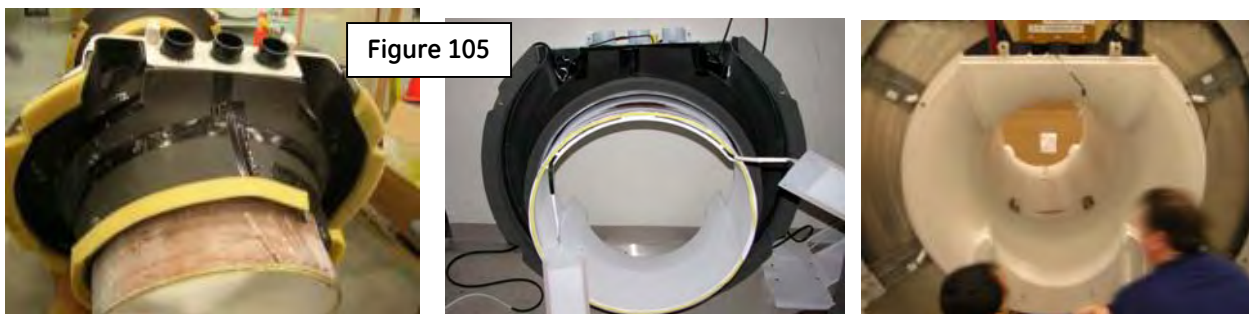
15. Rear End Bell Install

A. Components

- | | |
|--------------------------------------|--------------|
| a) Rear End Bell Assembly | 5173880 |
| b) M10 x 25 screws | 2109867-23 |
| c) Nylon Washer | 46-294167P73 |
| d) Comfort Module Speaker Link Cable | 5183628 |

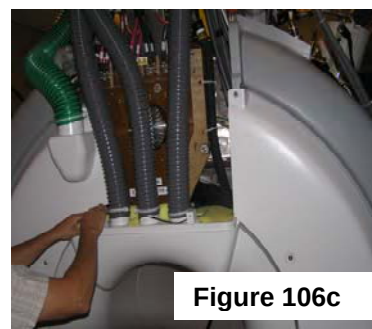
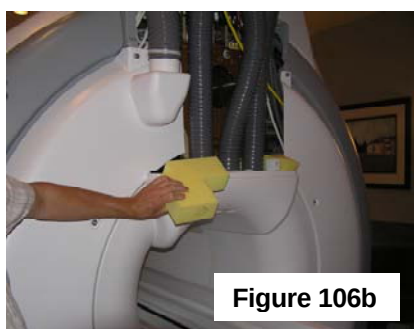
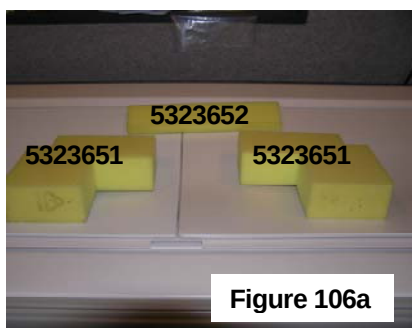
NOTE: Be sure the filter on the cable is on the *outside* of the turtle ring assy and that the cable in the gradient area is well taped in order to prevent damage to the cable. Once the RF has been tuned, add thermal joint compound to the tip of the sensor and place in the hole of the RF before sliding completely into the gradient bore.

- B. Feed the bore light rope through the rear endbell (**Fig 105**) and pull the two ropes through the top cut-outs in the rear endbell (pull back the extra tape and foam pieces while performing this step.) Adjust the speaker cable wire at the top of the rear end bell so that it will not be clamped during the endbell installation.



- C. Position the rear endbell into the bore of the magnet. Start the screws and washers (2109867-23 and 46-294167P73) on the front face of the endbell.
- D. Tighten screws down while adjusting in the X & Y direction to achieve a uniform concentricity (flush surface between endbell and RF). Top and sides are the priority. Tighten to a target gap of 2 +/- 1/2 mm. Using a wrench, tighten the nuts on the standoffs behind the endbell to secure.
- E. Press in foam insert (5323651) on either side of the endbell in the orientation shown. (**Fig 106a**)
- F. Press in foam insert (5323641) between the Rear Endbell and the Busbar assembly (**Fig 106a**). (**Fig 106b & Fig 106c**) show installation and final assembly.

Discovery MR450 1.5T Magnet Enclosure Upgrade



16. RF Tuning Process Check

- A. Remove the front endbell and set it aside for re-installation later in the process.
- B. Slide the RF forward (so that it is no longer contacting the rear endbell). Push back to re-engage the RF to the rear endbell.
- C. Perform the initial tuning check at 120 kHz. Refer to WSI 610-035 for details regarding the test and set-up.
- D. Remove the RF and change the capacitors specified by the ATE test program.
- E. Re-install the RF from the patient end push back into location to achieve a gap of ~ 2mm around the rear endbell. Verify the endbell remains flush with the RF in the top and side locations.
- F. Repeat the Body Coil Tuning test at 120 kHz with only the rear endbell in place. If the test fails, repeat steps D through F until the coil finally passes.
- G. Re-level RF Coil after tuning is complete.

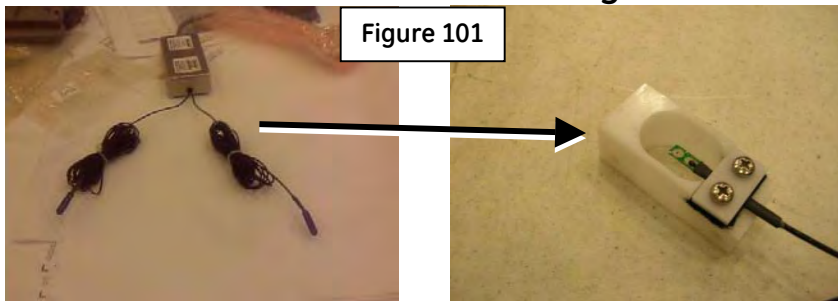
Discovery MR450 1.5T Magnet Enclosure Upgrade

17. Front Endbell Installation

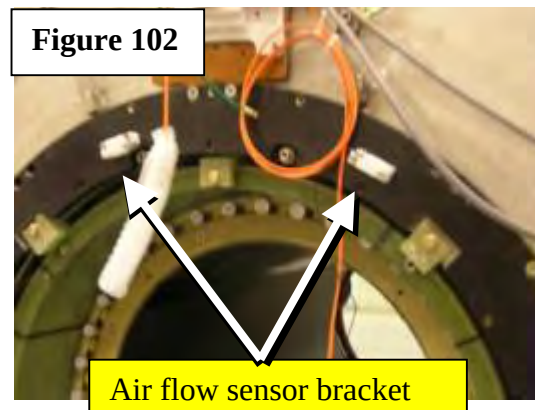
A. Components

- a) Front End Bell (HDv) 5173983
- b) M10x25 SHCS 2109867-23
- c) M10 nylon washer 46-294167P73

- B. Assemble the Airflow Sensor (526679) to AirFlow Brackets (5273219) by removing the screws from the bracket and placing one of the sensors in the center. Screw to bracket to secure. **(Fig 101)**



- C. Install airflow sensor brackets at front of magnet using (4) M6 x 30 pan head screws. **(Fig 102)**



- D. Loosen the nuts on the standoffs on the rear turtle ring arms (where the endbell will be attached to the turtle ring.) Adjust the standoff by turning them into or out of the turtle ring until the slit on the side of the standoff is no longer visible (screwed into the turtle ring region). Leave the nut loose.

Discovery MR450 1.5T Magnet Enclosure Upgrade

- E. Connect cable Run M3350 from SRI J10 to Bore Temperature Tip fits inside of RF tube (5180907). Tape cable along the inside of the gradient.
- F. Move the front endbell (5173983) in place. Start all M10 x 25 screws and nylon washers in place (2109867-23 and 46-294167P73).
- G. Starting at the top (12 o'clock position), adjust the endbell to be flush with the RF surface. Begin tightening screws as the area impacted is adjusted.
- H. Continue to adjust the sides to achieve a uniform concentricity, working around the endbell in and tightening all screws ~2 turns per rotation so the RF and endbell surfaces are flush. Tighten all screws using a T-handle tool (not the battery powered guns)

Note: There may be some variation on the bottom surface. The top and sides need to be as even as possible. Adjust the gap between the front endbell and the RF to a target value of 2mm (+/- ½ mm).



Front End Bell 5173983



Adjust concentricity and gap



Figure b

Figure a

18. Final RF Tuning Checks

- A. Insert the tuning fixture into the bore of the magnet and perform final Body Coil Tuning Test using the 200 kHz setting (since both the front and rear endbell are in place.)
- B. If the Body Coil Tuning test passes, continue to next install procedure.

Discovery MR450 1.5T Magnet Enclosure Upgrade

19. Bridge Installation Preparation

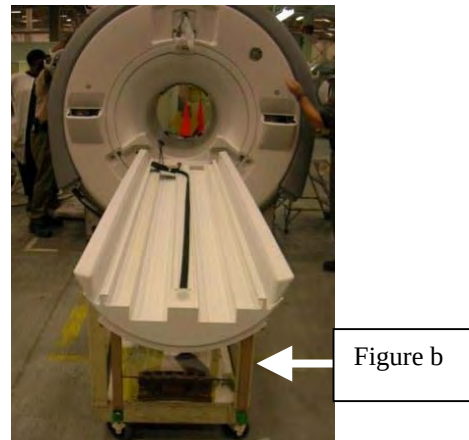
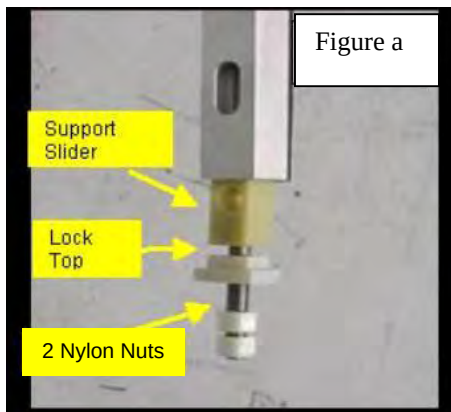
A. Components

- | | |
|-------------------------|------------|
| a) Front Bridge Support | 5168288 |
| b) Support Lock Top | 2293540 |
| c) Support Slider | 2293541 |
| d) Nylon Nut | 2293956 |
| e) M10 x 30mm SHCS | 2109867-24 |

B. Assemble the front bridge support hardware as shown. **(Fig a)**

C. Attach the front Bridge Support to the bottom side of the Bridge with M10x30 (2109867-24) socket head cap screw. **(Fig b)**

Note: *The bridge is heavy. At least two people must lift the bridge at all times – never attempt to lift this by yourself!*



Discovery MR450 1.5T Magnet Enclosure Upgrade

20. Bridge Installation

A. Components

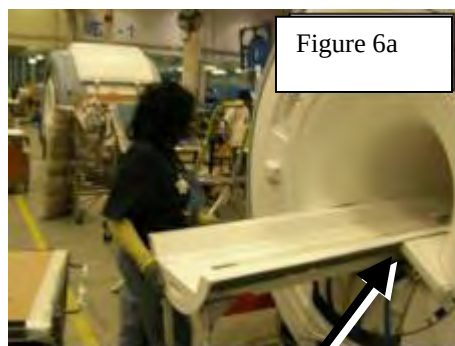
- a) M10 x 80mm screw 2109866-26
- b) Rubber Washer 2167321
- c) Washer 5146169
- d) Cable Ties 46-208758P5

- B. Lift the bridge thru the center of the gradient from the front and align the front bridge support screws with the corresponding holes in the front endbell. **(Fig 6a)**

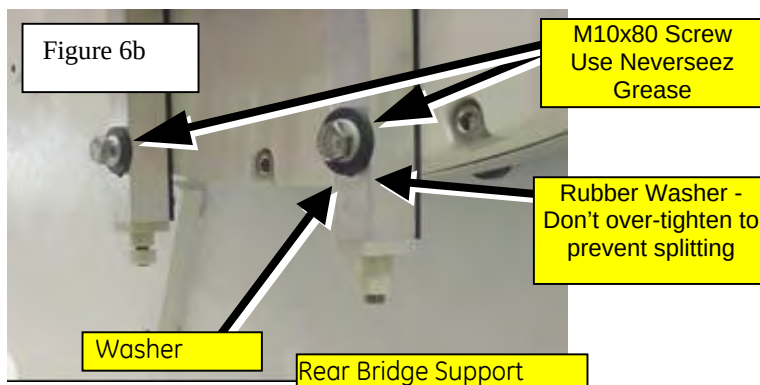
Note: *The HDv bridge is heavy. At least two people must lift the bridge at all times – never attempt to lift this by yourself*

- C. Start but don't tighten the bridge support mounting screws. **(Fig 6b)**

- D. Adjust the height of bridge to 3 mm above the adjacent surface on the end bell on both front and rear ends. **(Fig 6a)**



Adjust bridge 3mm
above front end bell



Discovery MR450 1.5T Magnet Enclosure Upgrade

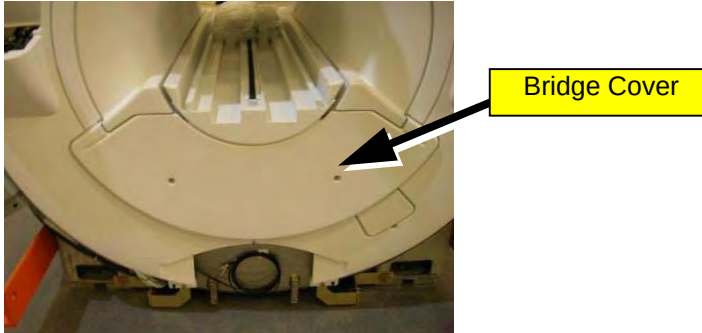
21. Bridge Cover Installation

A. Components

- a) M6 x 25mm SHCS 2109867-15
- b) Bridge Cover 5169762

- B. Attach the Front Bridge Cover with M6 socket head screws. Don't over-tighten.

Note: *RF Coil Bias cable should drape down underneath cover towards bottom of magnet.*



Discovery MR450 1.5T Magnet Enclosure Upgrade

22. Enclosure Tools and Parts for Installation:

Tool Requirements:

1. Slotted screwdriver
2. 3/16" nut driver

Consumable Material Requirements:

- | | |
|----------------------------|--------------|
| 1. Kimwipes™ or equivalent | |
| 2. Loctite 242 | 46-170684P2 |
| 3. Dual lock fastener (3M) | MP-3551/52 |
| 4. Cable ties | 46-252283P68 |

Cable Part Numbers		
Left Side Cable Runs		
5171941-2	1	Run M3339, MAG- J1 to MAG-SRI-J33, Console Intercom, 5600mm
5149629	1	Run M3366 ,Cable, Scan Room Monitor Control Y
5170192	1	Run M3369, MAG- J3 to MAG-SRI-J21, SRI Signal, 5600 mm
5170365	1	Run M3336, MAG- J4 to MAG-SRI-J1, SRI Power, 5600mm
5172994	1	Run M3364, MAG- J5 to MAG-DCK_L-J3, Dock Power, 4400mm
5194965	1	Run M3360 MAG- J6 to MAG-PAC-J2, PAC, HDv Option, 5600mm
5172512	1	Run M3352, MAG- J7 to PED-HUB-SLOT11-J10, Unblank, 8300mm
5166749-4	1	Run M1317, MAG- J8 to PED-HUB-Slot 11-J17, Rx Lo - Exciter Reference Clock,
Left Side and underneath Cable Routed Runs		
5172679-8	1	Run M1313 MAG- J9 to MAG-BORE-J6,6250mm
5172679-9	1	Run M1314, MAG- J10 to MAG-BORE-J7,6250mm
Right Side Cable Runs		
5176536-2	1	Run M1326, MAG- J12 to PED-IO-J1 ,P1-RF Transmit, 4088mm, Ref Dimension
5172955-4	1	Run M1316, MAG- J13 to PED-Hub-SLOT11-J16, NB Exciter, 8300mm
5176536	1	Run M1325, MAG- J14 to PED-IO-J2 , A RF Transmit 3747mm Ref, Phase Length
5172496	1	Run M3353, MAG- J15 to PED-HUB-SLOT11-J9, Transmit Switch, 8300mm
5177909	1	Run M3357 MAG- J16 to PED-J1, Long Drive Power, 8600mm
5172512-2	1	Run M3340, MAG- J17 to PED-TIF-J4, Hall's Signal, 8900mm
Hybrid Cable Runs		
5267796	1	Run Hybrid to Body Coil, 3446mm
5267796-2	1	Run Hybrid to Body Coil , 3446mm
5178044	1	Run M1319, Body Transmit, MAG-HYB-J1 to RUN M1001, 3.0T, 3065mm
5176853	1	Run M1324 MAG-HYB-J4 to MAG-DUM-J12, 3.0T, 1350mm

Discovery MR450 1.5T Magnet Enclosure Upgrade

PAC/SRI Cables		
5174620	1	RUN M2301, MAG-RX1-J9 to MAG-SRI-J17, Dab Out, Simplex F0, Plastic, 2000mm
5180923	1	Run M3342, MAG-SRI-J23 to MAG-BORE, Patient Alignment Lights
5181092	1	Run M3344, MAG-SRI-J22 to MAG-BORE-GRAD_TEMP and MAG-BORE-LIGHT,
5183773	1	Run M3348 MAG-SRI-J34 to Run M3347, Comfort Module Speaker Cable,
5180907	1	Run M3350 MAG-SRI-J10 to MAG-BORE, Bore Temperature, 3000mm
5169277-5	1	Run M3341 MAG-SRI-J20 to MAG-DCK_L-J1, Landmark, Touch and Go, 2500mm
5189419	1	Run M3370 MAG-SRI-J3 to MAG-DCK_L-J2, Dock Control, 2200mm
5170095	1	RUN M3337 MAG-SRI-J4 to MAG-LCP-J11, 5000mm and MAG-RCP-J11, 6600mm
TDM Cables		
5169566-2	1	Run M3368, MAG-TDM-J2 to MAG-RX2 , 44-pin HD Sub-D, 1360mm
5169566	1	Run M3367, MAG-TDM-J1 to MAG-RX1 , 44-pin HD Sub-D, 560mm

23. In Room Display Assembly Process (32 channel only)

A. Components

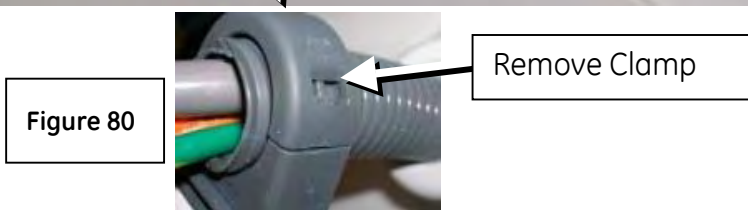
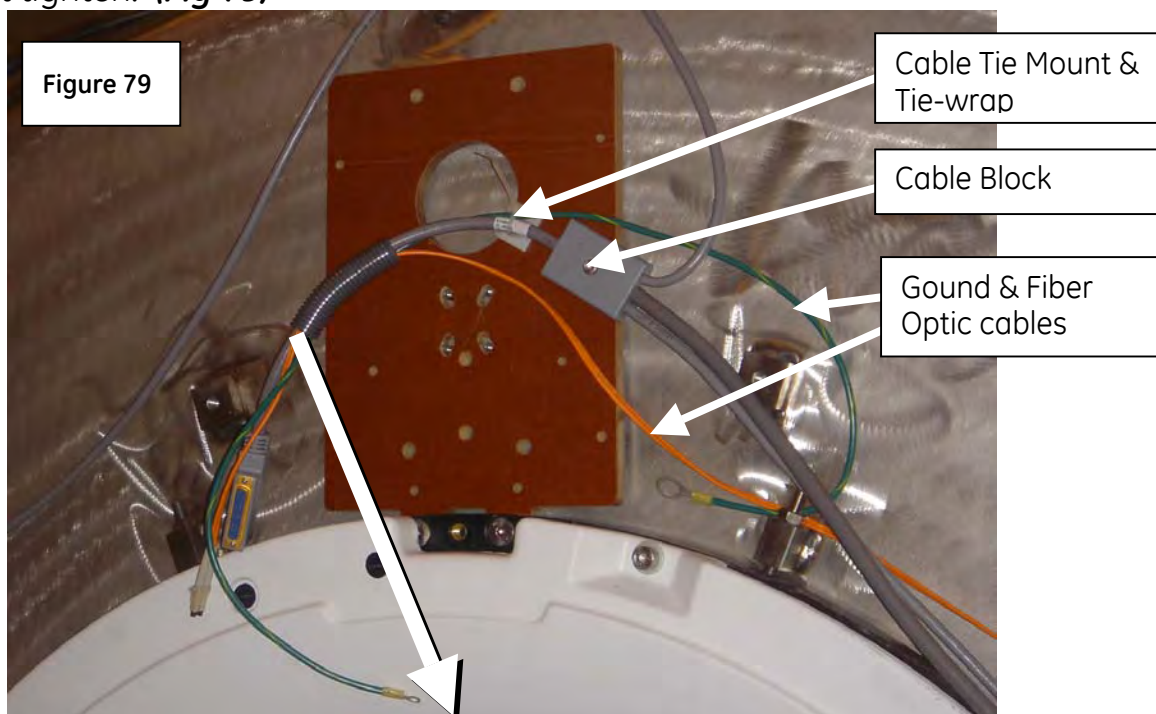
a) M6 x 16 screws (4)	2109867-13
b) Nylon washers (4)	2229794
c) M4 x 20mm screws (4)	2109867-6
d) In room display mount	5133768-3
e) Monitor interface cover	5182762
f) Laser mount bracket	5178328
g) M4 X 10 screws	2109873-14
h) M6 X 25 screw	2109867-15
i) M6 X 12 screw	2109873-24
j) M6 X 16 screw	2109867-13
k) M6 nylon washers	2229794
l) M10 X 16 screw	2109867-21
m) M10 SS washer	2109878-4
n) M4 washer	2109878-2

Discovery MR450 1.5T Magnet Enclosure Upgrade

- B. In Room Display is installed on the front turtle ring arm.
- C. Attach Cable Tie Mount base (46-208746P9) to Arm with M4 X 10 screw (2109873-14). Tighten the mount base with Philips screwdriver. **(Fig 79)**

Note: *The Mount should be angles at about 45 degrees as shown.*

- D. Attach Scan Room Monitor Control cable (5149629) to Arm with (1) M6 X 25 screw (2109867-15) Do not tighten at this time. **(Fig 79)**
- E. Route the ground wire and fiber optic cable adjacent to main cable, one on each side. Do not tie wrap either one to the tie mount. Both are to remain free sliding. Slide tie-wrap through the mount to hold cable in place,, but do not tighten. **(Fig 79)**

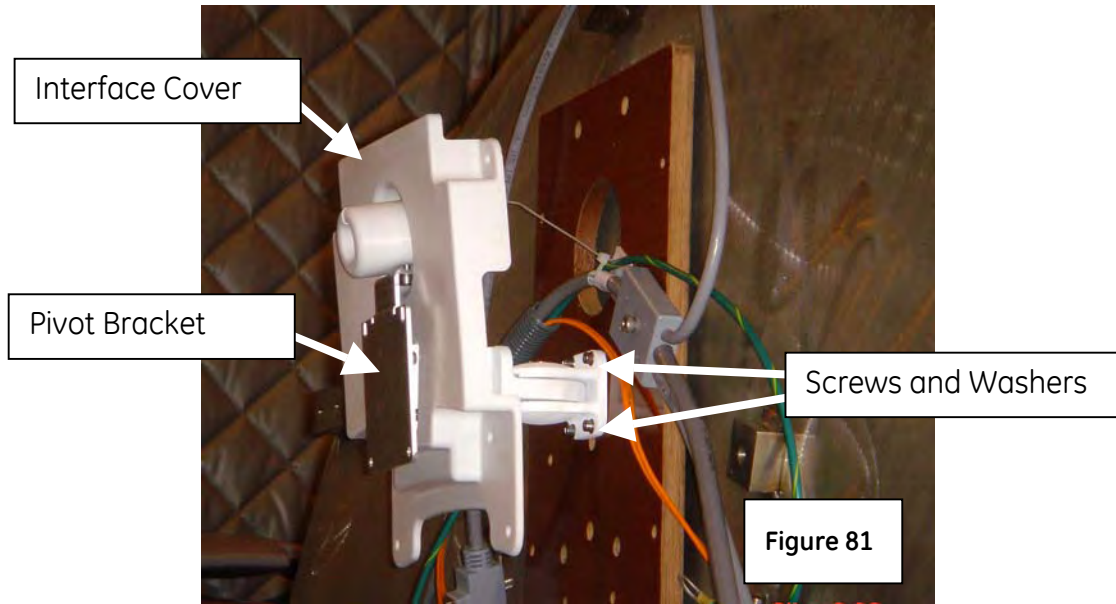


- F. Remove Cable Clamp from cable assembly (5149629). **(Fig 80)** Insert a small screwdriver into the side of the clamp that says "Open" until clamp releases.

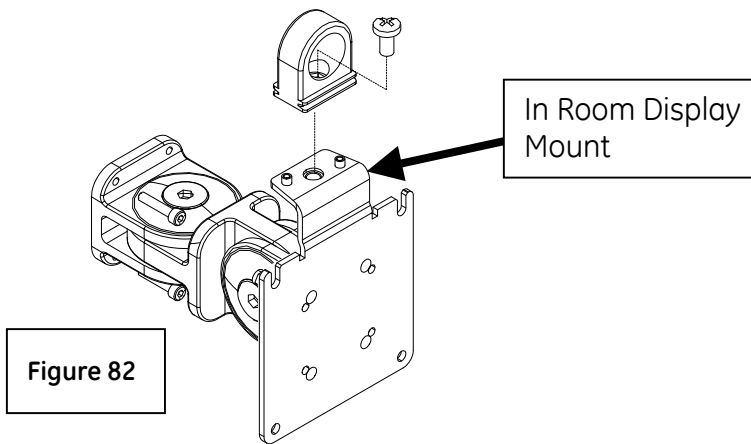
Discovery MR450 1.5T Magnet Enclosure Upgrade

- G. Pass Pivot Bracket through Interface Cover (5182762) and fasten with (4) M4 x 20mm screws (2109867-6) and (4) M4 washers (2109867-6). Add Loctite 242. Using a Number 3 Allen, tighten screws **(Fig 81)**.

Note: Using the single Allen with a 90 degree bend allows for proper tightening.)



- H. Attach the Cable Clamp, removed in Step G, to the In Room Display mount (5133768-3), which is attached to the Pivot Bracket, with M6 X 12 screw. (2109873-24) **(Fig 82)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

- I. On the Monitor Interface Cover, carefully turn inside-out the flexible tube housing to the other side of the cover. **(Fig 82)**

Note: Take special care, housing material can tear easily.

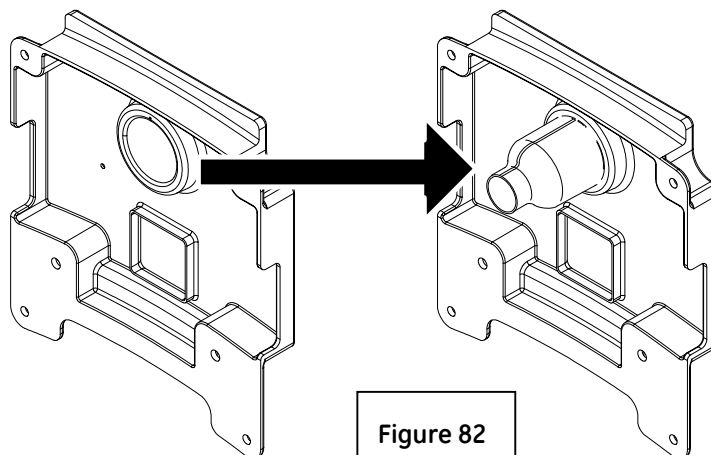


Figure 82

- J. Pass connectors and tubing of the Scan Room Monitor Control cable (5149629) thru the Monitor Interface Cover **(Fig 83)**. Fasten together with tie-wrap to secure the cable assembly to the interface cover. Trim excess from tie-wrap

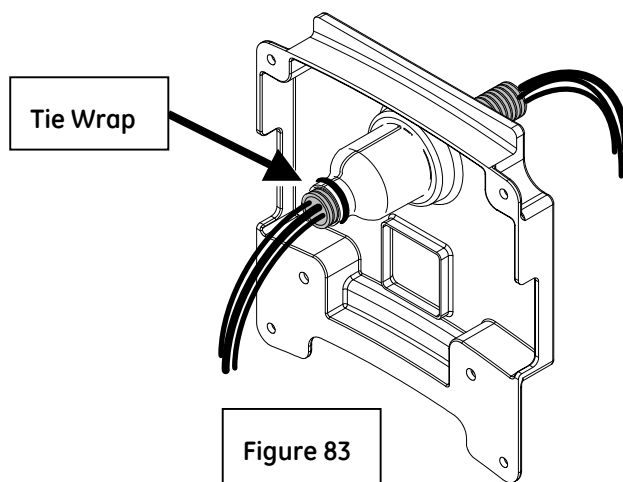


Figure 83

- K. After securing cables to the monitor interface cover, carefully return the flexible tube housing to its original position. **(Fig 84)**.

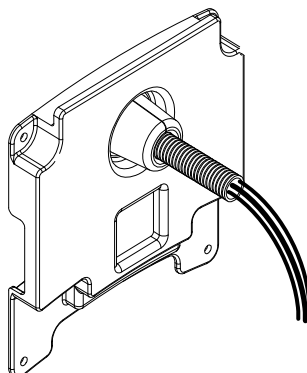
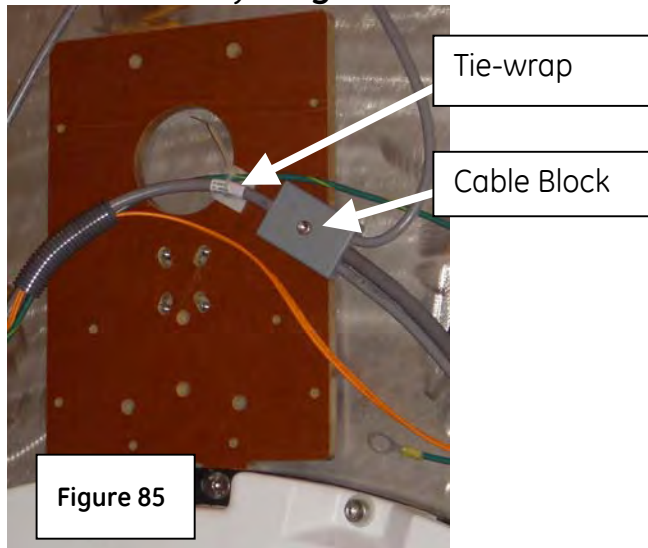


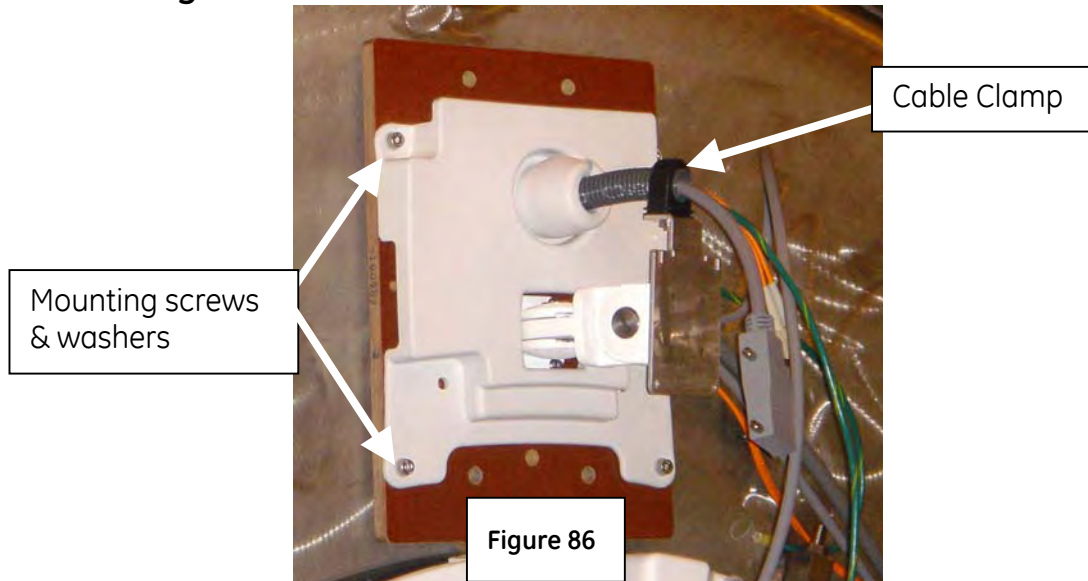
Figure 84

Discovery MR450 1.5T Magnet Enclosure Upgrade

- L. Tighten screw holding cable block to arm assembly. Also secure the tie-wrap around the gray cable only (through cable tie mount) of the Scan Room Monitor Control cable assembly. **(Fig 85)**



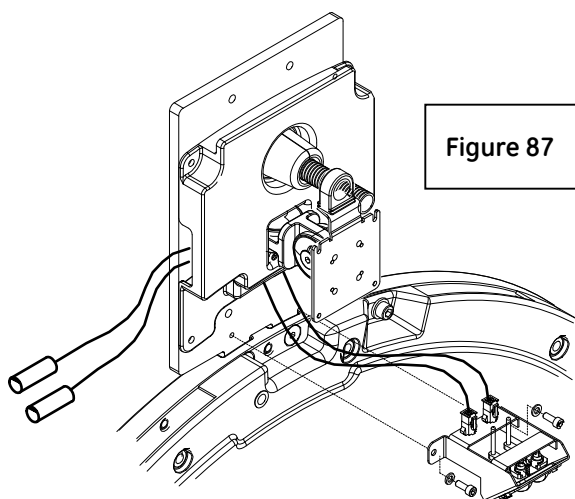
- M. Snap cable assembly into Cable Clamp on top of bracket and close clamp to secure. **(Fig 86)**



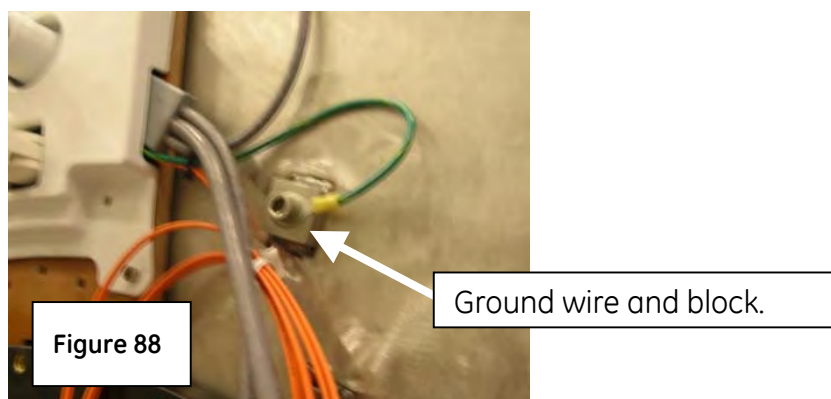
- N. Fasten the interface cover to the monitor mount board with (4) M6 x 16 screws (2109867-13) and 4 nylon washers (2229794) . **(Fig 86)**

Discovery MR450 1.5T Magnet Enclosure Upgrade

- O. Install the laser-mounting bracket (5178328) to the turtle ring arm using (2) M6 nylon washers (2229794) and (2) M6X16 screws (2109867-13). **(Fig 87)**
- a) Start right screw & washer. Slide bracket over slot. Add left screw. Be careful of wire position when adding screws so they aren't crimped behind bracket.
 - b) Route cable under IRD interface cover and out through the left side slot
 - c) Tighten bracket with #5 Allen wrench.

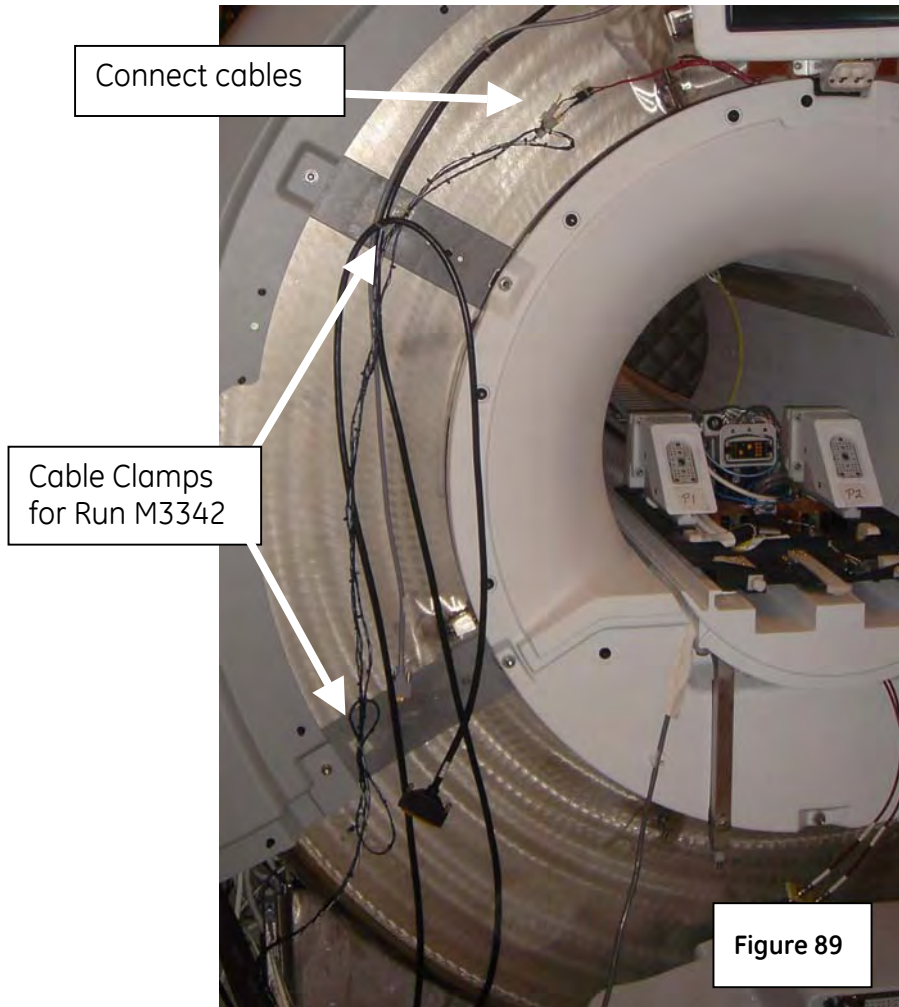


- P. Connect the Monitor Cable assembly ground wire to the front grounding block using (1) M10X16 screw (2109867-21) & (1) M10 SS washer (2109878-4) **(Fig 88)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

- Q. Install the laser light cable, Run M3342 (5180923.) Route laser light cable along the front, left face of the magnet (toward the SRI). **(Fig 89)**
- a) Attach cables to the magnet using cable clamps to ensure proper placement inside of the front covers.
 - b) Connect PAL-R to the right connector of laser bracket.
 - c) Connect PAL-L to left connector.



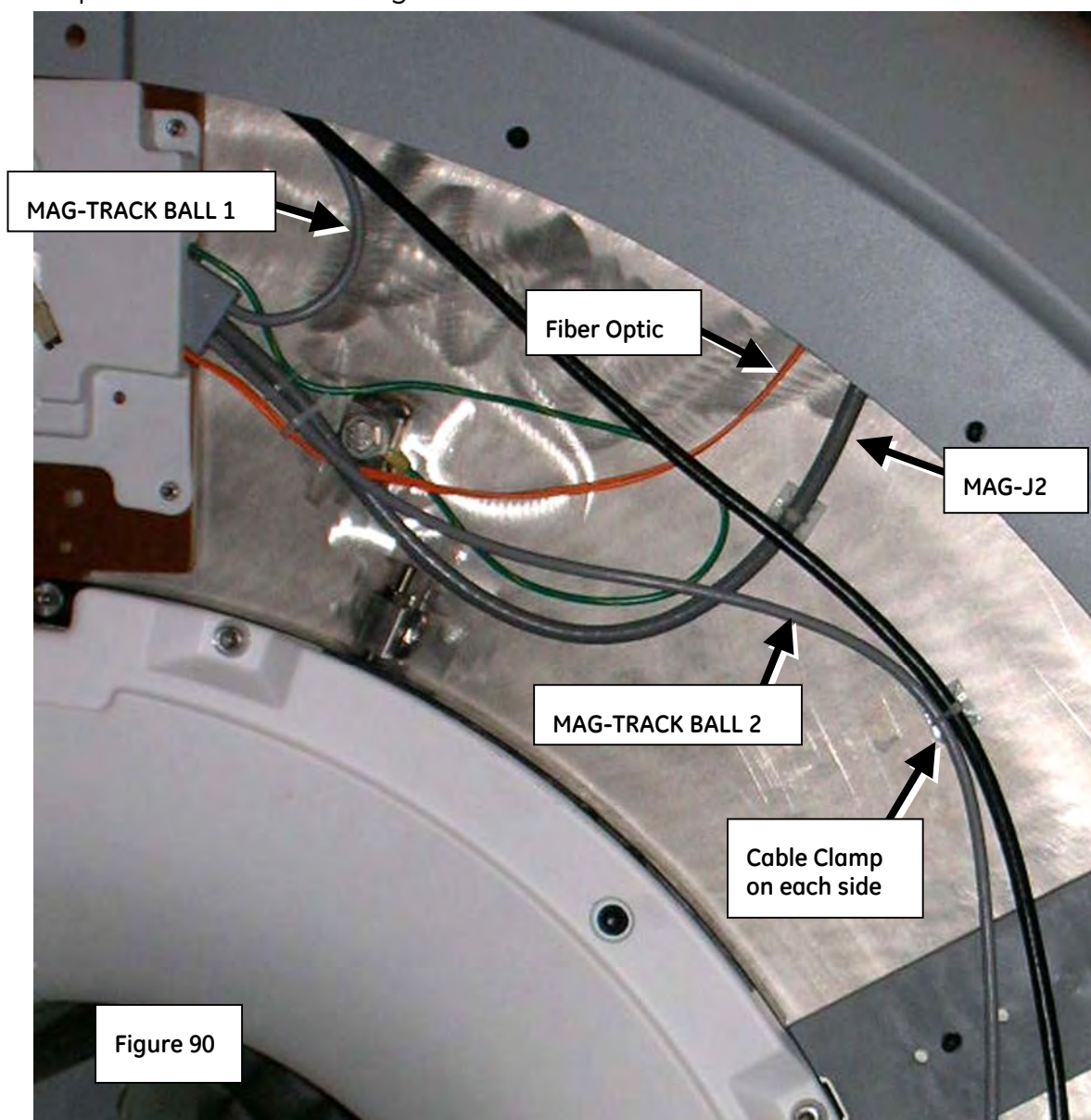
Discovery MR450 1.5T Magnet Enclosure Upgrade

R. Connect remaining cables from In-Room Display *(Fig 90)*

- a) Route cable marked MAG-J2 to the top of the magnet and connect to J2 on the magnet I/O board
- b) Route fiber optic cable to top of magnet and connect to J28 on the fiber optic I/O board

CAUTION: Do not crimp fiber optic cable.

- c) Route cable marked MAG-TRACK BALL 1 to the left side of the front of magnet
- d) Route cable marked MAG-TRACK BALL 2 to the right side of the front of magnet
- e) Attach cables to the magnet using cable clamps to ensure proper placement on the magnet.



Discovery MR450 1.5T Magnet Enclosure Upgrade

24. PAC / SRI Sled Assembly

A. Components

- | | |
|----------------------------|--------------|
| a) Cable ties | 46-252283P68 |
| b) Cable tie anchor | 46-208747P1 |
| c) Dual Lock fastener (3M) | 5110749-2 |
| d) Tray | 5245152 |
| e) SRI Module | 5160929 |
| f) PAC | 5112668-2 |

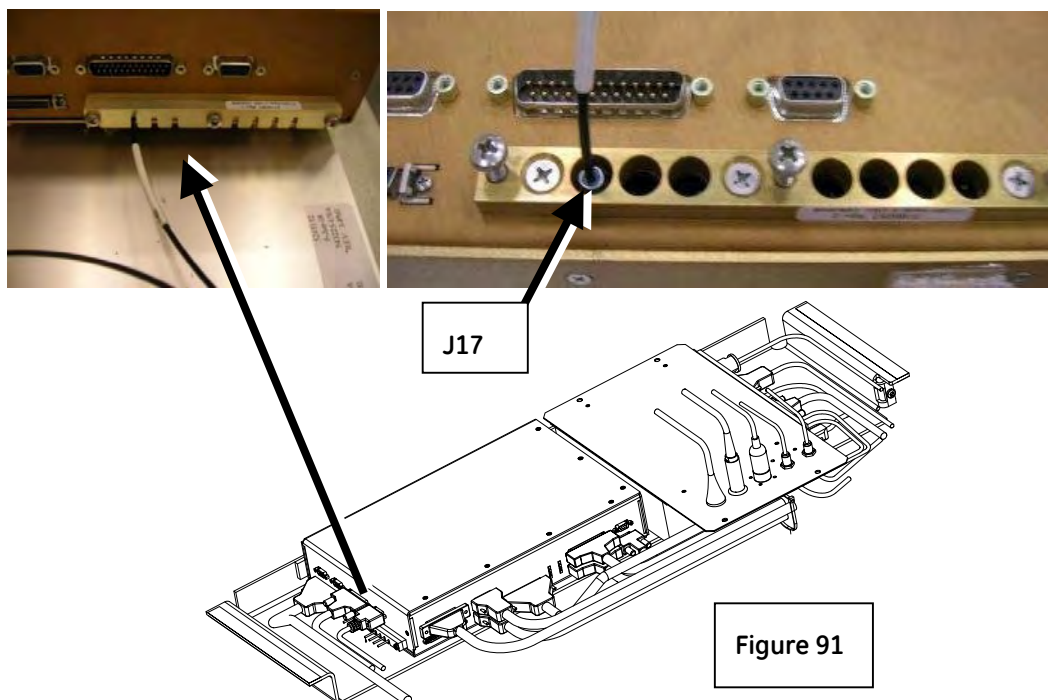
B. Connect the fiber optic cable (5174620) to the SRI **(Fig 91)**

(it is easier to do prior to locating the SRI to the sled, but can be done at any time during the PAC/SRI Sled Assembly or Installation process)

- Remove the cover plate from the side of the SRI by removing the three Phillips head screws.
- Insert the SRI cable into the center of the J17 port. The simplex connector should snap into the receptacle.

Note: Do not attempt to insert the connector at an angle. When properly installed, the connector should be straight into the port and secure.

- Slide the cover plate over the cable and re-secure using the three Phillips head screws.
- Gently coil the cable on top of the SRI. The other end will be connected to the Receiver (refer to Section 5.)



Discovery MR450 1.5T Magnet Enclosure Upgrade

- C. Place PAC/SRI components on the sled tray (5264147) using Dual-lock fasteners, configured as shown below. The Dual-lock is pre-applied to the sled tray assembly. **(Fig 92)**

Note: *Component dual lock strips should be perpendicular to sled dual lock strips*

- a) The SRI (5160929) is placed on the dual lock strips on the left side of the sled.
- b) The PAC (5112668-2) is placed on the right side.

Note: *The top plate of the PAC will overhang the sled.*



Note: *IMPORTANT – Complete steps 26 and 27 before starting Step 28*

Discovery MR450 1.5T Magnet Enclosure Upgrade

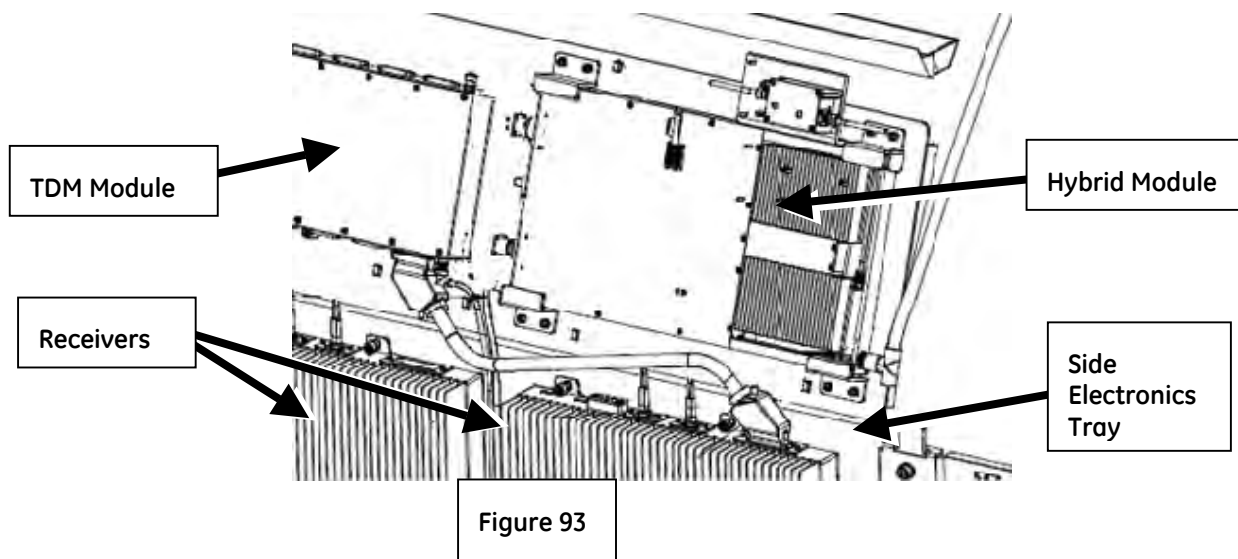
25. Install MR450 Modules on Side of Magnet

A. Components **(Fig 93)**

- | | |
|-----------------------------|------------|
| a) 1.5T Hybrid Assembly | 5328798 |
| b) TDM Module | 5165957 |
| c) M6x12 slotted screws (4) | 2109873-24 |
| d) Remote Receiver (2) | 5174169 |

- B. Attach the Hybrid assembly (5328798) to the top right position of the side electronics plate using (4) brackets on electronics tray.
- C. Install TDM module (5165957) on side electronics plate in left top location using (4) M6 x 12 Phillips screws (2109873-24).
- D. Install the receivers (5174169) onto the side electronics plate using the screws riveted to the receiver assy. The receiver on the lower right side is Receiver 1 and the receiver installed in the lower left position is designated as Receiver 2. LED and Fiber Optic connectors are at top of Receivers when installation is complete.

Note: Both receivers are identical components; the wiring is different



Discovery MR450 1.5T Magnet Enclosure Upgrade

26. Install Cables on Side Electronic Modules

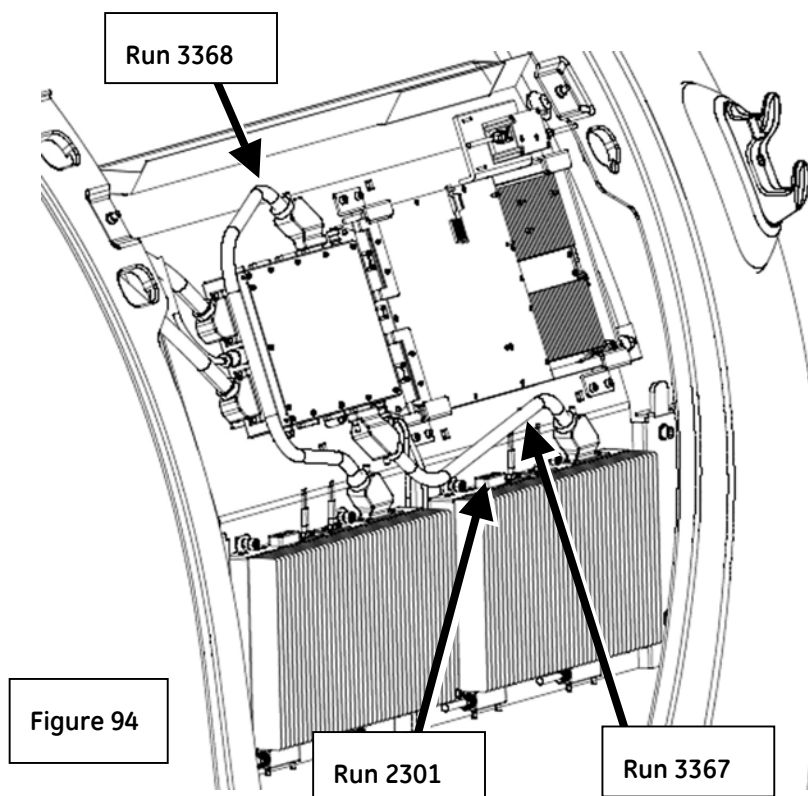
A. Components **(Fig 94)**

- a) Cable, Run M3367 5169566
- b) Cable, Run M3368 5169962-2
- c) Fiber Optic Cable, Run M2301 5174620

B. Install cable 5169566 Run M3367 from MAG-TDM-J1 to MAG-RX1-J7.

C. Install Cable 5169962-2 RUN M3368 from MAG-TDM-J2 to MAG- RX2-J7.

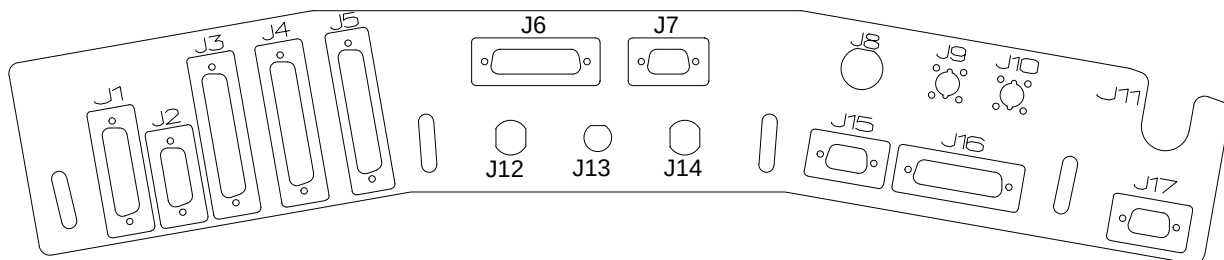
D. Install fiber optic cable 5174620 RUN M2301, MAG-RX1-J9 to MAG-SRI-J17 to Receiver 1. Remove the plastic protector from the J9 port and install the simplex connector into the center of the receptacle until it clicks into place.



Discovery MR450 1.5T Magnet Enclosure Upgrade

27. Cable Installation to magnet I/O Bracket

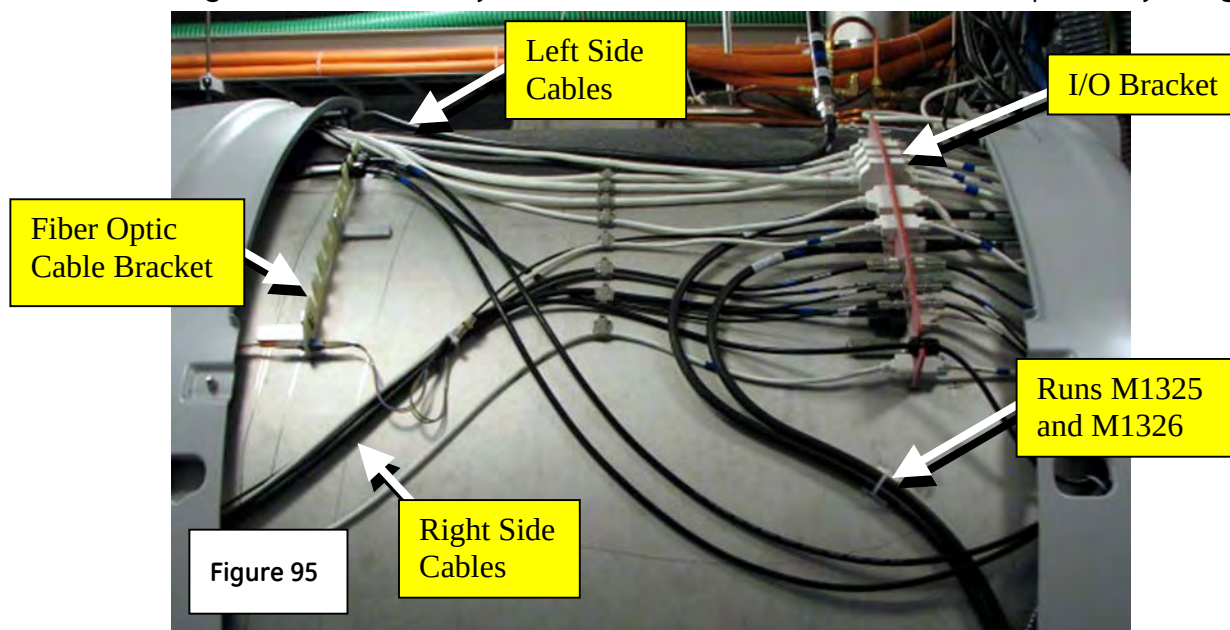
- A. Cables are to be connected to I/O Bracket as indicated in Table below.
- B. Use provided standoffs to secure cables to I/O panel.



Left Side Cable Runs	Right Side Cable Runs
J1: Run M3339 to SRI J33 (5171941-2)	J7: Run M3352 to PED-HUB-SLOT11-J10 (5172512)
J3: Run M3369 to SRI J21 (5170192)	J8: Run M1317 to PED-HUB-Slot11-J17 (5166749-4)
J4: Run M3336 to SRI J1 (5170365)	J12: Run M1325 to PED-IO-J2 (5176536)
J5: Run M3364 to MAG-DCK_L-J3 (5172994)	J13: Run M1316 to PED-HUB-SLOT11-J16 (5172955-4)
J6: Run M3360 to MAG-PAC J2 (5194965)	J14: Run M1326 to PED-IO-J1 (5176536-2)
Right Side Cable Runs to Front and Rear of Magnet Bore	J15: Run M3353 to PED-HUB-SLOT11-J9 (5172496)
J9: Run M1313 to MAG-BORE-J6 (5172679-8)	J16: Run M3357 to PED-MTR-J1 (5177909)
J10: Run M1314 to MAG-BORE-J7 (5172679-9)	J17: Run M3340 to PED-TIF-J4 (5172512-2)
J18: Run M1343 to MAG-BORE-J8 (5172679-18)	
J19: Run M1344 to MAG-BORE-J9 (5172679-19)	

Discovery MR450 1.5T Magnet Enclosure Upgrade

- C. Connect cables to be routed to left side of the magnet first. Route over top, towards front of magnet. **(Fig 95)**
- D. Then connect and route cables for right side of magnet. **(Fig 95)**
- E. **Note:** Runs M1325 and M1326, connected to **J12** and **J14**, should be first on the right side since they are less flexible and are routed separately. **(Fig 95)**



- F. Route left side cables along the top of; keep the cables in the area of 220mm width to ensure the proper fit of Arc cover. Secure with ~ 3 cable tie wraps. **(Fig 96)**
- G. For right side cables, route all cables underneath the side cover bracket. Secure with cable-tie wraps. **(Fig 96)**
- H. For M1325 and M1326, route cables in a U-turn from the I/O bracket and route along the rear magnet flange toward the lower part of the magnet. The cables will be connected to the I/O bracket on the Rear Pedestal. Make sure there is at least 2 feet of cable extending past the end of the magnet flange to make this connection. Secure with cable tie-wraps. **(Fig 97)**

Discovery MR450 1.5T Magnet Enclosure Upgrade

- I. Route and secure cables as described in the cable routing detail. Attach these cables to the magnet using the cable clamps and cable ties. **(Fig 96)**

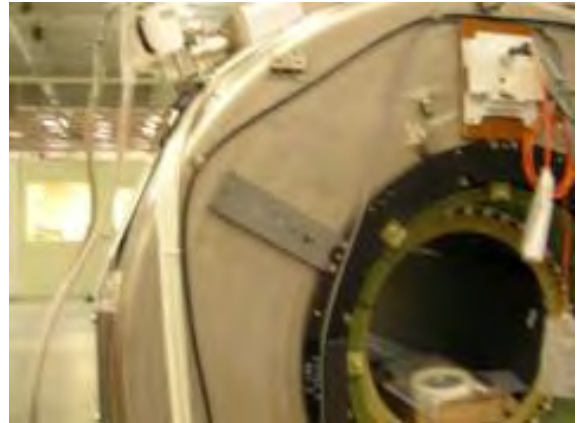


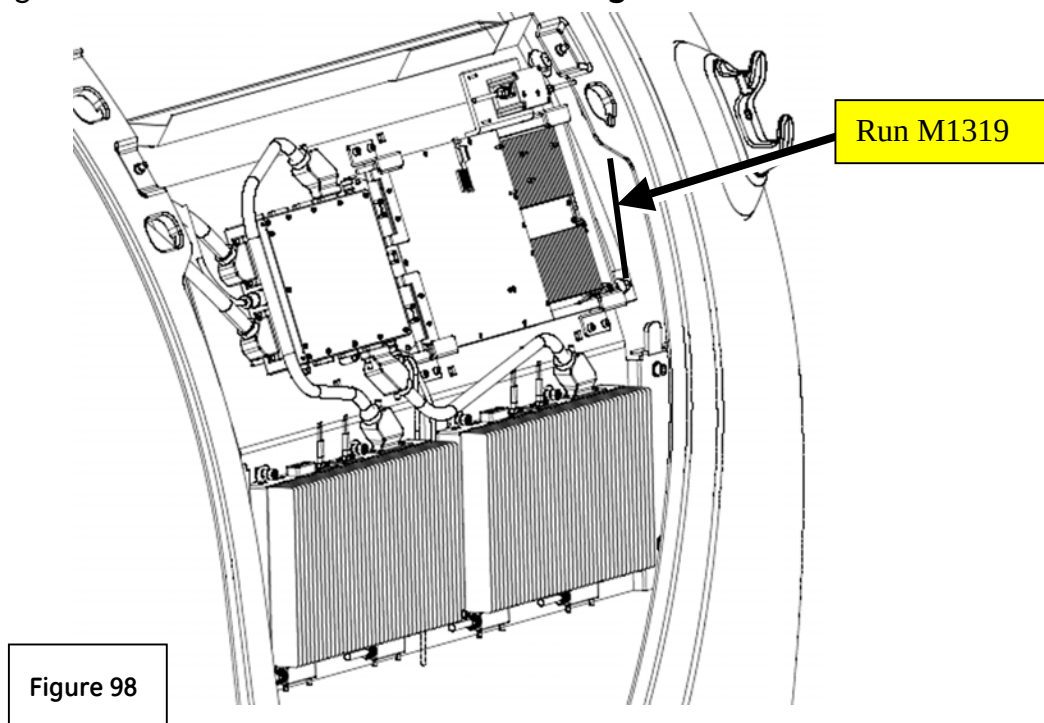
Figure 96

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- J. For M1225 and M1326, make sure there is at least 2 feet of cable extending past the end of the magnet flange to make this connection. Secure with cable tie-wraps. **(Fig 97)**



- K. Connect cable run M1319 to Hybrid J1. Route along top of magnet to rear of magnet for connection to Run M1001. **(Fig 98)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

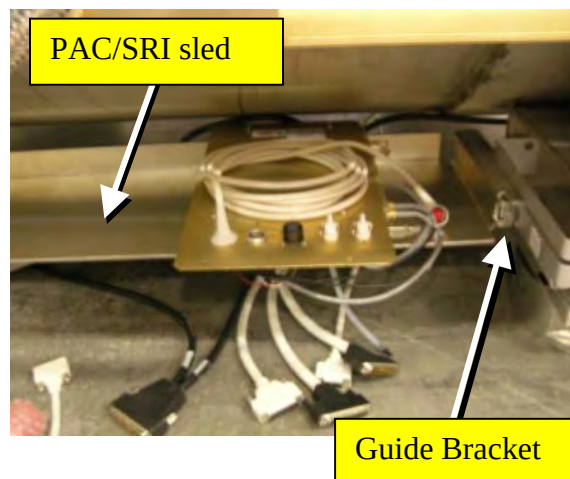
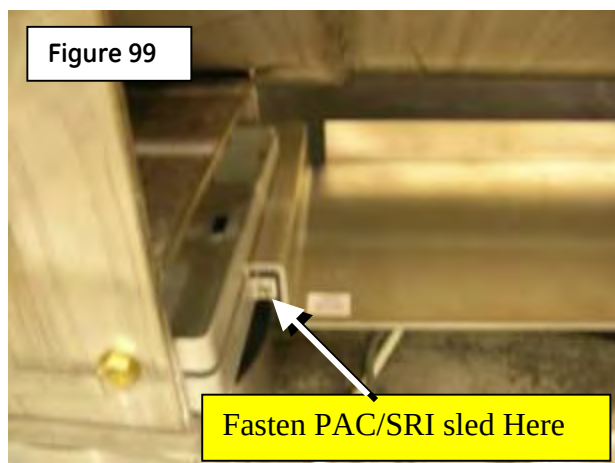
28. Install the PAC / SRI Sled

A. Components

- a) Bracket Shelf Guide 5245151
- b) PAC/SRI Sled 5245152
- c) Sled screw M6x12 46-312342P48

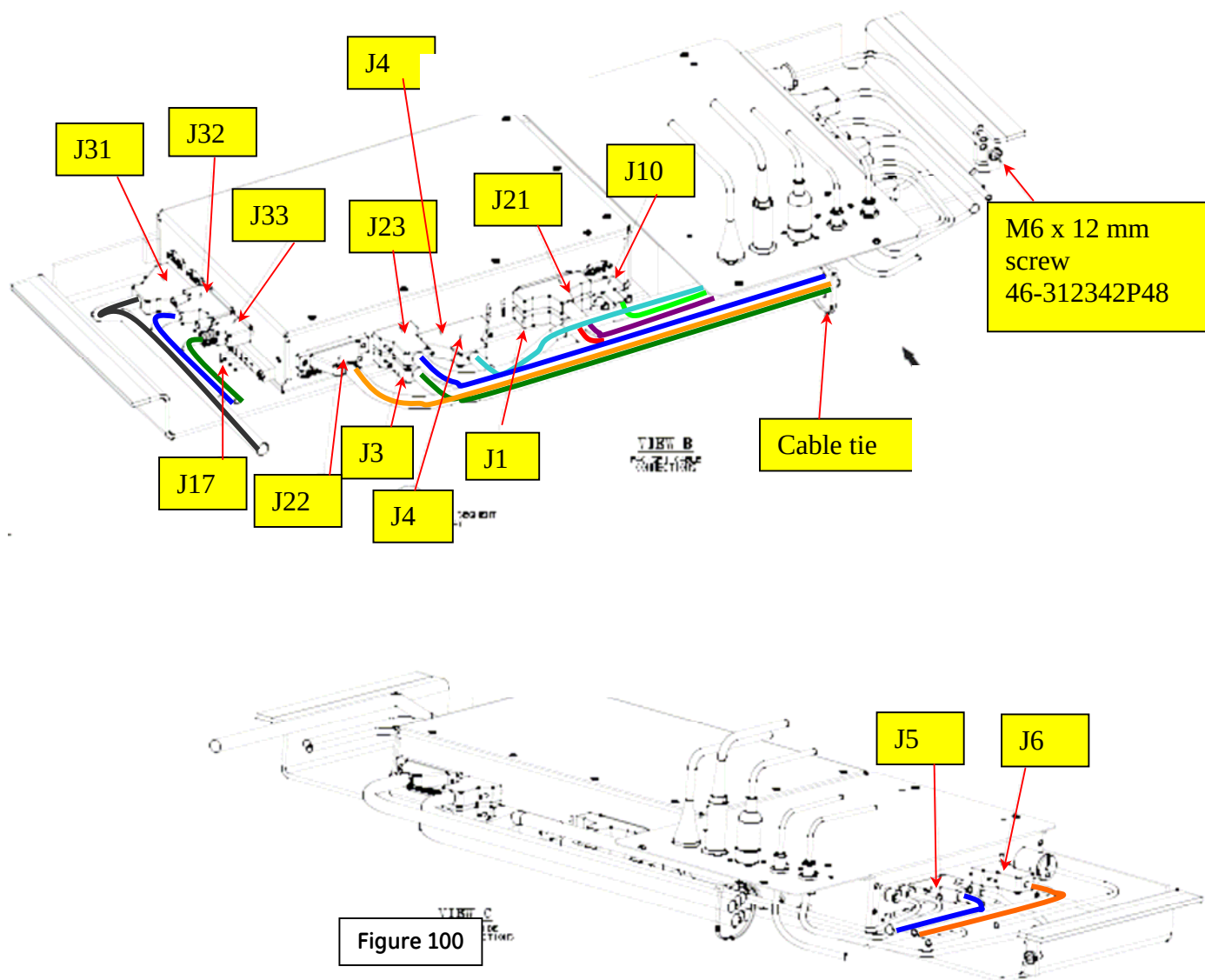
B. Install guide brackets. Place double-sided tape on bottom side. **(Fig 99)**

C. Install the PAC/SRI sled into guide brackets and connect the cables as shown in the following table. **(Fig 100)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

NOTE: PAC / SRI CABLE TERMINATIONS DETAIL



Discovery MR450 1.5T Magnet Enclosure Upgrade

29. Magnet Bore Cable Installation

A. Components

- | | |
|---------------------------|--------------|
| a) Cable Clamps | 5197229 |
| b) Dual Lock | 5110749-2 |
| c) Wire ties | 46-252283P68 |
| d) Wire tie anchor | 46-208747p1 |
| e) Air flow sensor | 5273219 |
| f) M6 x 30 pan-head screw | 2109873-28 |

- B. Route cable Run M1313 (5172679-8) from MAG J9 and Run 1314 (5172679-9) from MAG J10 to the lower center front of the magnet and secure the looped ends with cable clamps to the magnet. (These will be connected in the field to MAG-BORE-J6 and to MAG-BORE-J7.)

- C. Connect cable Run M3344 (5181092) from SRI J22 to MAG-BORE-GRAD_TEMP and MAG-BORE_LIGHT connectors. Route from the SRI to the service end of the magnet.

- D. Connect cable Run M3342 (5180923) from SRI J23 to MAG-BORE-PAL (Patient Alignment Lights) (The end that plugs into the laser alignment lights are marked with a -L and -R to designate the left and right connections.)

Discovery MR450 1.5T Magnet Enclosure Upgrade

- E. Connect cable Run M1322 (5267796) from Hybrid J4 to MAG-BORE-J2 (connector on service end of RF tube). Secure the Run 1322 cable (also referred to as "I" cable) to the side of the magnet with a cable tie. **(Fig 102)**
- F. Connect cable Run M1323 (5267796-2) from Hybrid J3 to MAG-BORE-J3 (connector on service end of RF tube). Secure the Run 1323 cable (also referred to as "Q" cable) to the side of the magnet with a cable tie as indicated in Figure 8a, Item 2. Route both the I & Q cables to the rear of the magnet and tie-wrap the cables together as shown in Item 3. When connecting the cables to the connections in the RF, make sure the cables are routed to the outside of the Rear Pedestal mounting studs on the rear endbell. **(Fig 102)**

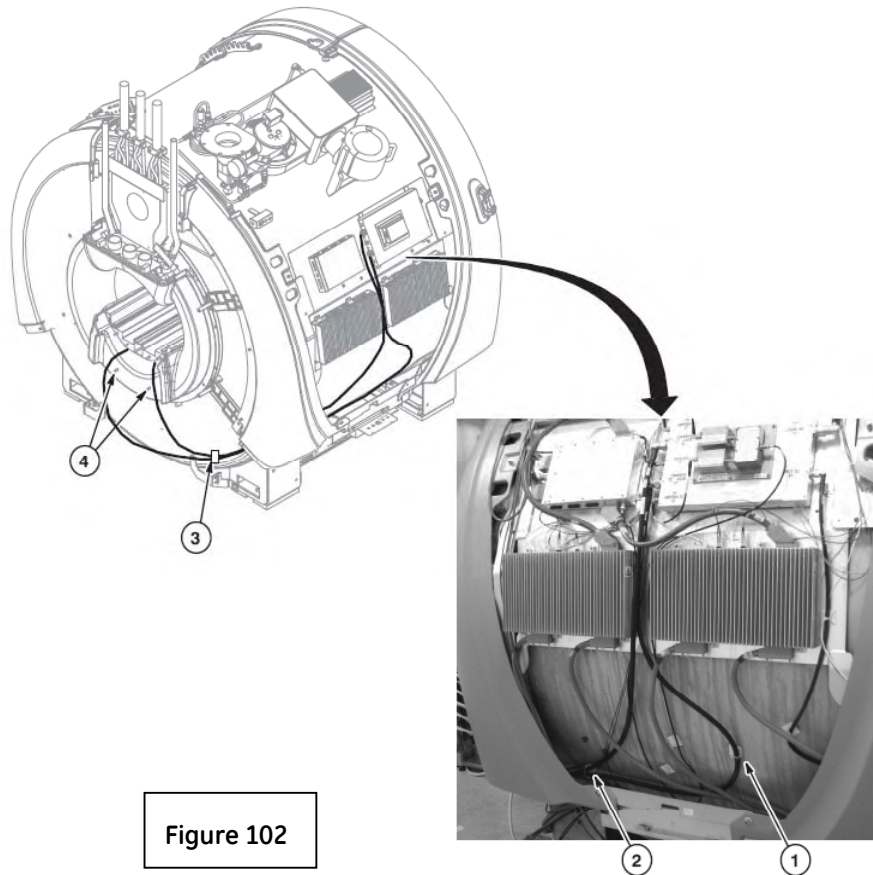


Figure 102

- G. Terminate the cables to the PAC and SRI modules on the sled.
- H. Attach the air flow sensor box to the top of the SRI using Dual Lock.

Discovery MR450 1.5T Magnet Enclosure Upgrade

- I. Connect cable Run M3348 (5183773) from SRI J34 to Run M3347 Comfort Module Speaker Cable. Route toward the service end of the magnet. Connect to the speaker cable that is part of the rear endbell assy once the rear endbell is installed.
- J. Connect cable Run M3337 (5170095) to MAG-SRI-J4. Cable is split and needs to be mounted so that it is accessible for the track ball assembly. Route MAG-RCP-J11 to the right side of the magnet and clamp near right flange. Route MAG-LCP-511 on left front Flange area.
- K. Connect cable Run M3366 (5149629) from the IRD bracket assy (on the front turtle ring arm) and route the section of cable labeled MAG-J2 to the top of the magnet and connect to the J2 slot on the I/O bracket. Remove the plastic plug from the fiber optic board (toward the front) and plug in the fiber optic cable portion to the J28 connector.
- L. Connect cable Run M3341 (5169277-5) to SRI J20 slot and route the cable toward the patient end of the magnet. Coil and clamp the cable to the magnet below the side electronics plate. This will need to be accessible for dock bracket connections (MAG-DCK_L-J1).
- M. Connect cable Run M3370 (5189419) to SRI J3 slot. Route the other end of the cable toward the front of the magnet. Coil and clamp the cable to the magnet below the side electronics plate. This will need to be accessible for dock bracket connections (MAG-DCK_L-J1) during the Staging and field installation processes.

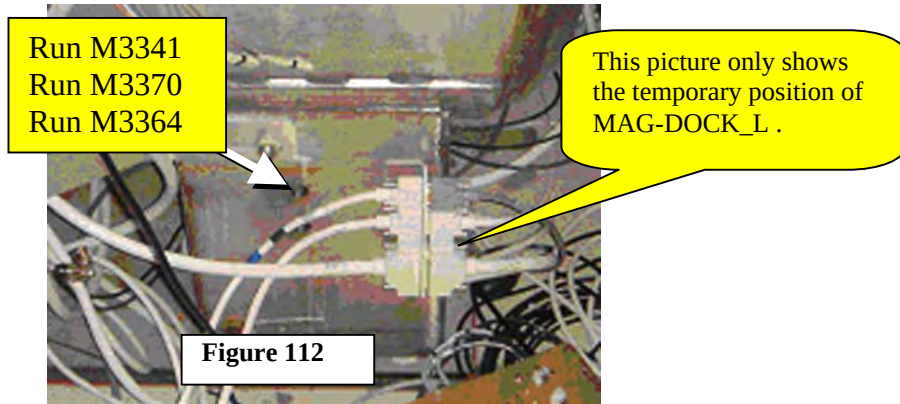
Discovery MR450 1.5T Magnet Enclosure Upgrade

30. Installation of cables for MAG- DOCK_L

A. Components

- a) Cable Clamps TOP 5197230
- b) Cable Ties 46-252283P68

- B. Connect cable Run M3370 to SRI J3, cable Run M3341 to SRI J20, then tie these 2 cables with cable Run M3364 which from MAG-J5 on Side Cover Bracket by Tie-wrap.



Discovery MR450 1.5T Magnet Enclosure Upgrade

31. Front and Rear Arc Installation

A. Components:

- | | |
|-----------------------|--------------|
| a) Right Front Arc | 5194882 |
| b) Left Front Arc | 5194883 |
| c) Right Rear Arc | 5191677 |
| d) Left Rear Arc | 5191116 |
| e) M10X25 SHC Screw | 2109867-23 |
| f) M10 Nylon Washer | 46-294167P73 |
| g) M10 x 16 SHC Screw | 2109867-21 |

B. Install the front right arc onto the arms of the turtle ring using M10 x 16 screws and nylon washers (2109867-21 and 46-294167P73.) (

C. Install the front left arc onto the arms of the turtle ring using M10 X 16 screws and nylon washers.

Note: Front Left Arc – Route cables over dock bracket assembly. Note: Air Sensor Cable mates with cable in Front Left Arc bundle in front of the magnet.

D. Re-install the tall standoffs on the rear turtle ring arms. Measure and set the standoffs to a target of 77 mm tall (3.03”) relative to the surface of the rear endbell. Adjust the standoffs for the rear arcs to be 22 mm tall relative to the rear endbell surface (25.75mm relative to the arms, or 1.01”). **(Fig 107c)**

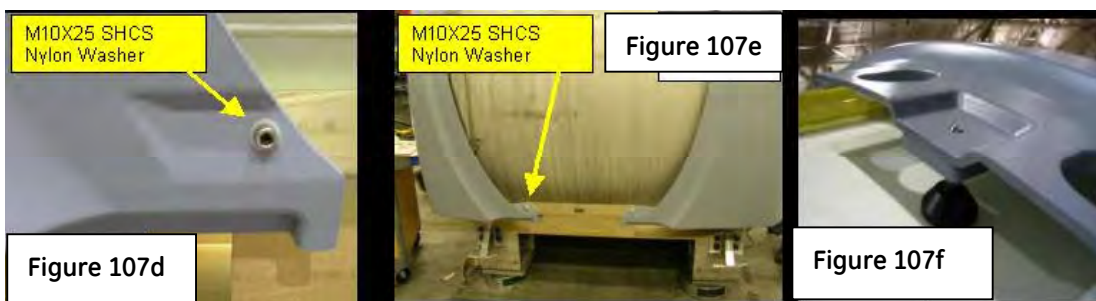
E. Install the right and left rear arcs as shown using M10 x 25 screws with nylon washers **(Fig 107a & b).**



F. Attach the front and rear arcs to the side brackets using M10 x 25 socket head screws and washers. Tighten all screws for the arcs. Adjust side brackets to minimize arc gaps or to avoid having the arcs rub against the radial pads. Remember to tighten the side brackets once arcs are completely adjusted.

Discovery MR450 1.5T Magnet Enclosure Upgrade

- G. Add a M10 x 25 screw and washer to connect the front right and left arcs at the top of the magnet. Tighten screw.
- H. Adjust the swivel pads to set the arc profile. Adjust to obtain a uniform surface for the enclosures and minimize gaps between components.

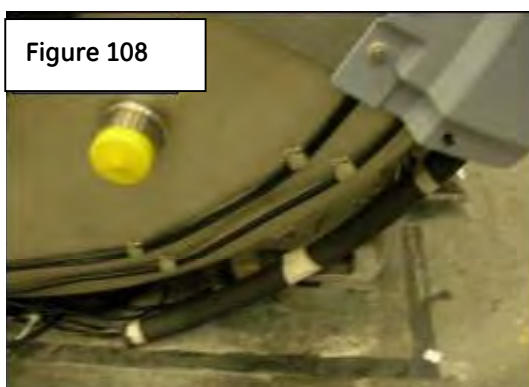


32. Front and Rear Cover Installation

A. Components:

- | | |
|------------------------------|--------------|
| a) Cable Clamp | 5197229 |
| b) Speaker Cable | 5183773 |
| c) Rear Cover right | 5175740 |
| d) Rear Cover left | 5175750 |
| e) M10X25 SHCS | 2109867-23 |
| f) Nylon Washer | 46-294167P73 |
| g) Phenolic Washer | 5146165 |
| h) Right rear cover bracket | 5160992 |
| i) Left rear cover bracket | 5160992-2 |
| j) Right front cover bracket | 5195461-2 |
| k) Left front cover bracket | 5195461 |
| l) M10 x 16 SHCS | 2109867-21 |

- B. Install cable clamps (5197229) to the rear turtle bracket and magnet end flange as shown (**Fig 108**)
- C. Route the cable on the turtle bracket using the cable clamps.



II.

Discovery MR450 1.5T Magnet Enclosure Upgrade

- D. Install the left and right rear covers using four button head screws. At the top of the rear covers, insert a M10 x 25mm screw (2109872-12) and nylon washer. **(Fig 109a)**
- E. Connect speaker cable **(Fig 109b)**. Check speaker cable using a multi-meter set to the ohms setting and checking pins 6 and 9.



Figure 109a



Figure 109b

- F. Push outer edge of the rear arc covers to engage the popper fasteners into the arcs. If these do not properly engage, adjust the standoffs in the arcs to stiffen the arc support. This should allow the rear covers to push into the poppers on the arcs.
- G. Attach the rear cover to the brackets using M10 x 25mm SHCS. **(Fig 110)**
- H. Adjust the rear arc at the bracket and the swivel pads
- I. Tighten and check the side bracket screws that attach the rear arcs.

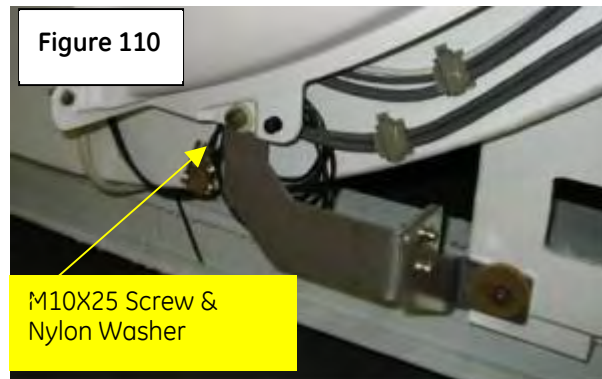
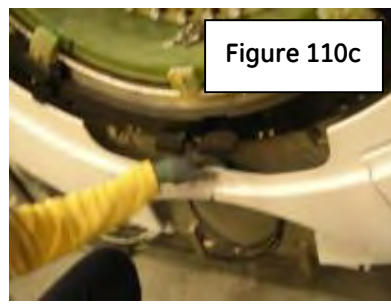
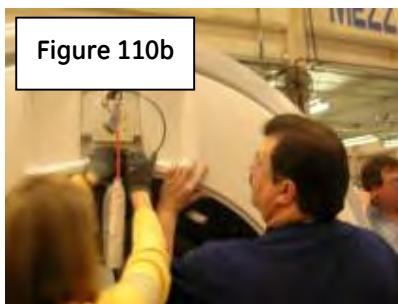
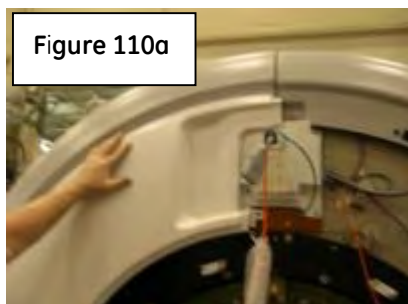


Figure 110

M10X25 Screw &
Nylon Washer

Discovery MR450 1.5T Magnet Enclosure Upgrade

- J. Install the right and left front covers using M10 x 16mm (2109867-21) screws and nylon washers (46-294167P73). Use caution when attaching the top screw so that the laser light wires or the fiber optic cable do not get crimped behind the cover. **(Fig 110a)**
- K. Push outer edge of the rear covers to engage the popper fasteners into the arcs. If these do not properly engage, adjust the standoffs in the arcs to stiffen the arc support. This should allow the front covers to push into the poppers on the arcs.
- L. Install the front cover brackets into the magnet feet using M10 x 25mm SHCS. **(Fig 110c)**
- M. Adjust the arcs at the bracket and the swivel feet.
- N.** Tighten and check side bracket screws that attach the front arcs.



Discovery MR450 1.5T Magnet Enclosure Upgrade

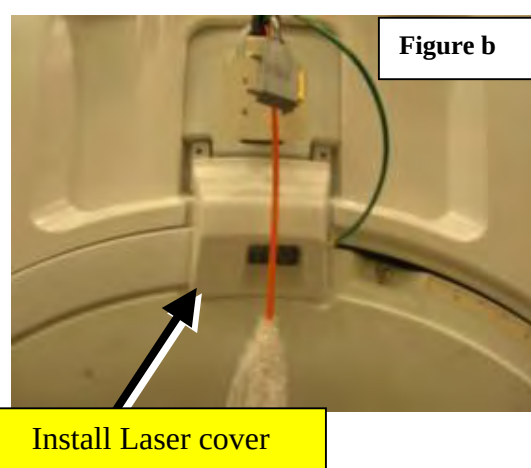
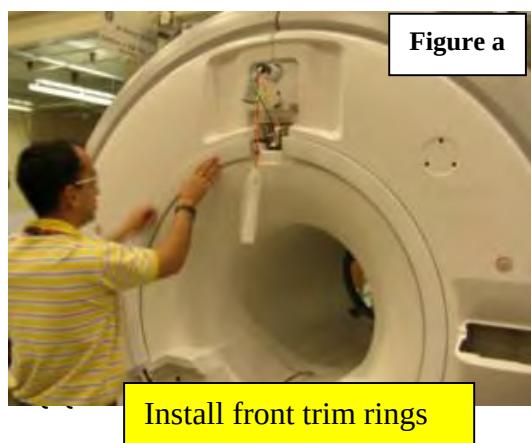
33. Front Cover Trim Installation

A. Components

- a) Laser Cover 5182894
- b) M6x16 SHCS 2109867-13
- c) Left Trim section 5177586
- d) Right Trim section 5177586-2

B. Install the front cosmetic cover left and right trim rings (5177586 and 5177586-2). **(Fig a)**

C. Install Laser bracket cover (5182894) using M6 x 16 socket head screws (2109867-13). **(Fig b)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

34. Logo Installation

A. Components:

- b) M10x25 SHCS 2109867-23
- c) M10 Nylon Washer 46-294167P73
- d) GE Logo 5125751

B. Apply the yellow laser-warning label above the top laser on the central control panel as shown. **(Fig 111)**

C. Verify the cover for the vacuum pump-out port is installed into the front cover. **(Fig 111)**



Discovery MR450 1.5T Magnet Enclosure Upgrade

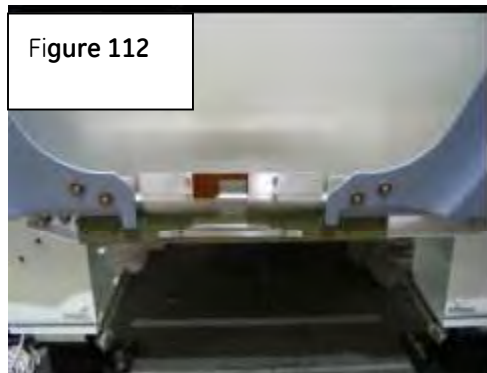
35. Side Cover Assembly

A. Components:

- | | |
|------------------|-----------|
| a) RH Top Cover | G6000_14 |
| b) RH Side Cover | 5151771-2 |

Note: The Right Side Top Cover (5169402), Left Side Top Cover (5170248-3), and the Left Side Cover (5151720-2) will be installed when the remaining magnet room components and cables are delivered to site for completion of the installation.

- B. Install the right side cover by engaging the spools in the arcs. Push the bottom of the cover down to the side bracket and engage the latch into the notch of the bracket. **(Fig 112)** Rotate the knob until it fully engages the catch (approx $\frac{3}{4}$ turn).
- C. Check the front-rear fit of the side covers and the arcs and also the fit of the rear cover to the rear arcs. Adjust the connection of the rear arcs to the lower support brackets to adjust the fit front-rear.
- D. Adjust the leveling pads on the front and rear arcs on the left side to minimize the gaps between the arcs and the other covers.



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